The Neurocognition of Longitudinal L3 Morphosyntactic Transfer: A Bayesian Single-Trial ERP Study of Three Grammatical Properties in Two Artificial Languages

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Abstract

Theories of third-language (L3) morphosyntactic transfer disagree over whether prior grammars are transferred whole or property by property. The Typological Primacy Model predicts that similarity in the lexicon selects the grammar that is transferred, whereas property-by-property accounts allow a prior-language property to be recruited whatever the lexicon resembles. We trained 65 Norwegian–English bilinguals on Mini-Norwegian or Mini-English, miniature languages sharing their grammatical morphemes but built on Norwegian or English words, and recorded grammaticality judgements and event-related potentials (ERPs) to three properties across four sessions, the last a retention test months later. For gender agreement, which Norwegian has and English lacks, signal-detection analyses of the judgements showed the Mini-Norwegian group to be no more sensitive than the Mini-English group at any session. In the ERPs, gender-agreement violations elicited a negativity that grew across sessions in the late window; neither it nor its growth differed credibly between groups. In the later sessions, Mini-Norwegian learners more readily judged gender-agreement sentences grammatical, ungrammatical ones included, a criterion difference not credible once each learner's change in criterion was modelled. Differential object marking and verb–object number agreement, absent from both prior languages, showed no credible group difference in sensitivity at any session. The absence of a Mini-Norwegian advantage is what property-by-property accounts allow and lexically cued wholesale transfer does not. Without a measurement before training or a comparison group lacking gender, the design narrows the debate without settling it. Lexical similarity brought no sensitivity advantage, and its influence on acceptance remains suggestive.

Keywords: L3 acquisition, morphosyntactic transfer, ERP, artificial language, signal detection theory, Bayesian multilevel models

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When multilingual speakers add a third or later language (L3/Ln), the grammars they already command shape how its morphosyntax is represented and processed. How that prior knowledge is recruited is a central question dividing contemporary models of L3 morphosyntactic transfer. Wholesale-transfer accounts hold that a single previously acquired grammar is selected as the basis for the initial L3 grammar, whether on typological grounds (the Typological Primacy Model: Rothman, 2011, 2015) or because the second language enjoys a privileged cognitive status (the L2 Status Factor: Bardel & Falk, 2007; Falk & Bardel, 2011). Property-by-property accounts predict instead that transfer is selective and incremental, with individual structures transferred (and, where helpful, "enhanced") one at a time as the learner detects structural similarity. The Linguistic Proximity Model (Westergaard et al., 2017, 2023; Westergaard, 2021), the Scalpel Model (Slabakova, 2017) and the Cumulative-Enhancement Model (Flynn et al., 2004) all take that form. The wholesale position has recent defenders who raise conceptual and empirical objections to selective transfer (Schwartz & Sprouse, 2021), and initial-stage evidence from early Catalan–Spanish bilinguals, whose performance on four properties was compatible with Catalan as the single source of their initial English grammar, has been read as support for it (González Alonso et al., 2021). Distinguishing these positions requires more than observing that prior-language congruence or a miniature language conditions performance. It requires contrasts for which the accounts make mutually exclusive predictions, measurements early enough to identify the initial representation before learning can obscure its provenance, and evidence that distinguishes representational transfer from transient effects of attention and processing (González Alonso et al., 2020, 2025). Longitudinal data are essential for characterising development, but later trajectories do not by themselves recover which representational state initiated that development (Puig-Mayenco, 2025).

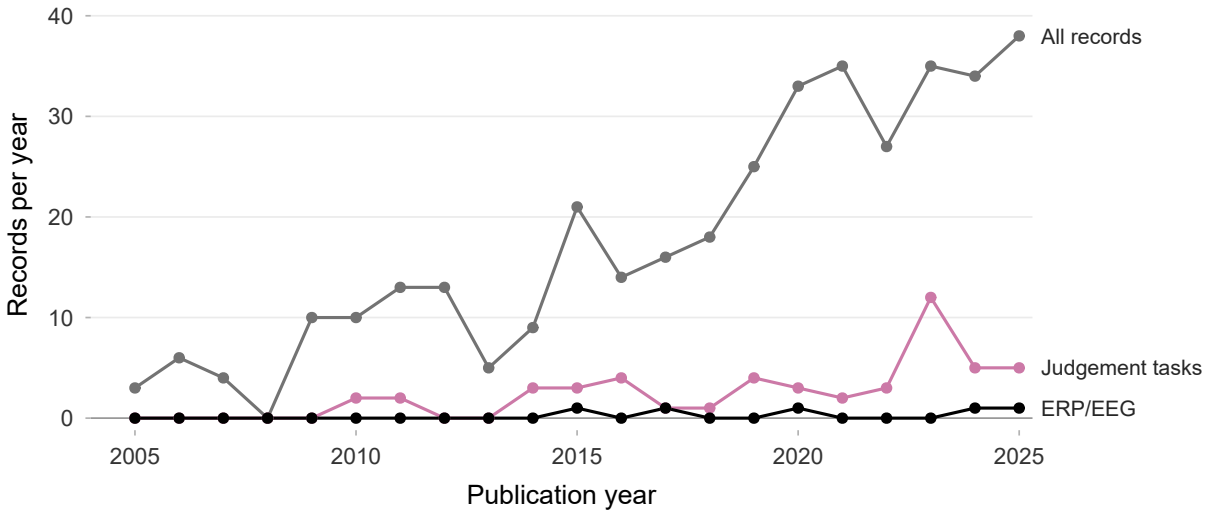
Electrophysiology of morphosyntactic processing

Event-related potentials (ERPs) provide a millisecond-resolution index of how the brain responds to grammatical violations and are therefore well suited to characterising when language-conditioned processing diverges, including before a difference surfaces in overt behaviour. They do not, on their own, establish that a difference reflects a copied grammatical representation rather than attention, anomaly detection or another process engaged by the task (González Alonso et al., 2020). Most evidence on L3 morphosyntactic transfer has come from offline, untimed acceptability judgements (Puig-Mayenco et al., 2020; Skałba et al., 2026). These index the outcome of learning. They leave untouched the moment-to-moment processing that ERPs expose (Angelovska & Roehm, 2023; Rothman et al., 2015). A single-trial ERP design can therefore test whether prior-language structure covaries with online comprehension. The imbalance is visible in the literature itself (Figure 1). In all, 369 records on L3 transfer were indexed between 2005 and 2025. Of those, 50 used acceptability or grammaticality judgements and 5 used electrophysiology, a ratio of roughly 10 to one. The field has grown steadily while its electrophysiological base has not. Online evidence has nonetheless begun to appear. Eye-tracking in early learners of Norwegian shows facilitative and non-facilitative influence from each prior language on different properties (Skałba et al., 2026).

Three components are relevant here. The left-anterior negativity (LAN) is often linked to automatic morphosyntactic agreement checking. The N400, a negativity sensitive to lexical–semantic and thematic integration (Frisch & Schlesewsky, 2001; Nieuwland et al., 2013), also marks the early stages of morphosyntactic learning, before a late positivity emerges with rising proficiency (Osterhout et al., 2006). That later positivity is the P600, first reported for words inconsistent with the preferred syntactic analysis (Osterhout & Holcomb, 1992). Over centro-posterior sites, it has since been associated with syntactic reanalysis and repair, and with the late integration of syntactic and semantic information (Friederici, 2002; Friederici et al., 2002). These map onto the three latency windows analysed here (200–500, 300–600 and 400–900 ms). Training studies show that the brain's response to second-language violations becomes more

Figure 1

Methodological Composition of the L3-Transfer Literature Indexed in Scopus, by Publication Year



Note. Each line gives the number of Scopus records per publication year. All records are those matching the L3-transfer query. Judgement tasks and ERP/EEG are the subsets whose title, abstract or keywords also mention acceptability or grammaticality judgements, or event-related potentials, ERP or EEG (electroencephalography). The two subsets can overlap, and none of the three is restricted to morphosyntax. Where the two subsets have the same count, only the ERP/EEG line is visible. The counts index relative composition and are not an exhaustive census: Scopus under-covers books and chapters, keyword matching errs in both directions, and records indexed late may still be added for the most recent years. The retrieval queries are recorded in the project repository.

native-like as proficiency increases (Morgan-Short, Steinhauer, et al., 2012), and again across a consolidation interval containing no further exposure (Morgan-Short, Finger, et al., 2012), with negativities and P600s emerging or shifting over time. Native speakers (Tanner & Van Hell, 2014) and learners (Tanner et al., 2013) also differ systematically from one another in whether they recruit a more negativity-dominated or a more P600-dominated response. Artificial-language paradigms, in which the input is fully controlled, have been particularly valuable for isolating these dynamics from the confounds of natural-language exposure (Friederici et al., 2002; González Alonso et al., 2020). Bilinguals learning such a language show a P600 to syntactic violations at a lower proficiency than monolinguals do (Grey et al., 2018).

The closest antecedent to the present design is the artificial-language L3 ERP work of González Alonso et al. (2020). Its findings shape both our predictions and our reading of the results,

so we set them out in full. Fifty first-language-Spanish, second-language-English bilinguals were given implicit training on one of two miniature languages, built on English and Spanish lexicons, and their ERPs to gender-agreement violations were recorded at first exposure. They named their two languages after those lexicons, and we refer to their groups by lexicon throughout, so that neither is confused with our own Mini-English and Mini-Norwegian conditions. The two groups responded differently. In their English-lexicon language, violations elicited an early (200–500 ms) fronto-lateral negativity, confined to electrodes in the right-anterior region and followed by nothing in later windows. In their Spanish-lexicon language, they elicited a broadly distributed positivity at 300–600 ms. Neither group produced a P600 or any other canonical index of grammatical processing, and judgement accuracy did not differ between the groups. The authors read the between-group difference as differential allocation of attention, driven by each miniature language's lexical similarity to a prior grammar, and therefore as a precursor of source selection, not as grammatical processing itself. The present study extends that logic in three ways. It goes beyond first exposure, tracking each property across up to four sessions, and it manipulates three properties whose relation to the prior grammars differs within the same learners. It also closes with an unrehearsed retention test after months without exposure, so the durability of the observed processing effects can be measured. Whether the initial-exposure pattern of González Alonso et al. (2020) reappears under these conditions is itself an empirical question, taken up in the Discussion.

Two behavioural studies share more of the present materials. Each trained speakers on a miniature language with a Norwegian lexicon and a case system, a property Norwegian lacks. Prior knowledge of a case language eased acquisition even though the lexicon pointed to Norwegian, but only where that language and the miniature language marked case in the same overt form, on the noun (Castle et al., 2026; Mitrofanova et al., 2023). Greek, which marks case on the determiner, gave no such advantage in the first study, and the second found the advantage subject to word-order bias and proficiency. The overt realisation of a property conditioned crosslinguistic influence, and the present gender manipulation varies exactly that realisation between the two miniature languages.

Recent syntheses of the transfer literature draw attention to the methodological choices that

constrain what its findings can show (González Alonso et al., 2026; Puig-Mayenco et al., 2020; Rothman et al., 2019), and longitudinal and electrophysiological evidence remains among its scarcest. An ERP successor to González Alonso et al. (2020) trained heritage and second-language bilinguals on a Latin mini-grammar and found violation responses that no current model had predicted, an outcome its authors took as a question about the paradigm itself (Pereira Soares et al., 2022). The recurring objection to miniature languages is that a grammar acquired in hours may not engage the mechanisms of natural morphosyntax. This literature answers it with the finding that artificial-language learners, once sufficiently proficient, show qualitatively native-like violation responses (Friederici et al., 2002; Morgan-Short, Steinhauer, et al., 2012). What the design buys in exchange is exact control over every learner's input history, which no natural-language design can offer.

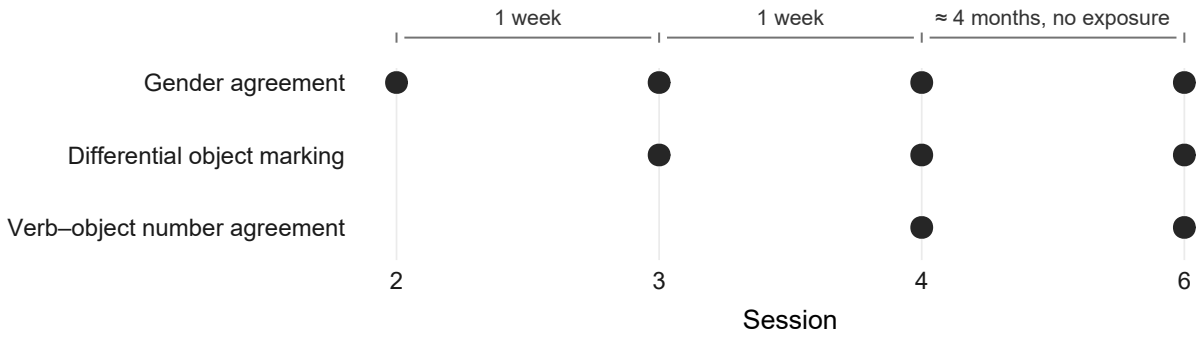
The present study

We use a controlled artificial-language design to ask whether learners' sensitivity to three morphosyntactic properties, in judgements and in the ERP response, differs between two miniature languages, and how that sensitivity develops from the first post-training session to a retention test months later. Norwegian–English bilinguals (first language Norwegian, second language English) acquired one of two miniature languages, Mini-English or Mini-Norwegian. Within each language, we manipulated three morphosyntactic properties that vary in their relationship to the learners' existing grammars: subject–verb gender agreement, differential object marking and verb–object number agreement. The two languages share their grammatical morphemes and differ in two respects. Their content words are built on English and on Norwegian stems respectively, and gender is marked by a free prenominal determiner in Mini-English but by a determiner suffixed to the noun in Mini-Norwegian, as the Norwegian definite article is (see Method).

Single-trial ERPs and grammaticality judgements were collected across four laboratory sessions (Sessions 2, 3, 4 and 6) that trace the learning trajectory from the first post-training session to a post-consolidation retention test. The properties enter the design at different points (gender agreement from Session 2, differential object marking from Session 3 and verb–object number

agreement from Session 4), yielding a cumulative, property-by-property exposure schedule (Figure 2).

Figure 2
Exposure and Measurement Schedule



Note. Points mark the sessions at which each grammatical property was presented and tested. Each property enters at a different session and recurs at every later one, so gender agreement is tested at four sessions, differential object marking at three and verb–object number agreement at two. Sessions 1 and 5 were home sessions of the cognitive battery analysed in a companion paper, and Session 1 also included the Language History Questionnaire. Neither session included training or testing in the miniature language, so both are omitted. The session axis is ordinal, and equal spacing does not imply equal elapsed time. The brackets above the grid give the intervals: Sessions 2 to 4 were one week apart, and Session 6, the unrehearsed retention test, followed Session 4 after about four months without exposure.

The transfer accounts make their clearest predictions for gender agreement, the only one of the three properties that Norwegian has and English lacks. The Typological Primacy Model holds that the parser selects a whole prior grammar on the basis of overall similarity, assessed from a hierarchy of cues in which the lexicon comes first (Rothman, 2015). A Mini-Norwegian learner should therefore draw on Norwegian, gender included, and a Mini-English learner on English, which has no grammatical gender. The model predicts greater sensitivity to gender-agreement violations in the Mini-Norwegian group, in the judgements and in the ERP response, at least at the earliest sessions, before further input could overcome a non-facilitative choice. This is the prediction that Rothman et al. (2015) derived for the analogous Mini-Spanish and Mini-English groups. The L2 Status Factor predicts instead that both groups draw on English (Bardel & Falk,

2007; Falk & Bardel, 2011), so that neither gains an advantage from Norwegian gender. Property-by-property accounts make no such prediction from grammatical knowledge alone. The Norwegian gender system remains available to every participant, and a learner of either miniature language can recruit it once the input reveals that the property is shared (Westergaard et al., 2017; Westergaard, 2021). On that derivation, the two groups need not differ.

Two features of the design limit what a group difference could show. First, the Linguistic Proximity Model allows superficial, especially lexical, similarity to sway influence at the earliest stages, before learners can detect abstract structural similarity (Westergaard et al., 2017; Westergaard, 2021). A Mini-Norwegian advantage would therefore not refute property-by-property transfer, a point made both about artificial-language evidence of this kind (Jensen & Westergaard, 2023) and about the closest ERP precedent (González Alonso et al., 2020). Second, the two miniature languages differ in structure as well as in lexicon, because Mini-Norwegian marks gender on a suffixed determiner, as Norwegian does. Structural similarity of that kind is what property-by-property accounts take to trigger selective transfer, so they too could expect Mini-Norwegian learners to recruit Norwegian gender more readily. A Mini-Norwegian advantage in gender-agreement sensitivity is therefore compatible with every account except the L2 Status Factor. The accounts separate on its absence, which the Typological Primacy Model does not allow and property-by-property accounts do. Differential object marking and verb–object number agreement are absent from both prior languages and realised identically in the two miniature languages, so no account predicts a group difference in sensitivity to either. They serve as controls for group differences that arise for reasons other than the prior grammars, such as a difference in how easily the two lexicons are learned.

These predictions concern the source of grammatical knowledge, but the measures record processing and decisions. A judgement also reflects how readily a learner accepts sentences at all, and a violation response can reflect attention or anomaly detection as well as grammatical computation (González Alonso et al., 2020). We therefore separate sensitivity to grammaticality from the criterion applied in the judgement, using signal detection theory (Huang & Ferreira, 2020;

Macmillan & Creelman, 2005), and read a group difference in sensitivity as the stronger evidence about grammatical knowledge in this design. The development of sensitivity across sessions, and its change over the retention interval, are characterised for each property. The transfer accounts make no distinct prediction about retention, for which the expectation comes from training studies in which a grammaticality response that emerged during training holds or grows across months without exposure (Morgan-Short, Finger, et al., 2012). Because the first measurement follows training, no observation here captures an initial state free of learning (González Alonso et al., 2025; González Alonso & Rothman, 2017).

Alongside the region-averaged models, we ask whether grammaticality is decodable from the whole-scalp voltage pattern at each moment. A multivariate decoder can detect a distributed violation response too weak to move any single region's mean. Its cross-language generalisation asks whether the pattern that separates the conditions in one mini-language also separates them in the other. Its cross-property generalisation tests a further prediction. If the grammaticality response were largely a domain-general salience or integration signal (Aurnhammer et al., 2023; Sassenhagen & Fiebach, 2019), a decoder trained on one property should transfer to the others. If each property recruits its own code, as the fractionation of the late positivity (Kaan & Swaab, 2003) and the separable developmental courses of agreement features (Alemán Bañón et al., 2018) suggest, it should not. No null distribution was computed for either generalisation analysis, so both are reported as descriptive probes.

Method

Participants

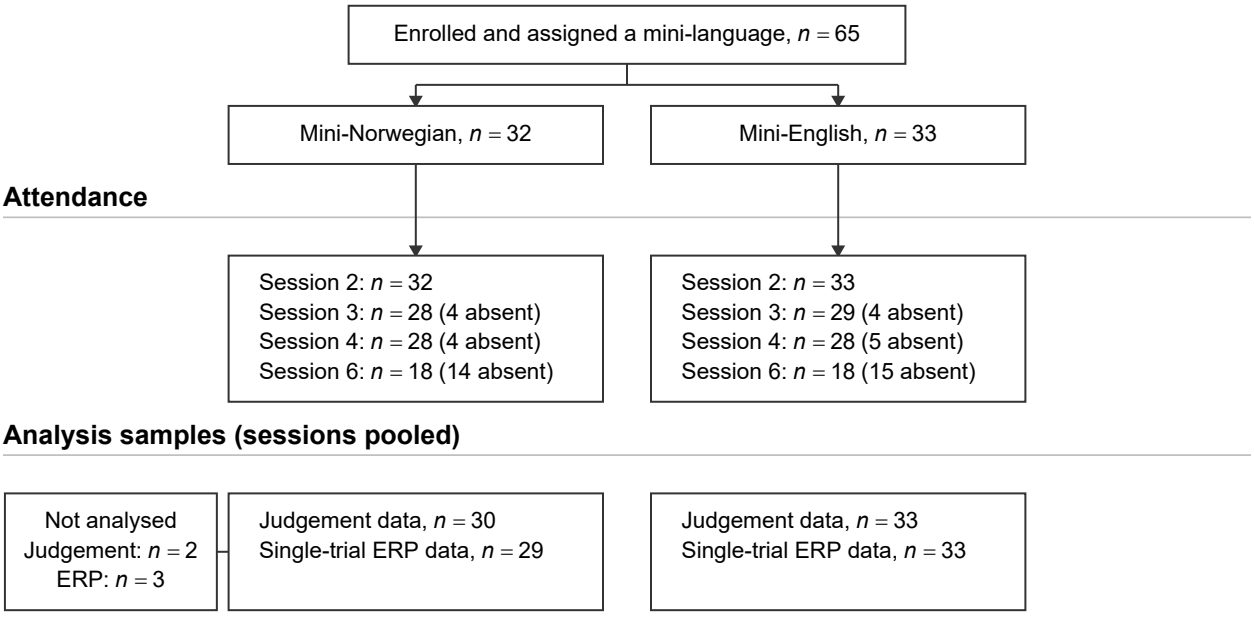
In total, 65 Norwegian–English bilinguals (first language Norwegian, second language English) took part, each assigned to learn one of the two miniature languages. At the Norwegian site, odd-numbered participant identifiers were assigned to Mini-English and even-numbered identifiers to Mini-Norwegian, a deterministic allocation and not a random one. That rule gave 33 Mini-English and 32 Mini-Norwegian learners, so language is a between-participants factor. The six-session protocol comprised an approximately 1.5-hour home baseline (Session 1), laboratory

sessions of approximately 2.5 hours (Session 2), 2.5 hours (Session 3), and 3 hours (Session 4), a 1-hour home retest (Session 5), and a 2.5-hour laboratory retention session (Session 6). Of the 64 participants with a Language History Questionnaire record (LHQ3: Li et al., 2020), the mean age was 25.7 years (SD = 7.4, range 16–54). The youngest participant was under 18 at the time of the questionnaire. All others were adults. Of the same record, 40 identified as female, 21 as male and 1 as non-binary (2 did not disclose). Among the 63 who reported handedness, 60 were right-handed and 3 left-handed. English had been acquired early (age of first exposure M = 6.9 years, SD = 3.9) and used for many years (M = 18.5, SD = 7.2), with high self-rated proficiency (M = 0.82 of 1, SD = 0.16). 48 of the 64 also reported knowledge of at least one further language, most often German or Spanish. Several of the languages named mark grammatical gender or case, and the design did not screen them out (see Limitations).

Attrition increased across the training: of the 65 who began, 57 remained at Session 3, 56 at Session 4, and 36 completed the Session 6 retention test. One participant assigned to Mini-Norwegian is excluded from every analysis, although they are counted among those enrolled and attending. At Session 2, the laboratory computer ran the other study site's experiment file, which assigned that participant's identifier to Mini-English, so they were trained and tested in Mini-English at that session and in Mini-Norwegian from Session 3 onwards. The presentation log and the EEG trigger record agree on what was shown, so the error lies in what was presented, not in how it was recorded. Because the lexicon and the first property are trained at Session 2, this participant's trajectory belongs to neither group. The exclusion rests on that design argument and not on this participant's results, which were not examined separately. The ERP analyses draw on the 62 participants with usable single-trial EEG (33 Mini-English, 29 Mini-Norwegian). The fitted ERP models also require the L1-dominance covariate described under *Data analysis*, and the trials of a participant without that score are dropped by listwise deletion, so the gender-agreement models were fitted to 62 participants, the differential-object-marking models to 56 and the verb–object number-agreement models to 55. The grammaticality-judgement analyses draw on 63 participants (Figure 3).

The project was registered with the Norwegian Agency for Shared Services in Education and Research (Sikt; project 2559611). *[The approving ethics committee, approval date and reference number are to be supplied.]* All procedures complied with the relevant laws and institutional guidelines, the privacy rights of the participants were observed, and every participant gave written informed consent. The 16-year-old participant was covered by the study's self-consent procedure. Participants were compensated with shopping vouchers worth NOK 250 per hour of participation, pro rata, and received an additional NOK 400 completion bonus after the sixth and final session.

Figure 3
Participant Flow



Note. Counts are participants. Each attendance line gives the number who attended that session and, in parentheses when any were absent, the number of the group's enrolled participants who did not. Sessions 1 and 5 were online cognitive sessions belonging to the companion paper and are omitted. The analysis samples pool data across sessions, so they are not subsets of any single session's attendance and exceed the Session 6 count. The box joined to a group's analysis samples counts that group's enrolled participants missing from each sample, who lacked usable data for that measure or were excluded by design (see Participants). ERP = event-related potential.

Design and materials

The longitudinal artificial-language paradigm and the participants are those of the project reported by González Alonso et al. (2025), restricted to its Norwegian site, so that project's Spanish site is not included here. Within each language, the design crosses grammaticality (contrast coded grammatical = +0.5, ungrammatical = −0.5) with session (the cumulative exposure schedule above) within participants, and language between participants (Mini-Norwegian = +0.5, Mini-English = −0.5), so a positive language-bearing coefficient indicates a larger value for the Mini-Norwegian group. The three properties were analysed separately. Each was probed with 48 grammatical and 48 ungrammatical sentences per session, built from a closed lexicon of twelve nouns, twelve adjectives, twelve transitive verbs and twelve proper names per language. A third, ancillary violation type was recorded but is not analysed here. Each sentence frame was presented in both a grammatical and an ungrammatical version. The two miniature languages draw on separate content lexicons, Mini-English building its nouns, adjectives, verbs and names on English stems and Mini-Norwegian on Norwegian stems. The critical word carried the EEG trigger that defined the single-trial time-lock.

The two miniature languages share the same invented grammatical morphemes (the determiners *je*/*ze*/*jer*/*zer*, the object marker *fi*, the verbal suffix *-evo* and the adjectival agreement suffixes). They differ in one structural dimension, determiner position, and that difference is the pivot of the transfer contrast. Subject–verb gender agreement is built on a two-way common/neuter distinction, as in real Norwegian. Each noun carries the gender of its Norwegian translation equivalent, and the same noun carries the same gender in both miniature languages, so gender assignment never conflicts with Norwegian and the two languages differ in where gender is marked and not in which gender a noun takes. In Mini-English, the determiner is a free prenominal word that marks gender and number (*je*/*ze* for common/neuter singular, *jer*/*zer* for the plural), as English articles are free-standing. In Mini-Norwegian, the same gender–number determiner is a bound suffix on the noun (e.g. *gateje*/*gatejer*), as the Norwegian definite article is suffixal. In both languages, the predicate adjective carries a matching agreement suffix, on which the

gender-agreement violation is realised. Differential object marking is a free particle *fi* placed before the object noun phrase, identical in the two languages, with the violation omitting it. Verb–object number agreement is a verbal suffix *-evo* that marks the verb to agree with a plural object, again identical across languages, with the violation a number mismatch between verb and object.

These three properties map onto the learners' existing grammars differently, which is what the language factor exploits. Grammatical gender expressed through a suffixed determiner is a property of the learners' first language, Norwegian, but not of their second language, English. The Mini-Norwegian realisation therefore aligns closely with the first-language grammar, whereas the Mini-English realisation expresses a feature that English does not encode (gender) through an English-like free determiner. Differential object marking and verb–object number agreement, by contrast, are absent from both Norwegian and English, so neither is supported by a prior grammar in either group.

Procedure

Sentences were presented one word at a time in the centre of the screen (rapid serial visual presentation, RSVP) using OpenSesame 3.3.14. Word duration was a deterministic function of word length. A non-critical word was shown for $250 + 35 \times (\text{characters} - 3)$ ms, floored at 250 ms, giving durations of 250–600 ms. The critical word was shown for 40 ms less than a non-critical word of the same length (210–560 ms). Non-critical words followed one another at a stimulus-onset asynchrony of approximately 505 ms. At the critical word, the asynchrony was held fixed at approximately 750 ms, with the following blank interval lengthened to compensate for the varying word duration. The timing of the post-critical region therefore did not depend on the length of the critical word. The critical word occupied the third or fourth position and was followed by three to six further words. Presentation was entirely experimenter-paced. Nothing in the sentence stream depended on the participant, so a condition difference in reading speed cannot shift the epoch. The grammaticality judgement was requested only after the final word of the sentence, never during it, and was made with a two-button response (left = grammatical, right = violation). No motor response therefore fell inside the analysed epoch. Participants were instructed to minimise

movement and blinking, especially during the presentation of words. During pauses in these long sessions, experimenters monitored the signal and intervened when necessary to maintain low electrode impedance and acceptable signal quality.

This timing has one consequence for the late window. The raw presentation logs record all 64,907 critical-word trials, a census taken before any EEG-side exclusion, so its count exceeds the modelled trial numbers. In that census, exactly one subsequent word onset falls within 900 ms of the critical word (the per-trial maximum is 1), at 743–864 ms (median 747 ms). The final roughly 150 ms of the 400–900 ms measurement window therefore contains the beginning of the response to the following word. This is unavoidable in connected-sentence RSVP at this rate, and it is constant across conditions because the post-critical asynchrony is held fixed, so it cannot manufacture a grammaticality effect. Late-window amplitudes should nonetheless be read as containing that overlap, not as a pure response to the critical word.

EEG acquisition and single-trial ERP quantification

Continuous EEG was recorded with a 32-channel Brain Products LiveAmp system, using active electrodes in an extended 10–20 layout with the CACS-32 cap montage recorded in the data headers. The sampling rate was 500 Hz and the online passband DC–131 Hz, with no online notch filter. The 131 Hz anti-aliasing low-pass is a causal acquisition filter, so it imposes no phase distortion of the kind that motivates the offline filter choices set out below.

Three channel counts appear in what follows, and they refer to different things, so we set them out once. Thirty-two electrodes were recorded. Two of them, TP9 and TP10, become the offline linked-mastoid reference and are therefore not analysed as signals. FCz served as the online reference and is recovered as a channel when the data are re-referenced. The remaining 30 electrodes are the ones assigned to the nine analysis regions, so they are what the region-averaged amplitude models use and the electrodes from which the scalp maps are estimated. The decoder ignores the region map and takes every EEG channel available in all contributing sessions, which is those 30 plus the recovered FCz, giving 31 features. The LiveAmp's three accelerometer traces (`x_dir`, `y_dir`, `z_dir`) are movement signals, not brain signals, and are excluded throughout. The

Session 2 exports additionally retain TP9, TP10 and the accelerometer channels, whereas the Session 3, 4 and 6 exports do not, which is why the decoder uses only the channels common to every session. Offline, in BrainVision Analyzer, the data were re-referenced to the mean of the two mastoids, with FCz recovered, and then band-pass filtered with zero-phase-shift (forward and reverse) Butterworth IIR filters, high-pass 0.1 Hz and low-pass 30 Hz, each specified in Analyzer as fourth order. Analyzer realises that specification as a second-order filter applied forwards and then backwards, so the combined zero-phase response is the fourth-order one, falling off at 24 dB per octave. A scripted reconstruction of the processing chain reproduced Analyzer's exported data to the precision of the export only under that reading. No notch filter was applied offline. Filtering preceded ocular correction, so the independent component analysis was trained on the same 0.1 Hz high-passed data that the analyses themselves use. It was computed on each whole recording, leaving out intervals marked as bad during raw-data inspection, and components were selected semi-automatically. In Sessions 3, 4 and 6, TP9 and TP10 had been discarded at re-referencing, so the decomposition used the 30 remaining electrodes (29 in one Session 3 recording in which CP2 had been interpolated). In Session 2 it used 32 electrodes, the 30 scalp electrodes and TP9 and TP10. The recovered FCz passed through it unchanged in every session.

In Session 2, the re-referencing step also included the online reference in the average, so those recordings were exported against the mean of TP9, TP10 and a zero-valued FCz instead of against the mean of the mastoids, and TP9 and TP10 were kept. We brought them to the reference of the other sessions offline, as each export was read. Subtracting a sixth of the sum of the exported TP9 and TP10 from each scalp electrode re-references the ocularly corrected scalp data to the ocularly corrected mastoids, an identity that holds on the exports to their precision. FCz bypassed the ocular correction, so 1.5 times its exported value is the linked-mastoid FCz exactly. Against a linked-mastoid branch that the analyst had built, but not exported, in one participant's Session 2 history, the correction was four times more accurate than a simpler alternative that adds half of the exported FCz to every channel. That alternative is exact before ocular correction but not after it, and it was applied in an earlier version of these analyses. Without any correction, the Session 2

grammaticality effects in four recordings rebuilt from the raw data would have been displaced by up to 2.3 µV relative to the other sessions.

Owing to a processing oversight, the Session 3 recordings of nine participants were high-pass filtered at 1 Hz rather than 0.1 Hz. The archived Analyzer histories show a single 1 Hz filter node from which their exports descend, and a scripted rebuild of those exports from the raw recordings reproduces them only at 1 Hz. A 1 Hz high-pass is a documented generator of artefacts in this kind of paradigm (Tanner et al., 2015; Widmann et al., 2015), so the 1 Hz data enter no analysis. The nine recordings were instead reprocessed from the raw data under a protocol fixed before any reprocessed output was inspected. A scripted reconstruction of the archived processing first reproduced every one of their 1 Hz exports to within the rounding of the export (root-mean-square error below 0.0004 µV), and only then was the high-pass cutoff changed to 0.1 Hz, with every other step and every retained trial kept as archived. The reprocessed data enter every fitted ERP model, the decoding and the ERP displays. A robustness analysis fixed in the same protocol refits the ERP models and the confirmatory decoding without them. It can qualify a primary result but not create one. The decision rests on the filter setting itself and on nothing observed in these participants' data, so it is not results-contingent. The argument is set out in Appendix A. The reprocessed datasets hold 16.0% of the retained Session 3 gender agreement trials and 16.6% of the retained Session 3 differential object marking trials.

Every exported trial was then traced to its source recording through the export's time stamps and the processing history, and every export was checked for a stimulus-locked response. Three exports failed, and each failure was confirmed on the EEG itself. The Session 3 export of one participant is a second copy of another participant's Session 3 recording. The Session 6 export of a second participant was segmented at the event times of that participant's Session 4 recording, and the Session 2 export of a third participant at the event times of a recording other than the one archived. The epochs of the last two are not time-locked to any stimulus and show no evoked response. None of the three exports measures the participant-session it is named for. The same protocol allows a misaligned record to be recovered only if its archived processing decisions were

made on the participant's own recording, which it tests by first reproducing the legacy export from that recording. The two misaligned records passed: each export was reproduced from the participant's own archived recording and the event markers its processing history carries, to the same precision as the filter reconstruction, so only the event markers had come from another recording. Each record was then re-segmented at its own events by the trigger rule described below, inside the stretch of recording the analyst had processed and inspected, with every epoch that overlaps a marked bad interval dropped and no manual epoch removal carried over. The recovered records enter the primary analysis, and the robustness analysis drops them together with the nine reprocessed ones. The duplicated Session 3 export cannot be recovered, because none of its processing decisions was made on that participant's recording, so it is excluded from every EEG analysis. The accuracy models retain all of these sessions, because neither the filter nor the segmentation bears on the judgement logs.

Each critical word was marked by four serial-port triggers sent in succession as the word was drawn: the word identity, which marks word onset, then the sentence identity, the grammatical property and the grammaticality condition. After ocular correction, each record was segmented in two steps. The first step cut 1,200 ms epochs beginning 120 ms before the property trigger, and these were screened semi-automatically: epochs containing a voltage step above 30 μV/ms, a range above 100 μV within 200 ms or activity below 0.5 μV within 100 ms were flagged, and the operator reviewed the flagged epochs, removing some and keeping most. The second step cut 1,200 ms epochs beginning 130 ms before the condition trigger, and kept an epoch only if it lay wholly within a retained first-step epoch. That holds only when the condition trigger fell exactly five samples after the property trigger. The triggers had a nominal spacing of 10 ms, but the realised spacing averaged about 10.7 ms, so at 500 Hz the two triggers were six samples apart for about a third of critical words, and those words yielded no epoch. An audit of the behavioural logs, marker files, processing histories and exports of every participant-session traced each loss to its cause: in the processing histories, 21,520 of the 64,287 critical-word events had an interval other than five samples, and none of them reached an export, whereas 473 first-step epochs were removed on review, every one

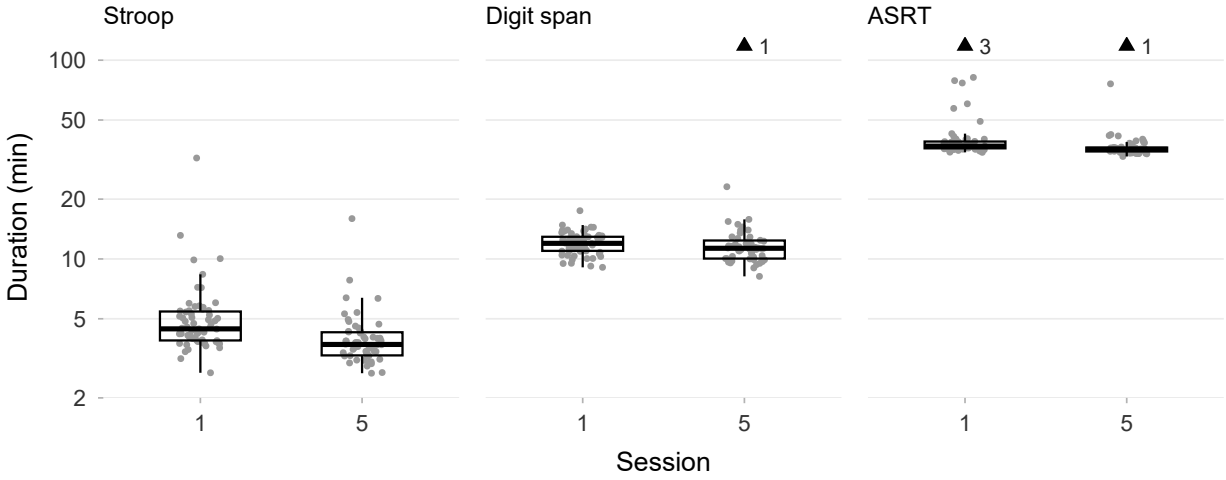
of them flagged by the criteria (5,679 were flagged in all). The share of words kept by the trigger rule ranged from 24% to 81% across recordings, mainly with how quickly the stimulus computer sent the triggers. Within recordings, loss was unrelated to grammaticality (odds of losing an ungrammatical rather than a grammatical trial 1.01, 95% CI [0.97, 1.06]), response accuracy and item, and only weakly and inconsistently related to response time and trial position, so it lowers precision without favouring either side of the grammaticality contrast. It was not perfectly even across groups and sessions, however. The largest imbalance was in the Mini-English group at Session 6, where grammatical retention fell below ungrammatical retention for every property, by 4.2, 1.1 and 6.7 percentage points of the 48 presented trials for gender agreement, differential object marking and verb–object number agreement, and by 4.0 points on average (odds ratio 0.83, 95% CI [0.73, 0.94]). A segmentation tolerant of one sample of trigger jitter would have kept 1.5 times as many trials, but projected onto the trial-level structure of the ERP models it would have narrowed the interval of the gender-agreement grammaticality × language term by only 9–10%, so the recordings were not re-exported. Differential trial loss between the compared conditions would make a grammaticality effect partly a difference in measurement precision and only partly one in brain response, so retention is reported per condition. On average, 31.5 of the 48 presented trials were retained per modelled cell ($SD = 5.8$), so 34.3% were not retained. These retention figures are counted from the single-trial data the models were fitted to, after every exclusion and with the reprocessed and recovered records as analysed. Table 1 gives the full per-condition breakdown, averaged over the two groups. Averaged over properties and sessions as well, a participant retained 31.48 of the 48 grammatical trials and 31.43 of the 48 ungrammatical ones, a difference of 0.05 trials. Epoch time is labelled from −100 to 1098 ms relative to word onset on the assumption of the nominal trigger spacing; relative to the recorded word-identity trigger, the labels understate elapsed time by 1.6 ms on average and by at most 8 ms, which is negligible for window mean amplitudes. The audit's scripts and the tables these figures are read from are in the project repository (`paper_1_transfer/audit/`).

The observed elapsed duration of each home task and of each phase of the laboratory

sessions is shown in Figure 4 and Figure 5. These are descriptive timing checks, not additional outcomes: home-task duration runs from the first to the last Gorilla event for each participant and task, whereas lab-phase duration uses the OpenSesame phase markers and the final event in a phase when the next phase was recorded in a separate export. The median home-task durations were 4.45 min for Stroop, 11.98 min for digit span and 36.94 min for ASRT in Session 1, and 3.71, 11.32 and 35.61 min, respectively, in Session 5. Median laboratory-phase durations (pretraining, training, test and experiment) were 4.21, 16.14, 9.93 and 47.49 min in Session 2; 1.57, 18.97, 7.18 and 52.54 min in Session 3; 16.73, 20.08 and 78.37 min for training, test and experiment in Session 4, where pretraining was not recorded; and 73.42 min for the experiment in Session 6, where the other phases were not recorded.

Figure 4

Duration of the Home Cognitive Tasks by Session

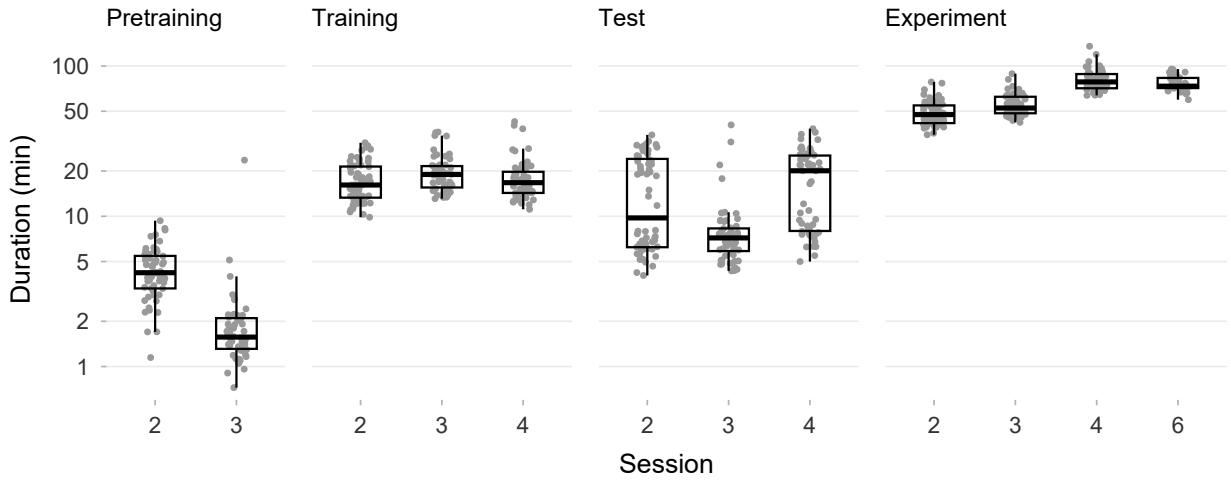


Note. Points are participants, and each box summarises their durations for one task and session: the thick line is the median, the box spans the interquartile range, and the whiskers reach the most extreme duration within 1.5 times the interquartile range of the box, computed on the logarithmic scale of the vertical axis. Duration is the elapsed time from the first to the last raw Gorilla event, so it includes instructions, practice and test material when those events are present in the export. Session 1 is the baseline and Session 5 the post-training session. Durations longer than 100 min, each produced by a multi-hour idle gap in the export, lie beyond the top of the axis; a triangle marks them, with their number beside it, and they are included in the box statistics. Participants without a linked enrolled-participant key are excluded. ASRT = alternating serial reaction time.

For each retained trial, we computed the mean amplitude in three theoretically motivated

Figure 5

Duration of the Laboratory-Session Phases



Note. Points are participants, and each box summarises their durations for one phase and session: the thick line is the median, the box spans the interquartile range, and the whiskers reach the most extreme duration within 1.5 times the interquartile range of the box, computed on the logarithmic scale of the vertical axis. A phase runs from its recorded start marker to the next phase's start marker or, when the next phase was exported separately, to the last event of the phase. Pretraining was not recorded at Session 4, and only the experiment phase was recorded at Session 6, so those sessions are absent from the other panels. Sessions 2, 3 and 4 are training sessions, one week apart, and Session 6 is the retention session, about four months after Session 4 without exposure. Sessions are equally spaced on the horizontal axis, which is therefore not proportional to elapsed time.

latency windows, 200–500 ms (the LAN/early-negativity range), 300–600 ms (the N400 range) and 400–900 ms (the P600 range), at lateral and midline electrode macroregions. The windows are named by their nominal bounds throughout. On the 2 ms acquisition grid, each is inclusive of samples up to 498, 598 and 898 ms respectively. That convention follows the validated import procedure and is noted here so that the figures and the deposited code can be read against the prose. The lateral region comprises six clusters, left- and right-hemisphere anterior, medial and posterior. The midline region comprises three clusters centred on the midline. Those clusters are parasagittal, so they are not strictly midline. Besides Fz, Cz, Pz and Oz they include the pairs FC1/FC2 and CP1/CP2, which the cap places about a quarter of the head radius either side of it. Because each pair is exactly symmetric, however, the cluster mean remains centred on the midline and has no net hemispheric bias. Each trial's amplitude in a macroregion is the mean over its clusters, each cluster

weighted equally whatever its number of electrodes, so every trial contributes one observation per macroregion. Two further choices, the within-participant z-scoring of single-trial amplitudes and the treatment of each trial's pre-stimulus mean amplitude (−100 to 0 ms) as a covariate instead of a subtracted baseline, are modelling decisions and are set out under *Data analysis*.

Table 1
Mean Number of EEG Trials Retained per Participant and Condition, by Grammatical Property and Session

	Session			
Condition	2	3	4	6
Gender agreement				
Grammatical	31.6	32.2	32.2	30.3
Ungrammatical	30.7	31.9	31.6	30.9
Differential object marking				
Grammatical		32.0	31.9	30.7
Ungrammatical		30.8	31.7	31.5
Verb–object number agreement				
Grammatical			31.7	30.8
Ungrammatical			32.1	31.7

Note. Values are means over participants of the trials retained out of the 48 presented per condition, averaged over the two language groups. Rows are split by grammaticality because differential trial loss across the compared conditions can bias both amplitude and decodability. The two conditions are retained almost equally here (see text). A blank cell marks a session at which the property had not yet been introduced. The counts are those of the single-trial data the models were fitted to.

Grammaticality-judgement task

The grammaticality judgement was collected concurrently with the ERP recording. On each experimental trial, participants judged whether the sentence was grammatical for the property under test, so the trial-level accuracy (correct/incorrect) and the single-trial ERP are two measures of the same event. The scored judgement is the sole behavioural measure, and it is analysed as sensitivity and criterion (see *Data analysis*). Rows that the presentation software logged twice under one trial number (68, each a re-logging of a file's last trial when the session closed) were first reduced to one per participant, session, property and trial. Every judgement was then analysed except the three without a recorded response time, leaving 41,181 trials. Two features of the Session 4 logs removed

further judgements. The runs of two participants were exported in two files whose names the extraction's file pattern does not match, and the logs of two participants were exported with a semicolon delimiter that the comma-delimited reader cannot parse. Those participants contribute no Session 4 judgements and enter the models through their other sessions only. The project's earlier preprocessing also discarded judgements of 200 ms or less or of 4,000 ms or more. That screen removes a further 8,159 trials (19.8%), and we do not apply it, for three reasons. Its bounds have no recorded rationale. It selects on the outcome. The judgement was requested only after the sentence had ended, so a response within 200 ms of the request was a decision already made, and these responses were the most accurate in the data (87.5% correct), whereas those of 4,000 ms or more were the least accurate (73.6%), against 78.6% for the judgements the screen keeps. And it does not fall on the two groups alike: it removes 22.9% of the Mini-English group's judgements and 16.5% of the Mini-Norwegian group's, mostly through the fast responses (11.9% against 6.9%). The Results report the accuracy models refitted with the screen as a sensitivity analysis. The unlabelled filler sentences shown at Session 2 carry no property and fall outside the modelled contrast. Every judgement sentence filled one of the sentence frames of its property, which the presentation software marked with the same trigger code in the EEG, so the judgement trials share the ERP models' item factor (see *Data analysis*).

Data analysis

All models were fitted in a fully Bayesian framework with `brms` (Bürkner, 2017, 2018) as a front end to the Stan No-U-Turn sampler (Carpenter et al., 2017). A Bayesian treatment is not incidental to the question here. The hypotheses concern the size and direction of effects that are expected to be small. Several of the comparisons that matter, such as whether a property's language-conditioned effect grows across sessions, are naturally expressed as statements about a posterior. Priors, sampler settings, diagnostics and inference rules are set out in the sections that follow.

The analysis and reporting code was written with the assistance of Claude Code (version 2.1, Anthropic), running primarily the Claude Opus and Claude Fable models, which drafted and revised

scripts under the authors' direction and checked them against the method reported here. The authors reviewed and ran every script, and the deposited code reproduces the analyses without the tool.

Predictor coding and standardisation

Every continuous predictor was z-standardised and every dichotomous predictor was ±0.5 contrast coded before model fitting (the ±0.5 codes are then z-standardised with the other predictors, which preserves their sign). On this common standardised scale, regression coefficients are directly comparable, and lower-order terms in interactive models retain interpretable main-effect meanings (Brauer & Curtin, 2018; Schielzeth, 2010). Standardising the inputs is also what makes a prior on a coefficient interpretable, because it fixes the scale that prior is placed on (Gelman et al., 2008). Session was recoded to consecutive integers in the order the four sessions occurred, Sessions 2, 3, 4 and 6 becoming 0–3, and then z-standardised (within property for the accuracy models, whose properties span different session ranges). The change in an effect between any two measured sessions, including across the retention interval, is then a single linear combination of the posterior draws. Besides grammaticality (grammatical = +0.5) and language (Mini-Norwegian = +0.5), stated with the design, the models carry no topographic predictors, because each trial enters once per macroregion (see *Model structure*).

One further covariate enters the ERP models. It is the L1 Dominance Score, the per-participant derived score of that name in the Language History Questionnaire's Aggregate Scores export (LHQ3: Li et al., 2020), read from that export and z-standardised with the other continuous predictors. It indexes how far the first language dominates the participant's repertoire, and it is carried in the ERP models as a control for prior language experience. In the extraction code, the variable carries the name of the multilingual language diversity score, a legacy of an earlier version of the export in which the two scores sat in different columns, so the label and the quantity modelled differ. The models use the score described here. One participant had no questionnaire record. Because the score serves only as a control, that participant was given the mean of the other participants' scores (zero after standardisation), so that none of their trials was lost to the missing covariate. Because single-trial amplitudes are z-scored within participant (below), this

between-participant score, like the language main effect, is identified only through the imbalance of trials across each participant's conditions and sessions, and because it enters no interaction it adjusts none of the grammaticality effects we interpret. The covariate is not carried into the accuracy models. That asymmetry has no justification, and we return to it in the Limitations.

The response side has two standardisation decisions of its own. Single-trial amplitudes were z-scored within participant. The reasoning is the one Faust et al. (1999) set out for response latencies. When individuals differ in the overall scale of a measure, and those differences are roughly linear, a group whose scale is larger will show a larger raw treatment effect, and the resulting overadditive interaction is an artefact of scale. They recommend a within-individual z transformation as the remedy. Single-trial ERP amplitude has the same property, and the risk applies directly here because the language contrast is between participants, so a scale difference between the two groups could otherwise present as a grammaticality × language interaction. We also entered the trial's own pre-stimulus mean amplitude (−100 to 0 ms) as a within-participant z-scored covariate. Subtraction of a baseline is the special case of this model in which the baseline coefficient is fixed at one. Estimating that coefficient lets the model absorb baseline variance that subtraction would leave in the residual, which Alday (2019) shows can raise power. It also avoids the assumption subtraction requires, that conditions do not differ systematically during the baseline interval.

Model structure

For each ERP cell (property × window × macroregion), we fitted a Gaussian model of single-trial mean amplitude:

$$\text{amplitude} \sim \text{baseline} + \text{Gram} \times \text{Session} \times \text{Language} + \text{L1 dominance} + (1 + \text{Gram} \times \text{Session} \mid \text{participant}) + (1 + \text{Gram} + \text{Session} \mid \text{item}).$$

Each trial enters once per model. The electrode clusters of a macroregion record the same trial, and their amplitudes are correlated (between .45 and .64 in these data), so entering them as separate rows would treat one trial as several independent observations. That pseudoreplication (Lazic, 2010) narrows the intervals of every trial-level term, and here it narrowed those of the

grammaticality × session interaction and its interaction with language by a factor of 1.4 to 2.0. Averaging the clusters within each trial removes it, and the scalp distribution of the effects is described separately (see *Topography*).

For each property, we fitted a Bernoulli (logit-link) model of judgement accuracy (Dixon, 2008; Jaeger, 2008):

$$\text{correct} \sim \text{Gram} \times \text{Session} \times \text{Language} + (1 + \text{Gram} + \text{Session} \mid \text{participant}).$$

Participants and items are crossed rather than hierarchically nested. Within a language, every participant sees every sentence frame, and every frame is seen by every participant in that language, so the ERP models carry crossed by-participant and by-item random effects (Baayen et al., 2008; Barr et al., 2013). Items are nested within language, because the two miniature languages draw on separate lexicons, so language does not vary within an item and no by-item language slope is estimable. Grammaticality and session, by contrast, vary within both grouping factors, since every frame is presented in both a grammatical and an ungrammatical version and is re-presented across sessions.

Random effects therefore follow the "keep it maximal *as justified by the design*" principle (Barr et al., 2013), so both grouping factors receive a random intercept plus grammaticality and session slopes, with correlations estimated under the shared $\mathrm{LKJ}(2)$ prior. Participants also receive a slope for the grammaticality × session interaction, because the development of each participant's grammaticality effect is the within-participant quantity on which the language-conditioned interaction rests, and omitting that slope would make the interval of the fixed interaction too narrow (Barr et al., 2013; Brauer & Curtin, 2018).

Session enters that specification as a single linear predictor, so the change in the grammaticality effect across the retention interval is read off the fitted trend rather than estimated for that interval on its own. Gender agreement and differential object marking were therefore also refitted with a binary indicator for the retention session, interacted with grammaticality and language and added to the linear session term. The linear term then carries the training trend and the indicator the deviation at Session 6. The indicator is coded $+0.5$ at Session 6 and -0.5

elsewhere and, unlike the other predictors, is not standardised, so its coefficient is the deviation at Session 6 on the response scale. The refits are otherwise identical to the models above and were run under both random-effect structures. Verb–object number agreement admits no such refit. It is sampled at the two bracketing sessions only, so an indicator for the later of them would be collinear with the linear term. Where the two derivations disagree, the freed estimate is the one that bears on consolidation, and both are reported (Table B1, Table B2).

The case for retaining the by-item grammaticality slope, and the power-based argument against maximal structures that it was weighed against (Barr et al., 2013; Matuschek et al., 2017), are set out in Appendix A. The Results report the whole grid refitted without that slope, so a reader can see which conclusions depend on it. The reported accuracy models carry by-participant random effects only, although the judgement trials share the ERP models' item factor, the sentence frame: each property has 288 frames, half in each miniature language, and every frame was judged under both grammaticality levels and by between 17 and 31 participants (medians 29, 24 and 18 for gender agreement, differential object marking and verb–object number agreement). Leaving that factor out treats the trials of one frame as independent, and because each frame belongs to one miniature language, the group contrasts in sensitivity and criterion are also contrasts between two sets of frames (Baayen et al., 2008; Barr et al., 2013). Each accuracy model was therefore refitted with the ERP models' by-item term, a by-frame intercept with grammaticality and session slopes, both on its own and together with the by-participant grammaticality × session slope described below. These refits are sensitivity analyses specified after the reported models had been fitted.

Signal-detection reading of the judgement models

A model of accuracy that includes grammaticality as a predictor is a logistic signal-detection model in another parameterisation (DeCarlo, 1998). Let H be the rate of answering "grammatical" to a grammatical sentence (a hit) and F the rate of answering "grammatical" to an ungrammatical one (a false alarm). Accuracy on grammatical sentences is H, and accuracy on ungrammatical sentences is $1 - F$. With grammaticality coded symmetrically, the terms of the accuracy model that do not contain grammaticality (the intercept, session, language and session ×

language) are half the logit-scale sensitivity, logit H − logit F, and its change with session and language. The terms that do contain grammaticality are half of logit H + logit F, a response bias: a learner inclined to answer "grammatical" gains accuracy on grammatical sentences and loses it on ungrammatical ones without discriminating any better. The grammaticality × language and grammaticality × session × language terms therefore index a between-language difference in bias and in its development, not in discrimination. We report both quantities, derived for each language and session as linear combinations of the posterior draws: sensitivity d_L = logit H − logit F, and the criterion $c_L = -(\text{logit}\, H + \text{logit}\, F)/2$, which is negative for a readiness to answer "grammatical" (Macmillan & Creelman, 2005). On the logit scale, d_L is about 1.7 times the familiar probit d' (DeCarlo, 1998). For display, d' and c were also computed per participant and session with the log-linear correction for extreme rates (Hautus, 1995). The models were specified with the grammaticality terms read as discrimination, and *Transparency, preregistration and analysis status* records when and why that reading was replaced.

The reported accuracy models include by-participant slopes for grammaticality and for session but not for their interaction, although that interaction, the within-participant change in bias, varies within participants. Omitting such a slope can make the corresponding fixed effect's interval too narrow (Barr et al., 2013). Each accuracy model was therefore refitted with the interaction slope added, as a sensitivity analysis specified after the signal-detection reading above. With that slope, the refit is the signal-detection model of the response itself in a different parameterisation: its by-participant intercept and session slope are each learner's sensitivity and its change, its by-participant grammaticality and grammaticality × session slopes are each learner's criterion and its change, and the map between the two parameterisations is linear and invertible, so the two models can differ only through their priors.

Priors

We placed informative priors derived from the morphosyntactic-ERP and behavioural literature, in place of the common weakly informative default, following a selective, sign-informed and magnitude-regularising strategy (Schad et al., 2021). Because the response and all predictors

are standardised, each coefficient is expressed in standard-deviation units (within participant, for the ERP models). A prior *mean* can therefore encode the direction of a well-established effect, while its *standard deviation* stays wide enough to let the data dominate any well-identified effect (Gelman et al., 2008).

The one firmly established directional effect is the P600: words that violate the syntactic structure of a sentence elicit a late positivity, larger for ungrammatical than for grammatical sentences (Osterhout & Holcomb, 1992). For morphosyntactic violations specifically, that positivity is posterior and centroparietal, and it is obtained in native speakers (Chen et al., 2007; Mueller et al., 2005, 2007) and, once some proficiency has been reached, in artificial-language learners (Morgan-Short et al., 2010; Morgan-Short, Steinhauer, et al., 2012). A P600 also arises to structural violations in implicit artificial-grammar learning, where the sequences carry no morphology at all (Silva et al., 2017). Because grammaticality is coded grammatical = +0.5, this implies a *negative* prior mean on the grammaticality slope in the P600 window, carried more mildly into the 300–600 ms window for the two agreement properties, whose late signature is a positivity rather than an N400. For differential object marking, the specification differs in both windows, being widened in the later window and left zero-centred in the earlier one, because case-marking violations are configuration-dependent and elicit a P600, an N400 or a biphasic response according to animacy and markedness (Frisch & Schlesewsky, 2001; Mueller et al., 2007; Nieuwland et al., 2013). The early (200–500 ms) anterior-negativity window was left zero-centred, because the LAN is small and present in only roughly half of participants and items, so its sign is not safely predictable (Caffarra et al., 2019). The grammaticality × session × language interaction, the language-conditioned contrast under test, was kept zero-centred, because the second-language signature of these effects is qualitatively contested (Chen et al., 2007; Mueller et al., 2009) and proficiency-dependent (Caffarra et al., 2015). The priors thus inform the model where the evidence is strong without biasing the effects we set out to estimate.

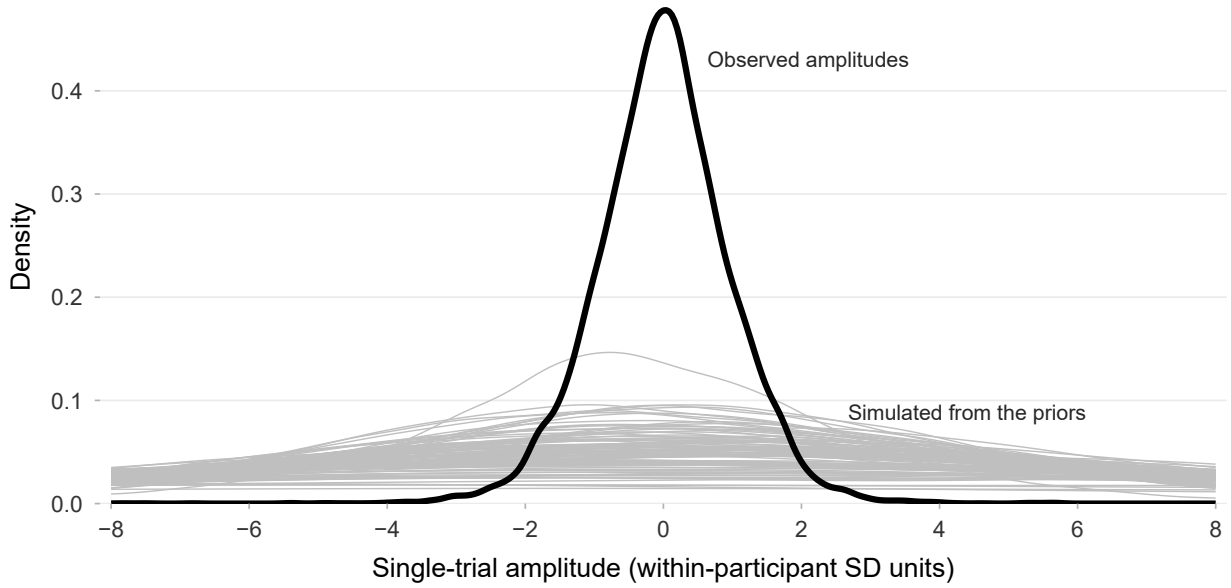
For the accuracy models, artificial-language learners judge grammaticality well above chance and improve across successive test blocks even under mere exposure, without explicit

instruction or feedback (Mueller et al., 2009), motivating a positive intercept and a positive session slope. In signal-detection terms, these encode above-chance sensitivity that increases with training. Every term that carries grammaticality, and so indexes response bias, received the zero-centred default. Random-effect standard deviations, and the residual standard deviation of the ERP models, received a generous half–Student-t prior, a weakly informative default for variance parameters in hierarchical models (Gelman, 2006). A generous prior is warranted here because between-participant variation in the morphosyntactic ERP response is not incidental noise. Tanner and Van Hell (2014) found that although the grand mean showed the usual biphasic LAN–P600 pattern, individual responses varied systematically along a continuum between negativity-dominant and positivity-dominant, so participants differ in the form of the response and not only in its size. Modelling that variation as a random effect over single trials, instead of averaging across it, is the approach Volpert-Esmond et al. (2021) set out for ERP data. Random-effect correlation matrices received an LKJ(2) prior (Lewandowski et al., 2009). Table C1 in Appendix C gives the distribution placed on every parameter. The literature does not supply a defensible microvolt effect size for these components (even systematic resampling finds the magnitude of the P600 to vary widely across samples, Kim et al. (2024)). Prior means therefore encode only sign and a regularised magnitude, and their influence is bounded by deliberately wide standard deviations. Prior predictive simulation confirmed that the priors imply amplitude and accuracy distributions that encompass the observed data (Figure 6). Every model in the grid, under both random-effect structures, and every accuracy model was also refitted on the same data, as a sensitivity analysis, under the former weakly informative priors of $\mathcal{N}(0, 1)$ and $\mathcal{N}(0, 1.5)$, and the estimates were essentially unchanged. The retention-free refits were fitted under the informative priors only. The largest shift in any grammaticality term was 0.0004 on the within-participant standardised scale for the ERP models and 0.059 in log-odds for the accuracy models. No grammaticality term's credible interval changed width by more than 3.3% in the ERP models or 2.7% in the accuracy models. Taken over every reported parameter, the largest width change was 6.5%. The informative priors therefore regularise without driving the conclusions (Depaoli & van de Schoot, 2017; Schad et al., 2021; van de Schoot

et al., 2021). Under both random-effect structures, each grammaticality term is credible in the same ERP cells, with the same sign, under the weak priors as under the informative ones. The credible effect whose interval lies closest to zero (gender agreement, grammaticality × session, midline 400–900 ms) is $b = 0.021$, 95% CrI $[0.002, 0.041]$, $p_d = .99$ under the weak priors, so its credibility does not depend on them.

Figure 6

Prior Predictive Check for a Representative ERP Model



Note. The model is subject–verb gender agreement, 400–900 ms, midline, with its informative priors and the maximal random-effect structure. Grey lines are the densities of 100 datasets simulated from the priors alone, with the likelihood switched off, and the black line is that of the 12,908 observed single-trial amplitudes, z-scored within participant. Each density is a Gaussian kernel estimate. The horizontal axis is amplitude in within-participant standard deviations and the vertical axis is density. The prior predictive distribution comfortably encompasses the observed data, so the priors are not in conflict with it. The horizontal axis is limited to 8 standard deviations either side of zero, which holds 66% of the simulated values. Wide priors on a standardised scale are meant to reach far: 1% of the simulated values lie beyond 38, and on that range the observed density would collapse to a spike at zero. The check therefore establishes that the priors generate data of a plausible order of magnitude, not that they are tightly calibrated to the observed distribution.

Sampler settings and convergence

Each model was run with four chains of 10,000 iterations, the first 2,000 of each discarded as warm-up, with `adapt_delta = 0.95` and `max_treedepth = 12` to accommodate the challenging geometry of maximal multilevel posteriors. One fit was run longer, with the same seed and control settings, because its first run missed the $\widehat{R}$ criterion (Vehtari et al., 2021): gender agreement, 200–500 ms, lateral, base structure, weak prior (4,000 warm-up and 16,000 sampling iterations per chain). Convergence was assessed with the rank-normalised split-$\widehat{R}$ and bulk/tail effective sample sizes (Vehtari et al., 2021), requiring $\widehat{R} < 1.01$, bulk- and tail-ESS > 400, and no divergent transitions. Model fit was checked with posterior predictive checks (Gabry et al., 2019). These two thresholds do different jobs. An effective sample size of 400 (at least 100 per chain) is the point at which the $\widehat{R}$ and ESS diagnostics are themselves reliable (Vehtari et al., 2021). The stability of a 95% interval limit is a more demanding requirement, because extreme quantiles are estimated far less efficiently than a posterior mean (Kruschke, 2021). Over all parameters of all models, the smallest bulk effective sample size is 785 and the smallest tail effective sample size is 1,900, with the per-model values in Table D1 and Table D2 in Appendix D. At the lower end of that range, the margin over the floor of 400 is narrow. The interval limits of those models are therefore estimated less precisely than their medians, and coefficients whose intervals lie close to zero should be treated as borderline.

Inference

We summarise each effect by its posterior median, 95% credible interval (CrI) and probability of direction (p_d, the posterior probability that the effect has its median's sign), an estimation-first approach that avoids dichotomous null-hypothesis testing (Makowski et al., 2019). We describe an effect as *credible* when its 95% credible interval excludes zero, and otherwise read p_d as a continuous index of directional certainty (Kruschke & Liddell, 2018). All credible intervals are equal-tailed quantile intervals bounded by the 2.5th and 97.5th percentiles of the posterior draws. For a skewed posterior, an equal-tailed and a highest-density interval can differ in whether they contain zero, so the choice is stated (Kruschke, 2021). Within each model, hierarchical shrinkage

across participants and items already regularises the coefficient estimates (Gelman et al., 2012; Kruschke & Liddell, 2018). The eighteen cells are fitted separately and are not pooled towards one another, so that shrinkage is not itself an argument about the grid. The grid (3 properties × 3 a priori windows × 2 macroregions) is fixed before fitting, and no cell is screened for significance before it is reported (Figure 7 reports all 18 cells for the four grammaticality terms, and Table B1 the derived retention contrasts). No correction across cells is applied. Where a conclusion does rest on particular cells crossing the credibility threshold, we treat it as provisional on that basis (see *Strength of evidence* in the Discussion). We do not claim the grid is immune to selection effects (Rubin, 2021). The derived retention effect (Session 4 → Session 6) and its language-scoped contrasts are posterior quantities computed from the same models, summarised and disclosed in full the same way (Table B1). For gender agreement and differential object marking, they are also re-estimated in refits that free the Session-6 step from the training trend, described under *Model structure* (Table B2).

Multivariate decoding of grammaticality

The Bayesian models above summarise each trial by its mean amplitude over a macroregion, which could miss a grammaticality signal that is distributed across the scalp too weakly to move any single cluster's mean but is nonetheless jointly decodable from the whole-scalp pattern (Haxby et al., 2001). As a complementary, classical (non-Bayesian) analysis, we decoded Grammatical from Ungrammatical trials directly from the single-trial amplitude pattern across all EEG sensors at every time sample, downsampled to a 4 ms grid, after subtracting from each sensor the trial's mean over the 100 ms before the critical word. The amplitude models enter that mean as a covariate instead (Alday, 2019). Subtraction moves the conditions' difference in that mean into every later sample of the decoder's input, with its sign reversed. The classifier was a regularised linear discriminant analysis under leave-one-participant-out cross-validation, a time-resolved multivariate pattern analysis (MVPA) approach for EEG/MEG (Grootswagers et al., 2017). Leave-one-participant-out is the generalisation test this question calls for, since what is at stake is whether the pattern transfers to an unseen learner. Varoquaux et al. (2017) show that leave-one-out

schemes give more variable accuracy estimates than repeated random splits, so we read the AUC time-course below as a profile whose individual point estimates carry limited precision. Decoding performance is summarised as the area under the ROC curve (AUC) at each time point. Chance-level calibration was established with a permutation test that shuffles the grammaticality labels once per permutation, holding the shuffle constant across the whole time-course so that the null preserves each trial's temporal autocorrelation. Multiple comparisons across time points were corrected with cluster-based permutation inference (Maris & Oostenveld, 2007). Clusters were formed from temporally contiguous samples whose AUC exceeded the 95th percentile of the pooled permutation null, the cluster statistic was the summed excess of AUC over that threshold, and each cluster was assessed against the distribution of the largest cluster statistic across permutations. A second correction layer, for the false-discovery rate (FDR), is applied across clusters, with the family defined per property. Benjamini–Hochberg correction ($q < .05$) is applied across the clusters of a property's pooled overall time-course, and each property's family is corrected separately. A Benjamini–Hochberg adjustment never lowers a p value, so this second layer can only remove clusters that pass the maximum-cluster test, and because that test already controls the family-wise error rate across time, the layer adds no error control of its own. We therefore report the clusters that pass the first test alongside those that survive both. The session-resolved and session-by-language breakdowns carry only their own cluster-level p values. Appendix A sets out why the family is defined this way. Grammatical and ungrammatical trial counts differ across participants and decline across sessions, so we balanced the classes by undersampling the majority class within each participant of the training fold. The discriminant's pooled within-class covariance was ridged by one thousandth of its mean diagonal for numerical stability. The time-course AUC, computed within each held-out participant, is in any case insensitive to class prevalence, so neither the classifier nor the metric can exploit a prevalence shortcut there, and chance was set by the empirical within-participant permutation null (Combrisson & Jerbi, 2015). The labels were shuffled within participant and not within each session, which the design allows because the two conditions were presented in near-equal numbers at every session. The permutation null comprised 1,000

shufflings, so the smallest attainable cluster p is 1/1,001. This bounds the resolution of near-threshold cluster p-values, and the surviving clusters are reported with that granularity in mind (T. E. Nichols & Holmes, 2002; Phipson & Smyth, 2010). The decoder is given the EEG sensors alone. The amplifier's built-in accelerometer channels are excluded, as is any channel not exported for every session a property was recorded in, so that each property's pattern is brain-derived and no sensor is present for one session's trials but structurally absent for another's. Beyond the overall time-course, five further analyses were run. Decoding separately by session, and by session × mini-language, gives a multivariate analogue of the ERP models' grammaticality × session and × language terms. A cross-language generalisation analysis trains the decoder on one mini-language and tests it on the other. A cross-property generalisation analysis trains on one grammatical property's Grammatical/Ungrammatical contrast and tests on another. Finally, temporal generalisation trains a decoder at every time point and tests it at every other, on a grid coarsened by a factor of two on both axes to keep the computation tractable. It characterises whether the same neural code is stable across the epoch or evolves (King & Dehaene, 2014). The two generalisation analyses carry no permutation null and are reported as descriptive comparisons. Cross-validating strictly out-of-participant throughout guards against the circular, non-independent analysis that would otherwise inflate apparent decoding accuracy (Kriegeskorte et al., 2009).

Transparency, preregistration and analysis status

The study design, artificial-language materials and session protocol were preregistered before data collection (https://osf.io/tjr54). The preregistration specified the design, the miniature-language materials, the six-session schedule and the EEG acquisition, but it did not commit to a statistical analysis plan. The Bayesian multilevel models, the literature-derived informative priors, the latency windows and macroregions, the Session 4 → 6 retention contrasts and the multivariate decoding analyses reported here were therefore specified after the data were collected. None was preregistered, so all are exploratory in the registered sense, and we report them as estimation-oriented analyses, since none tests a preregistered hypothesis (Nosek et al., 2018; Wagenmakers et al., 2012). Within that frame, we distinguish *primary* from *secondary* analyses.

The primary analyses, fixed before any model was fitted, comprise the ERP grid of properties, windows, macroregions and priors together with the pooled, all-sessions decoding time-course for each property. The secondary analyses are the session-resolved and session-by-language decoding breakdowns, the cross-language and cross-property generalisation analyses, and the temporal-generalisation matrix. What the paper reports of them varies, and we say which is which. The cross-language and cross-property peaks are quoted in the Results. The session-resolved and session-by-language breakdowns and the temporal-generalisation matrices are not summarised numerically here and are available in the deposited decoding artefacts. Where a secondary analysis carries a cluster-level p value, that value is family-wise-error-controlled over time in the usual way, but none of them is treated as a test of the paper's claims. Where the decoding analyses speak of a "confirmatory family", the term names the false-discovery-rate correction family (the clusters of each property's pooled overall time-course), not preregistered confirmation. On a minor point of adherence, a third, ancillary violation type was recorded per property but, as noted above, is not analysed here.

Several decisions were made after results had been inspected, and we label them as such. First, peak decoding accuracies are quoted on the post-stimulus axis, a restriction adopted once the pre-stimulus interval was found not to be at chance for differential object marking. The pre-stimulus interval is retained and reported alongside as a negative control. Second, the temporal-generalisation and cross-property analyses were reclassified as descriptive after it became clear that no null distribution had been computed for them. Third, the judgement models were first read with their grammaticality terms as discrimination. We later recognised that, with accuracy as the response, those terms index response bias (see *Signal-detection reading of the judgement models*), and the behavioural results are reported in signal-detection terms throughout. The models themselves were not changed. The refit adding a by-participant slope for the change in bias was specified at the same point, as a sensitivity analysis. Fourth, an audit of the archived EEG processing histories found five Session 3 recordings, beyond the four already excluded, that had been filtered at 1 Hz. All nine were reprocessed at 0.1 Hz under a protocol fixed before any

reprocessed output was inspected, and the protocol's robustness analysis refits the ERP models without them (see *EEG acquisition and single-trial ERP quantification*). The same audit found three exports that do not contain the participant-session they are named for, of which two were recovered under that protocol and the third is excluded. It also found that the Session 2 recordings had been re-referenced differently from the others, which we corrected, and identified the participant trained in the wrong miniature language at Session 2, who is excluded (see *Participants*). All of these decisions rest on how the data were collected or processed, not on the values of the data. Fifth, the Introduction's statement of what the transfer accounts predict was revised after the results were known, when it became clear that the lexical and structural differences between the two miniature languages do not separate the accounts as the original framing assumed. The revised statement sets out the predictions each account makes for the design as it was run, including those our results contradict. Sixth, the ERP models first entered each trial once per electrode cluster. A check of their residual structure after the first fits showed that one trial's observations across clusters were correlated, so those models overstated the precision of the trial-level terms, and the models were respecified with one observation per trial per macroregion and a by-participant slope for the grammaticality × session interaction (see *Model structure*). The change was made on statistical grounds and before the Bayesian models were refitted. In frequentist (lme4) fits of the same data, it widened the intervals of the language-conditioned terms, which the earlier specification had reported as credible in several cells. In the respecified models, fitted to the corrected sample, no language-conditioned term is credible in any cell. Seventh, the false-discovery-rate family of the decoding analyses was narrowed in July 2026 to the clusters of each property's pooled time-course, after a run under a wider family that also spanned the session-resolved breakdowns had been inspected. Correcting across those nested re-analyses of the same trials leaves the number of hypotheses undefined (see Appendix A). The second layer itself was kept, after the corrected sample's results were seen, as it had been specified, and the clusters that pass the first layer alone are reported beside it. Eighth, the judgements were first analysed under a reaction-time screen carried over from the project's earlier preprocessing. After the accuracy models had been fitted both with

and without it, we examined what the screen removes. It removes the most and the least accurate judgements, and a larger share of the Mini-English group's judgements than of the Mini-Norwegian group's (see *Grammaticality-judgement task*). We therefore stopped applying it. The fits with it are reported as a sensitivity analysis, and the results that only one of the two fits supports are graded as suggestive (see *Strength of evidence*). The refits on every judgement with the other random-effect structures, priors and covariate were run after this decision. In the refit that adds a by-participant slope for the change in bias, the gender-agreement criterion difference was credible at no session, which is why that difference is graded as suggestive. None of these decisions creates a new test of the paper's claims.

The preregistration planned data collection at two sites, in Norway and Spain. The present analyses concern only the Norwegian cohort (first-language Norwegian, second-language English learners of Mini-English or Mini-Norwegian). The Spanish-site data are not analysed here. The findings therefore speak to first-language-Norwegian learners, and generalisation to the other registered populations is an open question.

The preregistered protocol required participants to score above 80% on the post-training behavioural test in each grammar session before proceeding to the EEG experiment. After a few participants had been tested, we recognised that this criterion was too strict and modified the procedure: when the first test score was below 80%, pretraining, training and the test were repeated; after the second attempt, participants proceeded to the experiment regardless of their score. Participants were shown their percentage correct at the end of each test attempt. The 80% threshold therefore describes the first-attempt gate, not a requirement imposed on the second attempt. This performance threshold is both an inclusion criterion and a performance-conditional missingness mechanism, since it conditions which participant-sessions contribute EEG and behaviour. Its effect is therefore folded into the attendance counts in Figure 3, which record who contributed data at each stage but not why. Whether each participant met the threshold at a given session is not reported. It can only be reconstructed from the final pass record in the training logs, and that reconstruction does not reconcile with the attendance counts at Session 4, so we do not quote numbers from it.

Results

All 135 fitted models met the convergence criteria. Taken over every parameter of every model, the largest rank-normalised $\widehat{R}$ was 1.009, the smallest bulk and tail effective sample sizes were 785 and 1,900, and no model produced a divergent transition. Per-model values are given in Table D1 and Table D2.

ERP signatures of grammatical processing and their development

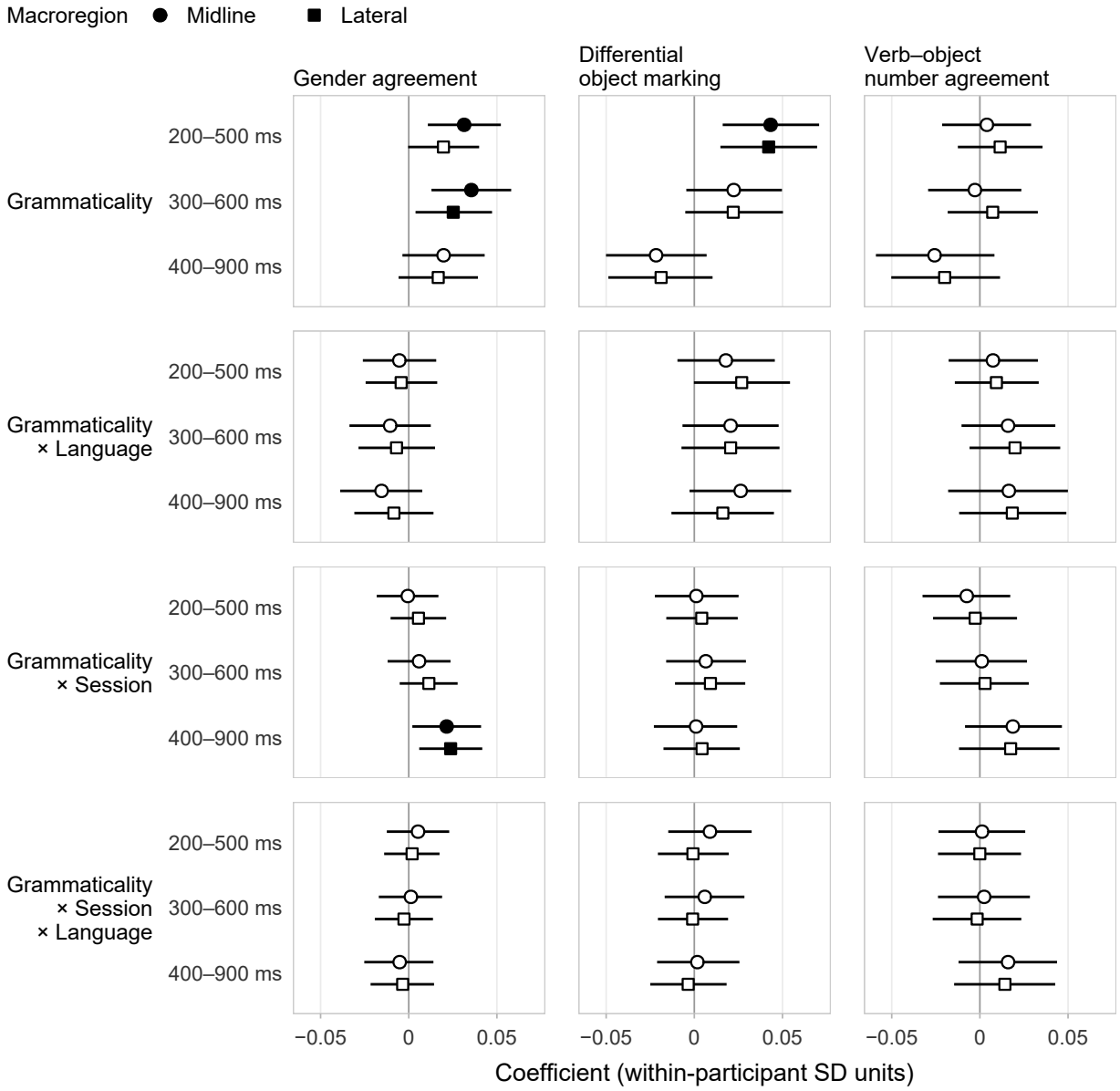
We assess, for each property, whether it elicited a grammaticality effect, whether that effect changed across sessions (the grammaticality × session interaction), and whether the effect or its development was modulated by language (the grammaticality × language and grammaticality × session × language interactions). The pattern differs by property. Gender agreement showed a grammaticality effect in the early and middle windows that grew across sessions in the late window, where the grammaticality × session interaction was credible in 2 of 6 cells; differential object marking showed a transient early grammaticality effect; and verb–object number agreement showed no credible effect (Figure 7). No term involving language was credible for any property: the grammaticality × language interaction was credible in 0 of 18 cells and the grammaticality × session × language interaction in 0 of 18. The same terms are credible, with the same signs, when the reprocessed and recovered records are left out (Table A1). The condition averages are shown for each miniature-language group in Figure 8, the two groups' difference waves in Figure 9, and the scalp distribution of the effect in each group and window in Figure 10.

One coding fact governs how every coefficient below should be read. A positive grammaticality coefficient means that grammatical trials were more positive than ungrammatical ones, because grammaticality is coded grammatical = +0.5. Violations therefore appear as a relative negativity. The canonical P600 would produce a negative coefficient. Every credible grammaticality effect below is positive, so the dominant signature in these learners is a violation-related negativity where our priors anticipated a late positivity, a point we return to in the Discussion. Results are organised by property.

The grammaticality effect is easier to compare between the groups once the shared evoked

Figure 7

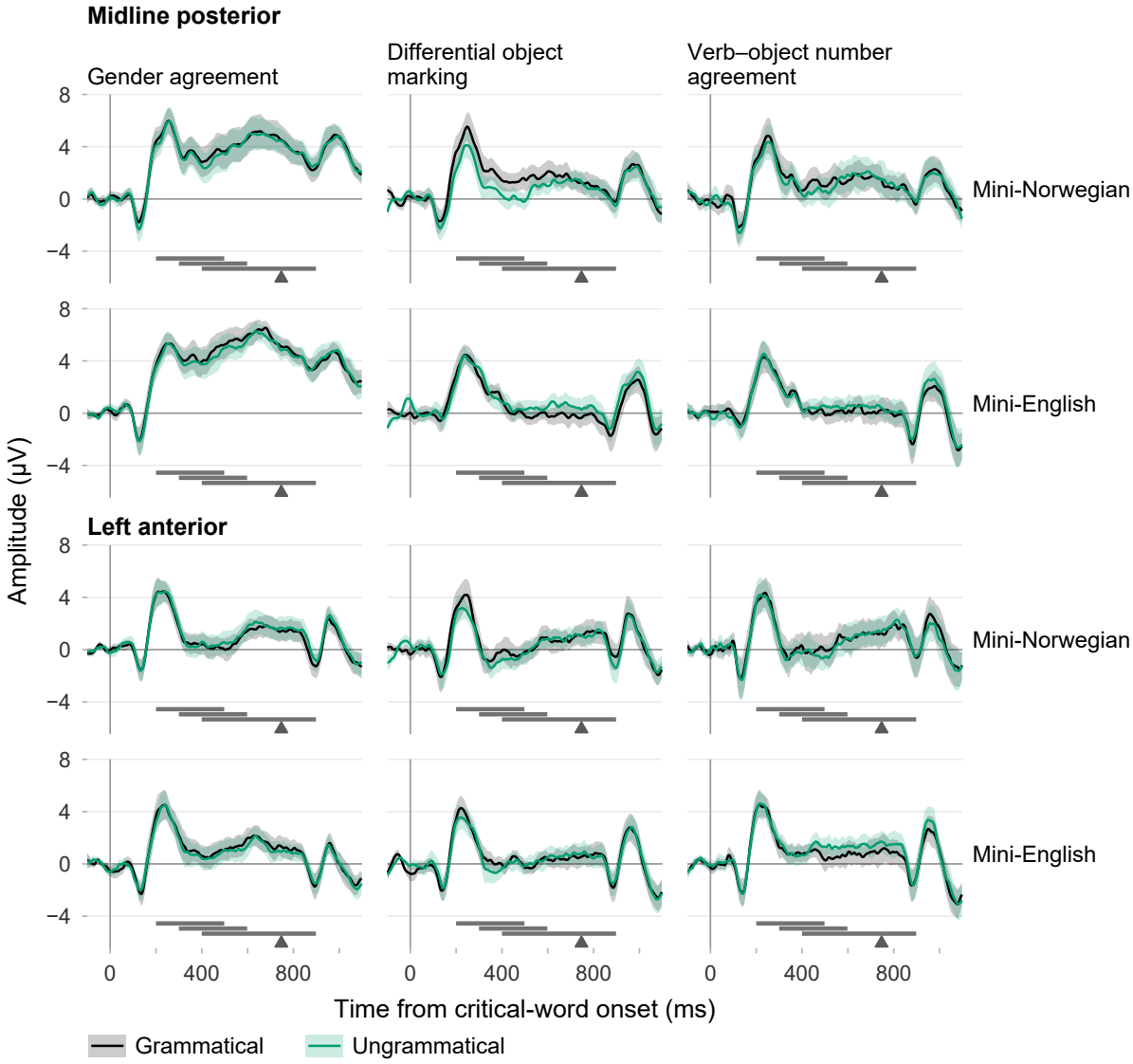
Posterior Medians and 95% Credible Intervals of the Grammaticality Terms in Every ERP Cell



Note. Points are posterior medians and horizontal lines 95% equal-tailed credible intervals (CrI) for the four grammaticality terms of each single-trial ERP model (maximal random-effect structure, informative priors). The vertical grey line marks zero. Columns are grammatical properties and rows are model terms. Within each panel the three a priori windows run from earliest (top) to latest (bottom), with the midline estimate (circle) above the lateral one (square). Filled symbols mark intervals that exclude zero; open symbols, intervals that include it. All panels share one horizontal scale, in within-participant standard deviations of single-trial amplitude. Grammaticality (grammatical = +0.5) and language (Mini-Norwegian = +0.5) were coded before standardisation, so a positive grammaticality coefficient is a relative negativity for violations, and a positive term involving language is a larger value of the corresponding grammaticality term in the Mini-Norwegian group. Verb–object number agreement was recorded at Sessions 4 and 6 only, so its session terms index change across the retention interval.

Figure 8

Grand-Average ERPs by Electrode Cluster, Miniature Language and Property

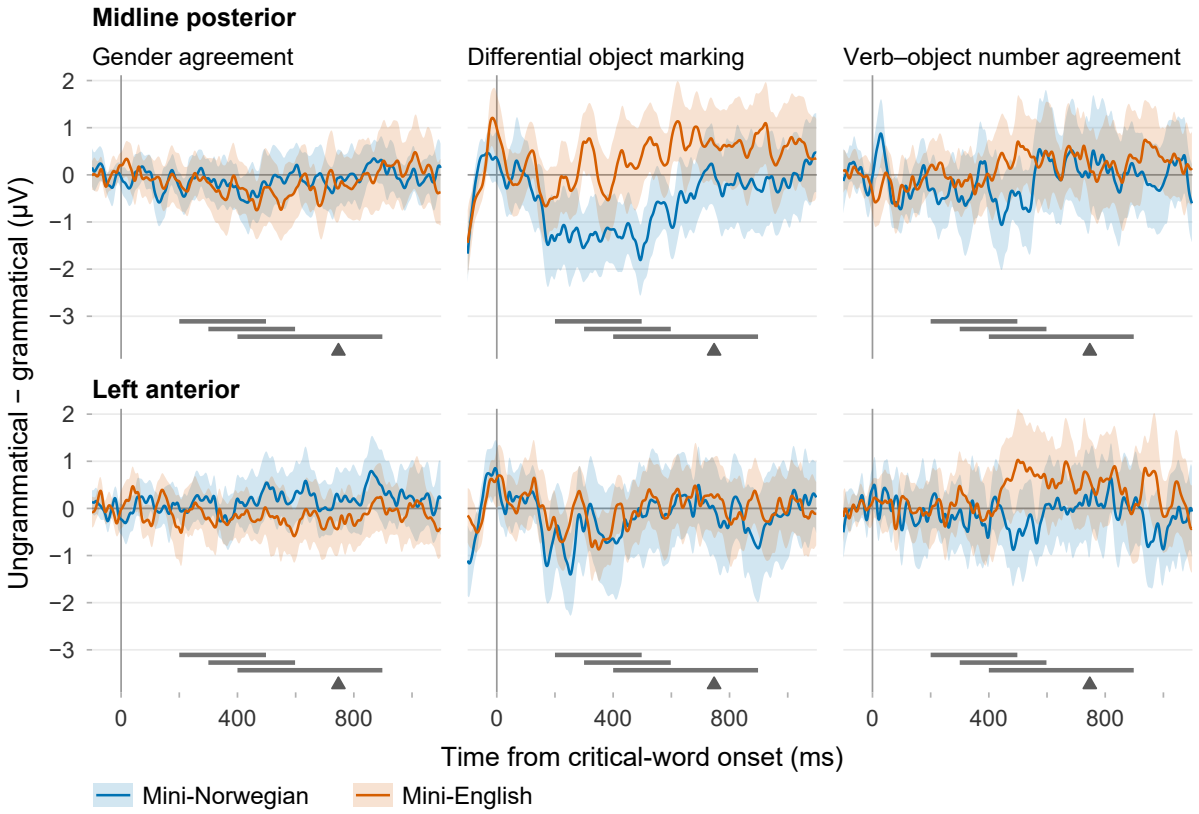


Note. The upper block is a midline posterior electrode cluster, where the P600 is typically largest, and the lower block a left anterior cluster, where the LAN is typically largest (the models use macroregion means). Black lines are grammatical and green lines ungrammatical sentences, pooled over each property's sessions. Each participant's waveform was baseline-corrected (−100 to 0 ms) and averaged over their sessions before participants were averaged, so every participant carries equal weight. Ribbons are descriptive pointwise 95% normal intervals from the participant-level standard error. Positive is plotted up, and all panels share one scale. Grey bars mark the a priori windows (from the top, 200–500, 300–600 and 400–900 ms) and the triangle the median onset of the next word. The sample is the modelled one, and inferences rest on the models summarised in the forest plot of posterior estimates.

response is subtracted. Figure 9 overlays the two groups' difference waves (ungrammatical minus grammatical) at the same two electrode clusters.

Figure 9

Difference Waves (Ungrammatical Minus Grammatical) by Electrode Cluster, Miniature Language and Property



Note. Each line is the participant-paired difference between ungrammatical and grammatical baseline-corrected waveforms, averaged over each participant's sessions and then over participants, for the Mini-Norwegian (blue) and Mini-English (vermilion) groups at the two electrode clusters of the grand-average figure. Values below zero mark a relative negativity for violations, the opposite sign to the grammaticality coefficient in the models, where grammatical is coded +0.5. Ribbons are pointwise 95% normal intervals from the participant-level standard error of the paired difference. Positive is plotted up. Grey bars mark the a priori windows and the triangle the median onset of the next word, as in the grand-average figure. Session-resolved difference waves are given in the appendix on group displays by session.

Subject–verb gender agreement

Gender agreement showed the clearest profile. The grammaticality effect was positive in direction in every window and macroregion (p_d from .93 to .998; Figure 7) and credible in 3 of 6 cells, all in the two earlier windows (midline 200–500 ms, $b = 0.031$, 95% CrI $[0.011, 0.052]$, $p_d = .998$; lateral 300–600 ms, $b = 0.025$, 95% CrI $[0.004, 0.047]$, $p_d = .99$; midline 300–600 ms, $b = 0.036$, 95% CrI $[0.013, 0.058]$, $p_d = .998$). Averaged over sessions, the effect was weaker in the late window (midline 400–900 ms, $b = 0.020$, 95% CrI $[-0.004, 0.043]$, $p_d = .95$), but that is where it grew. The grammaticality × session interaction was credible in the late window in both macroregions (lateral, $b = 0.024$, 95% CrI $[0.006, 0.042]$, $p_d = .995$; midline, $b = 0.021$, 95% CrI $[0.002, 0.041]$, $p_d = .98$) and in neither earlier window (lateral 300–600 ms, $b = 0.011$, 95% CrI $[-0.005, 0.028]$, $p_d = .91$). Language modulated neither the effect nor its development. The grammaticality × language interaction was credible in 0 of 6 gender-agreement cells and the grammaticality × session × language interaction in 0 of 6, and in the late window, where the effect developed, the latter sat near zero (midline, $b = -0.005$, 95% CrI $[-0.025, 0.014]$, $p_d = .70$).

Differential object marking

Differential object marking showed an early, transient grammaticality effect (grammatical trials more positive, a relative violation negativity), and no developing late effect. The effect was credible in the early window in both macroregions (Figure 7; lateral 200–500 ms, $b = 0.042$, 95% CrI $[0.015, 0.070]$, $p_d = .998$; midline, $b = 0.043$, 95% CrI $[0.016, 0.071]$, $p_d = .999$), positive but not credible in the middle window (midline 300–600 ms, $b = 0.022$, 95% CrI $[-0.005, 0.050]$, $p_d = .95$), and, if anything, slightly negative by the late window (midline 400–900 ms, $b = -0.022$, 95% CrI $[-0.050, 0.007]$, $p_d = .93$). One caveat attaches to this early effect specifically. Because the violation omits the particle *fi*, the two conditions differ physically in the region preceding the critical word. At the roughly 505 ms word-onset asynchrony, the response to that region overlaps the 200–500 ms window of the critical epoch. The same low-level difference that renders the pre-stimulus decoding baseline non-chance (see below) could therefore contribute here, so we read the early effect as a violation-related response with a possible stimulus-driven component. The

effect did not change credibly across sessions (the grammaticality × session interaction was credible in 0 of 6 cells), and no term involving language was credible (grammaticality × language, 0 of 6; grammaticality × session × language, 0 of 6). The grammaticality × language interaction came closest in the early lateral window ($b = 0.0269$, 95% CrI $[-0.0002, 0.0543]$, $p_d = .97$), where it is credible under the simpler random-effect structure (see *Dependence on the by-item random-effect structure*).

Verb–object number agreement

Verb–object number agreement, the latest-introduced property, showed no credible term in any cell (Figure 7). Its grammaticality effect was small and positive in the early window and negative in the late one (midline 400–900 ms, $b = -0.026$, 95% CrI $[-0.059, 0.008]$, $p_d = .93$). The terms involving session leaned positive in the late window without reaching credibility, both the grammaticality × session interaction (midline, $b = 0.019$, 95% CrI $[-0.008, 0.047]$, $p_d = .91$) and the grammaticality × session × language interaction ($b = 0.016$, 95% CrI $[-0.012, 0.044]$, $p_d = .87$).

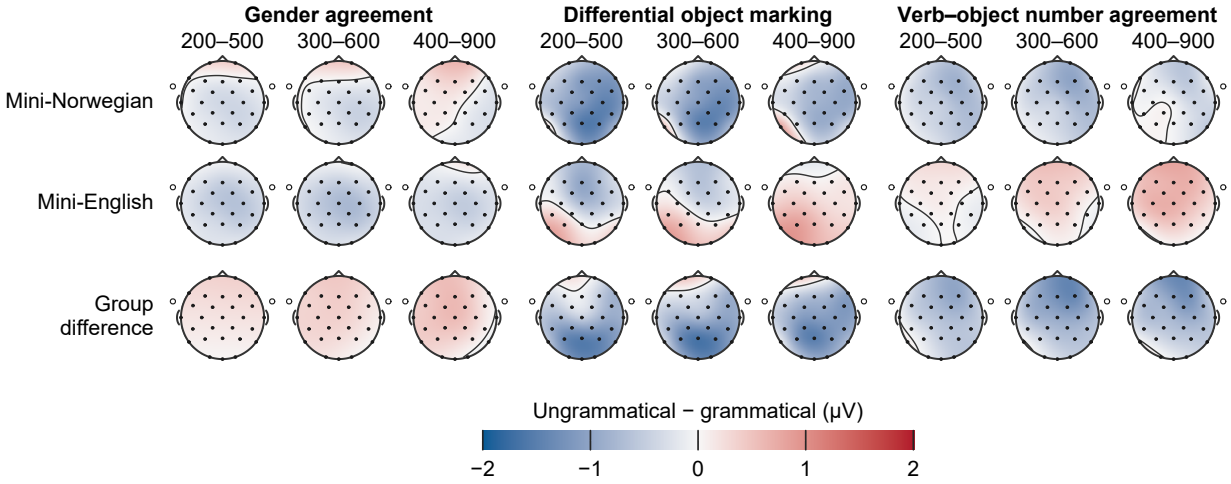
One reading caveat applies to this property alone. Verb–object number agreement is measured at Sessions 4 and 6 only, and the interval between them is the unrehearsed retention gap, so for this property the session term is the change across that gap and contains no further exposure. Its grammaticality × session and three-way terms are therefore offline-interval changes, and the *learning* and *language-conditioned development* glosses that the operationalisation gives those terms for the other two properties do not carry over unaltered here. We read them accordingly throughout.

Topography

The models average over the clusters of each macroregion and carry no topographic terms, so the scalp distribution of the grammaticality effect is described rather than tested. Figure 10 shows it for each language, property and window.

Figure 10

Scalp Topography of the Grammaticality Effect by Property, Analysis Window and Miniature Language



Note. Maps show the participant-paired ungrammatical-minus-grammatical mean amplitude in each a priori analysis window (column headers, in ms from critical-word onset), baseline-corrected within participant and averaged over each participant's sessions before the group mean. Within each property, the top two rows are the Mini-Norwegian and Mini-English groups and the bottom row, set apart, is their difference (Mini-Norwegian minus Mini-English). Negative values (blue) mark a relative negativity for violations. Each map is a penalised thin-plate spline surface fitted to the 30 electrodes assigned to the analysis regions, at the recorded montage positions (dots; nose at the top), and masked to the head, so no value is shown for scalp the montage did not sample. The two low-temporal sites below the equator, FT9 and FT10 (open circles beside the ears), shape the surface at the rim. The thin dark line is the zero contour. The colour scale is symmetric about zero and shared by every map here and in the session-resolved maps in the appendix on group displays by session. The sample is the one the models are fitted to. The maps are descriptive and are not electrode-wise tests.

Dependence on the by-item random-effect structure

Everything above comes from the maximal structure, with crossed by-participant and by-item random effects each carrying a grammaticality slope. Refitting the whole grid without the by-item grammaticality slope changes almost nothing. At the median, the grammaticality main effect's intervals under the maximal structure are 1.01 times their width without the slope, and the same cells are credible under both structures for every grammaticality term but one. That one is the grammaticality × language interaction for differential object marking in the early lateral window, credible without the by-item slope ($b = 0.0270$, 95% CrI $[0.0001, 0.0542]$, $p_d = .98$) and just

short of credibility with it, as reported above. No conclusion here rests on that cell. The comparison is set out in full in Appendix A.

Retention (Session 4 to Session 6)

The retention contrast quantifies the change in the grammaticality effect between Sessions 4 and 6, positive values indicating an effect that strengthened over the interval and values near zero an effect that held steady. Its derivation bounds what it can show. Session enters the models as a single linear predictor, so the contrast is the per-session change in the grammaticality effect rescaled to the width of the retention step. Every session in the series informs that estimate, including those outside the retention interval. For gender agreement, sampled from Session 2, and for differential object marking, sampled from Session 3, the contrast describes the fitted trajectory evaluated over the retention interval. It cannot by itself separate an offline gain specific to that interval from continued improvement at the rate the training sessions established. Verb–object number agreement is the exception, being present at Sessions 4 and 6 only, so for that property the linear session term is precisely the change across those two sessions and the contrast is estimated directly from them. Table B1, in Appendix B, reports every contrast overall, within each language and, where derived, as the between-language difference; the text below reports the ones that carry the argument.

For gender agreement and differential object marking, that limitation is addressed directly. Both properties were refitted with an indicator for the retention session, interacted with grammaticality and language, so that the linear term carries the training trend and the indicator the deviation at Session 6. Table B2, in the same appendix, reports those freed contrasts. They answer a different question from Table B1, and where the two disagree, the freed estimate is the one that bears on consolidation. Verb–object number agreement admits no such refit, for the reason given under *Model structure*.

A retention contrast was credible for gender agreement alone, in the late window where its development across sessions was also credible. There the grammaticality effect strengthened across the interval in both macroregions (lateral 400–900 ms, $\Delta = 0.022$, 95% CrI $[0.006, 0.039]$, $p_d = .995$; midline, $\Delta = 0.020$, 95% CrI $[0.002, 0.038]$, $p_d = .98$). Within the groups, the

strengthening was credible for Mini-English in the lateral cell ($\Delta = 0.025$, 95% CrI $[0.002, 0.048]$, $p_d = .99$) and for neither group elsewhere, and the difference between the groups was not credible in any cell (lateral, $\Delta = -0.006$, 95% CrI $[-0.041, 0.027]$, $p_d = .64$). A credible change in one group and not in the other is not evidence that the groups differ (Gelman & Stern, 2006; Nieuwenhuis et al., 2011), so we read the gender-agreement result as a strengthening common to both. No retention contrast was credible for differential object marking or for verb–object number agreement (Table B1). For verb–object number agreement, whose session term is itself the change across the interval, the late-window contrast leaned positive in the Mini-Norwegian group without reaching credibility (midline 400–900 ms, $\Delta = 0.07$, 95% CrI $[-0.01, 0.15]$, $p_d = .96$).

The refits that free the Session 6 step qualify the gender-agreement result. In them, the deviation of the Session 6 effect from the training trend was credible in none of the six gender-agreement cells, and the difference between the groups in that deviation in none of the six (midline 400–900 ms, $\Delta = -0.019$, 95% CrI $[-0.081, 0.042]$, $p_d = .73$; Mini-Norwegian minus Mini-English $\Delta = 0.07$, 95% CrI $[-0.05, 0.20]$, $p_d = .88$). The freed deviation is estimated far less precisely than the linear contrast, so it cannot tell whether the late-window effect went on growing over the interval at the training rate, held or fell back somewhat. What it shows is that the Session 6 effect did not depart credibly from the trend the training sessions set. For differential object marking, the freed deviation was credible overall in none of the six cells, but in the Mini-Norwegian group the Session 6 effect fell credibly below its training trend in the two early midline cells (200–500 ms, $\Delta = -0.13$, 95% CrI $[-0.23, -0.02]$, $p_d = .99$; 300–600 ms, $\Delta = -0.14$, 95% CrI $[-0.25, -0.03]$, $p_d = .99$), and the difference between the groups was credible in the same cells ($\Delta = -0.19$, 95% CrI $[-0.34, -0.03]$, $p_d = .99$; $\Delta = -0.18$, 95% CrI $[-0.34, -0.03]$, $p_d = .99$). For this property, the training trend rests on two sessions a week apart, so the projection the deviation is measured from is itself uncertain, and the early window carries the possible stimulus-driven component described above. Without the reprocessed and recovered records, every contrast credible in the primary analysis remains credible with the same sign, and one further contrast becomes credible, the late midline gender-agreement strengthening within Mini-English,

which under the protocol cannot become a claim (Table A2).

Grammaticality judgements

Accuracy on grammatical and on ungrammatical sentences reflects two separable quantities (see *Signal-detection reading of the judgement models*). Sensitivity is how well a learner told the two sentence types apart. The criterion is how readily they judged a sentence grammatical, whichever type it was. Figure 11 shows both for each group, and the accuracies they are computed from are given in Figure E3.

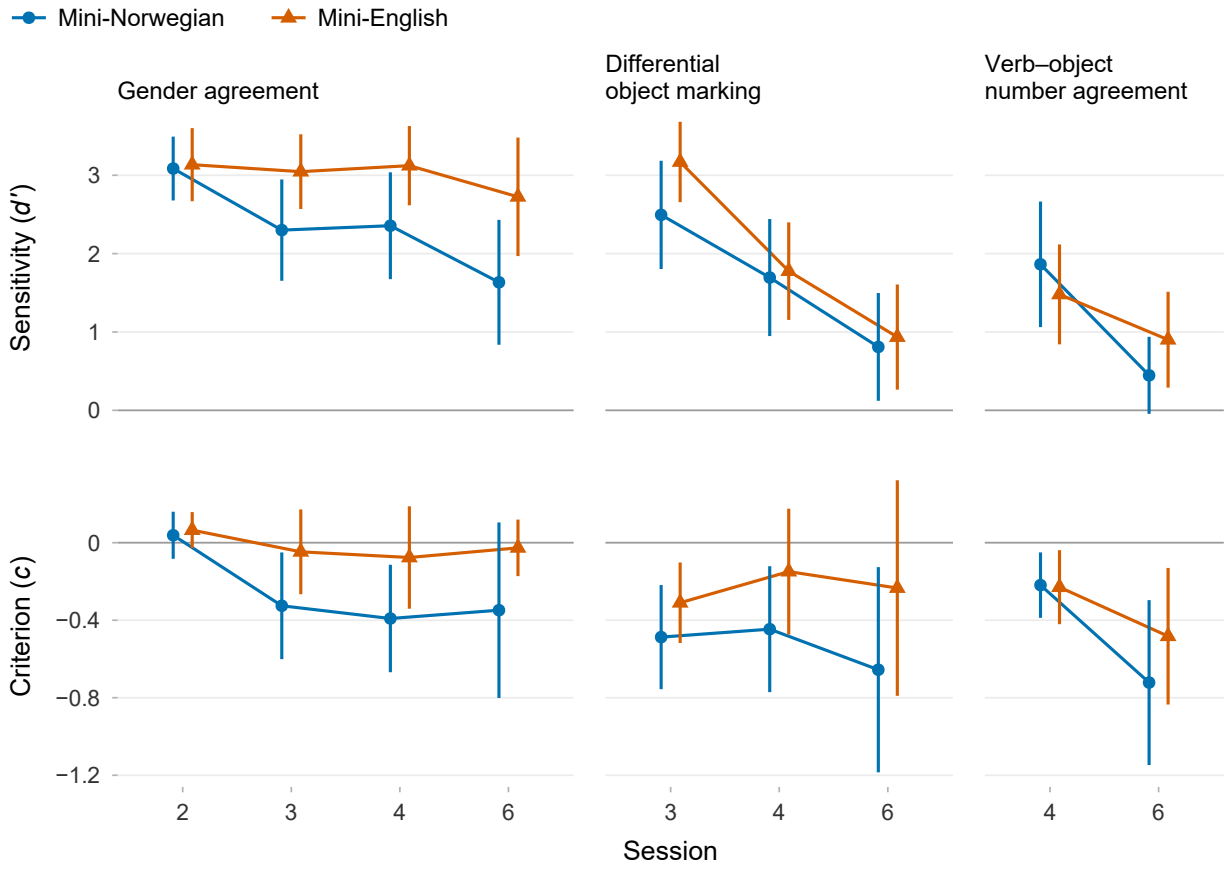
Sensitivity was high at each property's first session and declined from there in both groups (Table B3). Between Sessions 2 and 6, sensitivity to gender agreement fell in the Mini-Norwegian group ($\Delta = -3.48$, 95% CrI $[-4.81, -2.15]$, $p_d > .999$) and in the Mini-English group ($\Delta = -1.67$, 95% CrI $[-2.95, -0.31]$, $p_d = .99$). The declines for differential object marking were larger, and the one change that was not credible was that of the Mini-English group for verb–object number agreement ($\Delta = -1.42$, 95% CrI $[-3.20, 0.39]$, $p_d = .94$). The decline runs against the improvement that motivated the positive prior on the session term. Two features of the design bear on it. Properties accumulate, so a judgement made at Session 4 is made against a grammar with three manipulated properties instead of one, and Session 6 follows about four months without exposure. The design has no condition that holds task demands constant while sessions advance, so we do not separate these two sources of decline.

A group difference in sensitivity was credible for gender agreement at Session 6. For gender agreement, the only property with a Norwegian analogue, the difference between the groups averaged over sessions was $\Delta = -0.97$, 95% CrI $[-2.52, 0.59]$, $p_d = .89$ (Mini-Norwegian minus Mini-English), and at the retention session it was $\Delta = -2.01$, 95% CrI $[-3.93, -0.11]$, $p_d = .98$. The Mini-Norwegian estimate was the lower of the two at every session.

In the reported models, the groups diverged more clearly in their criterion. The difference between the groups' criteria was credible for gender agreement at Sessions 4 and 6. For gender agreement, the Mini-Norwegian group's criterion became more liberal between Sessions 2 and 6 ($\Delta = -1.04$, 95% CrI $[-1.27, -0.82]$, $p_d > .999$), whereas the Mini-English group's did not

Figure 11

Sensitivity and Criterion in the Grammaticality Judgements by Miniature Language, Property and Session



Note. Points are group means of participant-level d' (upper row) and criterion c (lower row), computed from each participant's hit rate (answering grammatical to a grammatical sentence) and false-alarm rate (answering grammatical to an ungrammatical one) at each session, with the log-linear correction for extreme rates. The horizontal line marks zero, which is chance-level sensitivity in the upper row and an unbiased criterion in the lower row, where a negative value is a readiness to judge sentences grammatical. Vertical lines are 95% t intervals over participants, and every participant receives equal weight. Each property is plotted from the session at which it entered the design: gender agreement at Session 2, differential object marking at Session 3 and verb–object number agreement at Session 4. These are descriptive, probit-scale summaries; the inferential estimates in the text come from the trial-level models, whose logit-scale sensitivity is about 1.7 times d'. Sessions are equally spaced, as in the models, although Sessions 2 to 4 are a week apart and Session 6 follows about four months without exposure.

change credibly ($\Delta = -0.06$, 95% CrI $[-0.33, 0.20]$, $p_d = .68$). At retention, the difference between the groups was $\Delta = -1.11$, 95% CrI $[-1.66, -0.57]$, $p_d > .999$, and the difference in how far the criterion moved between the first and last sessions was $\Delta = -0.98$, 95% CrI $[-1.32, -0.64]$, $p_d > .999$. At Sessions 4 and 6, then, Mini-Norwegian learners judged gender-agreement sentences grammatical more readily than Mini-English learners did, ungrammatical ones included, without telling the two types apart any better. For differential object marking, the criterion became more conservative in both groups (Mini-Norwegian $\Delta = 0.49$, 95% CrI $[0.26, 0.72]$, $p_d > .999$; Mini-English $\Delta = 0.60$, 95% CrI $[0.34, 0.87]$, $p_d > .999$), and for verb–object number agreement it did not change credibly in either.

In the fitted models, the gender-agreement criterion difference is carried by the grammaticality × language and grammaticality × session × language terms ($b = 0.27$, 95% CrI $[0.02, 0.53]$, $p_d = .98$; $b = 0.17$, 95% CrI $[0.11, 0.24]$, $p_d > .999$), and the sensitivity difference by the language and session × language terms ($b = -0.24$, 95% CrI $[-0.63, 0.15]$, $p_d = .89$; $b = -0.1624$, 95% CrI $[-0.3309, 0.0000]$, $p_d = .97$). The reported models carry by-participant slopes for grammaticality and for session but not for their interaction, which is the within-participant change in criterion. Refitted with that slope, the estimates of the difference between the gender-agreement groups' criteria moved by no more than 0.06 on the logit scale, but their intervals widened, and the difference was credible at no session. At retention it was $\Delta = -1.05$, 95% CrI $[-2.18, 0.11]$, $p_d = .96$, and averaged over sessions it was $\Delta = -0.55$, 95% CrI $[-1.13, 0.03]$, $p_d = .97$. The difference in how far the criterion moved between Sessions 2 and 6 was not credible in the refit either ($\Delta = -0.87$, 95% CrI $[-2.04, 0.33]$, $p_d = .93$). We therefore grade the criterion difference as suggestive (see *Strength of evidence*). Refitted with the ERP models' crossed by-item term as well, the 54 signal-detection contrasts of the three properties, compared under each by-participant structure in turn, kept their credibility in all but six of the 108 comparisons: the difference between the groups' changes in sensitivity between Sessions 2 and 6 for gender agreement ceased to be credible (with the reported by-participant structure); the difference between the groups' changes in sensitivity between Sessions 4 and 6 for gender agreement ceased to

be credible (with the reported by-participant structure); the Mini-English group's change in sensitivity between Sessions 2 and 6 for gender agreement became credible (with the added slope); the Mini-English group's change in sensitivity between Sessions 4 and 6 for gender agreement became credible (with the added slope); the difference between the groups' sensitivity for gender agreement at Session 6 ceased to be credible (with the added slope); and the Mini-Norwegian group's change in criterion between Sessions 4 and 6 for verb–object number agreement ceased to be credible (with the added slope). In each, the interval that excluded zero did so by no more than 0.06 on the logit scale. With both terms, the difference between the gender-agreement groups' criteria averaged over sessions was $\Delta = -0.55$, 95% CrI $[-1.12, 0.03]$, $p_d = .97$.

For differential object marking, sensitivity declined less in the Mini-Norwegian group than in the Mini-English group between Sessions 3 and 6 ($\Delta = 2.64$, 95% CrI $[-0.91, 6.19]$, $p_d = .93$). Its interval included zero, as it did in the refit with the added slope ($\Delta = 2.03$, 95% CrI $[-0.81, 4.85]$, $p_d = .92$) and in the refit that adds the by-item term as well ($\Delta = 2.07$, 95% CrI $[-0.82, 4.90]$, $p_d = .92$). Differential object marking is absent from both prior languages, so no account predicts this difference, and we treat it with the other unpredicted control-property effects in the Discussion.

We also fitted the accuracy models with the reaction-time screen of the project's earlier preprocessing, which we do not apply (see *Grammaticality-judgement task*). On the 33,022 judgement trials the screen keeps (8,159 fewer), the 54 signal-detection contrasts of the three properties kept their credibility in all but six: the difference between the groups' criterion for gender agreement at Session 3 became credible; the difference between the groups' changes in sensitivity between Sessions 2 and 6 for gender agreement ceased to be credible; the difference between the groups' changes in sensitivity between Sessions 4 and 6 for gender agreement ceased to be credible; the difference between the groups' sensitivity for gender agreement at Session 6 ceased to be credible; the difference between the groups' changes in sensitivity between Sessions 3 and 6 for differential object marking became credible; and the difference between the groups' changes in sensitivity between Sessions 4 and 6 for differential object marking became credible. In each, the

interval that excluded zero did so by no more than 0.11 on the logit scale. Two of the 24 population-level terms of the three models changed its relation to zero. With the screen, the difference between the gender-agreement groups' criteria averaged over sessions was $\Delta = -0.76$, 95% CrI $[-1.36, -0.17]$, $p_d = .99$, against $\Delta = -0.55$ without it.

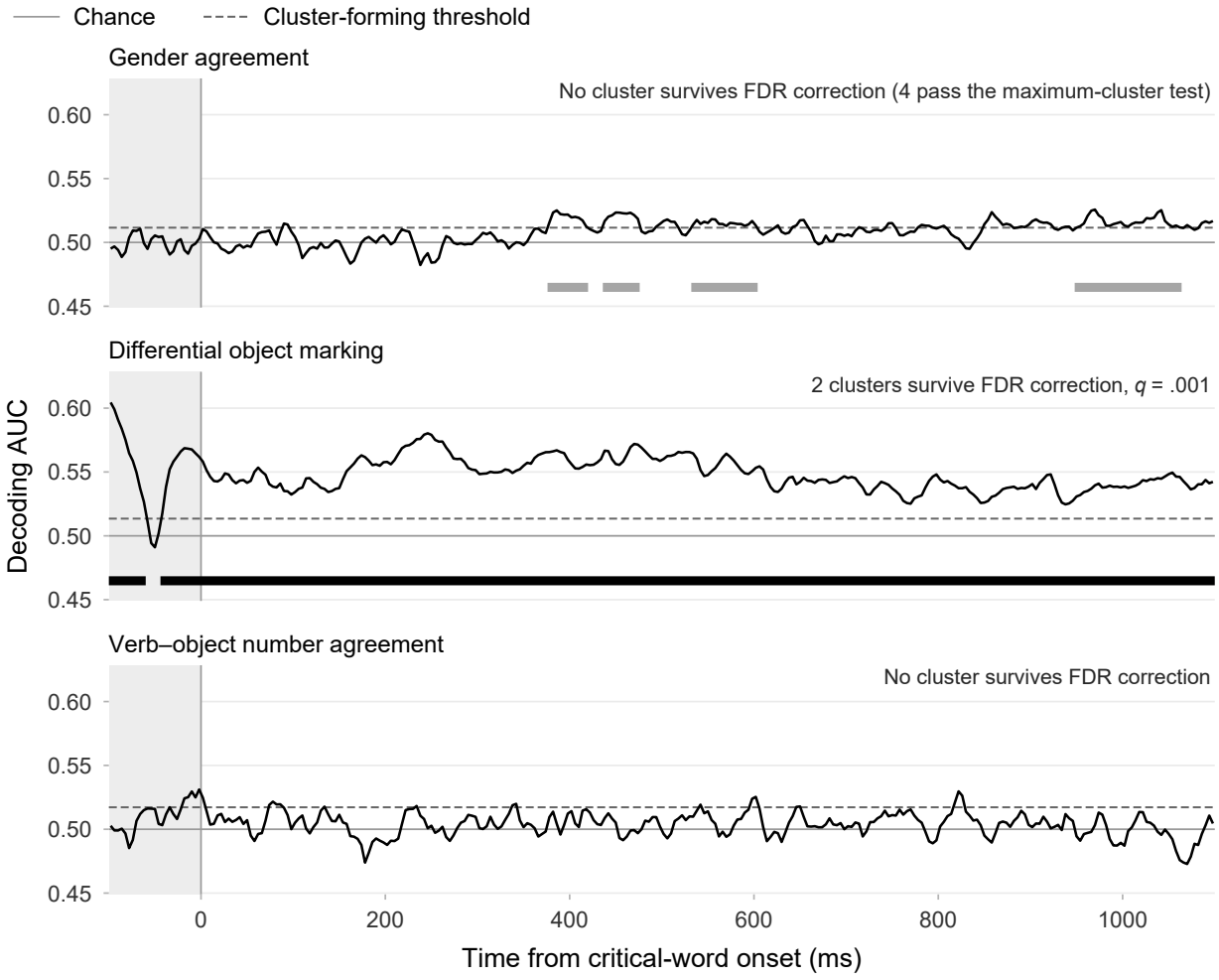
Multivariate decoding of grammaticality

We decoded Grammatical from Ungrammatical trials at each time sample (leave-one-participant-out AUC; Figure 12). Error was controlled across time within each analysis by cluster-based permutation and, within each property, across the clusters of its pooled overall time-course by Benjamini–Hochberg FDR (the confirmatory correction family; see *Transparency, preregistration and analysis status*). A significant cluster licenses only that the grammaticality time-course as a whole departs from chance. The temporal boundaries of a surviving cluster are not themselves inferential claims about the onset, offset or duration of the effect (Sassenhagen & Draschkow, 2019), and we do not read them as such.

The three properties differed here too, though the ordering of their raw separability has to be read against the pre-stimulus baseline. Differential object marking, whose univariate signature was the weakest and shortest-lived, was by far the most separable, though not necessarily by a violation response: its overall time-course carried two clusters surviving FDR correction, peaking post-stimulus at AUC 0.58 (at 246 ms; Figure 12). The pre-stimulus baseline should sit at chance. For the other properties, no cluster reaching into it passed the maximum-cluster test, but differential object marking's pre-stimulus AUC averaged 0.55 with 21 samples surviving FDR before the critical word. Its highest AUC anywhere in the epoch falls in that baseline (0.60), above its post-stimulus peak. Part of the differential-object-marking separability therefore precedes the stimulus and may reflect a low-level difference between conditions, for instance the physical presence of the object marker in the preceding region, and need not reflect a violation response. Peak AUCs are quoted on the post-stimulus axis for this reason. The extents of the surviving clusters make the limit concrete: one lies wholly before the critical word (−98 to −62 ms) and the other begins just before onset and spans the epoch (−42 to 1098 ms). The cluster test cannot

Figure 12

Time-Resolved Decoding of Grammaticality for Each Property



Note. Lines are leave-one-participant-out decoding AUC (area under the receiver operating characteristic curve) for grammatical against ungrammatical trials at each 4-ms sample, all sessions pooled (the confirmatory analysis). The solid grey horizontal line marks chance (AUC = 0.5) and the dashed line the cluster-forming threshold, the 95th percentile of the pooled within-participant permutation null. Black bars below each time-course mark the extent of the clusters that survive Benjamini–Hochberg false-discovery-rate (FDR) correction ($q < .05$) across the clusters of that property's time-course, and the text in each panel gives their number and adjusted cluster-level p values (q). Grey bars mark clusters that pass the maximum-cluster permutation test ($p < .05$), which controls the family-wise error rate across time, but not the second, FDR layer. A bar marks cluster membership, not sample-by-sample significance, and its edges are not estimates of onset or offset. The vertical grey line marks critical-word onset (0 ms), and the shaded interval before it is a negative control, in which decoding should sit at chance. The AUC is a point estimate: the pipeline records no per-fold dispersion, so no uncertainty band is drawn. All panels share one vertical scale.

attribute any part of the surviving separability specifically to a violation response. What it establishes is that the conditions are separable throughout the trial. Nor does the pre-stimulus level bound how much that difference contributes after onset, because the response to the preceding word continues into the epoch and need not keep its pre-stimulus strength or pattern there. The temporal-generalisation matrix shows how this separability is organised in time. Decoders trained in the first third of the baseline classified its last third at a mean AUC of 0.44, below chance, so the pattern separating the conditions reversed within the baseline. Each trial's baseline mean is subtracted from the decoder's input, which makes the conditions' difference average to zero over that interval, so a difference that changes across it must reverse there. Decoders trained in the last third of the baseline classified the first 100 ms after onset at 0.52, so that pattern carried across onset. From 200 ms to the end of the epoch a different pattern held. Decoders trained there classified the other latencies of that span at 0.54, whereas those trained in the last third of the baseline classified it at 0.50. The response to the differing preceding word overlaps the epoch, and the subtraction carries the conditions' difference in the baseline mean into every later sample, so the matrix cannot attribute that later pattern to a violation response. No null distribution was computed for the matrix, so these values are descriptive. Gender agreement, whose amplitude effect was credible in the region-averaged models, produced no clusters in its pooled time-course surviving the confirmatory correction (Figure 12). Of its 17 clusters, four passed the maximum-cluster test on its own (378 to 418 ms, 438 to 474 ms, 534 to 602 ms and 950 to 1062 ms, with cluster-level p from .003 to .030), but the second layer removed them all, the smallest adjusted value being $q = .051$. It peaked post-stimulus at AUC 0.53 (at 970 ms). Its pre-stimulus baseline sat at chance, so, unlike differential object marking, it carries no separability that precedes the critical word, and none of its clusters can be attributed to a difference in the region before it. Verb–object number agreement produced no clusters surviving correction in its pooled time-course (smallest cluster-level $p = .085$ across all its analyses, before or after correction, and peak overall AUC 0.53). In the robustness analysis, which leaves out the reprocessed and recovered records, the confirmatory decisions were the same. The numbers of clusters surviving both layers were zero, two and zero for gender agreement, differential

object marking and verb–object number agreement, and the numbers passing the maximum-cluster test, whether or not they survived the second layer, were two, two and zero (four, two and zero in the primary analysis). Cross-language generalisation, in which the decoder is trained on one mini-language and tested on the other, peaked post-stimulus at AUC 0.53 for gender agreement, 0.58 for differential object marking and 0.53 for verb–object number agreement. Those values, and the temporal-generalisation ones, are descriptive. The generalisation AUCs are computed over test trials pooled across participants, so, unlike the time-course AUC, they can be moved by differences between participants in class balance or in their overall classifier scores. No permutation null was computed for them, so we treat them as converging context and do not read the transfer question off them. Cross-property generalisation stayed close to chance (peak post-stimulus AUC 0.54 across all six train–test pairs, reached when a decoder trained on gender agreement was tested on differential object marking). This too is descriptive. It has no permutation null, and because every participant contributes trials to all three properties, the train and test sets are not participant-independent. Its theoretical reading is taken up in the Discussion. Because each per-cell decoding score is a mean cross-validated AUC estimated on cells of differing and sometimes small participant numbers, the ordering across properties is provisional, and every cluster-level decision reported here rests on the well-populated pooled time-course (Varoquaux, 2018).

Discussion

Two groups of Norwegian–English bilinguals learned miniature languages that shared their grammatical morphemes but differed in lexicon and in where gender was marked. For gender agreement, the one property Norwegian has and English lacks, the Mini-Norwegian group was not more sensitive than the Mini-English group. In the judgements, the Mini-Norwegian group was credibly less sensitive to gender agreement only at Session 6. In the ERP response, the grammaticality × language term was credible in 0 of 6 gender-agreement cells (Figure 7, second row), and its development across sessions, the grammaticality × session × language term, in 0 of 6. In the reported models, the groups differed more clearly in their response criterion. In the later sessions, Mini-Norwegian learners judged gender-agreement sentences grammatical more readily

than Mini-English learners did, ungrammatical ones included, although that difference was not credible once each learner's change in criterion was modelled, and we grade it as suggestive. Neither control property showed a credible group difference in sensitivity, and no ERP grammaticality effect, or change in one across the training sessions, was credibly modulated by language. We take these findings in turn, beginning with what they imply for the transfer accounts.

Gender agreement and the transfer accounts

The Typological Primacy Model predicted that the Norwegian lexicon would lead Mini-Norwegian learners to Norwegian grammar, gender included, and the English lexicon would lead Mini-English learners to English, with no gender to draw on (Rothman, 2015; Rothman et al., 2015). Sensitivity to gender agreement should then have been greater in the Mini-Norwegian group, at least at the earliest sessions. It was not. At Session 2, the first measurement, the difference in sensitivity between the groups was $\Delta = -0.19$, 95% CrI $[-1.93, 1.54]$, $p_d = .58$ (Mini-Norwegian minus Mini-English), and the posterior probability that the Mini-Norwegian group was the more sensitive was .42. Averaged over sessions, that probability was .11. The ERP evidence is consistent with this but less informative. The grammaticality × language term for gender agreement was not credible in any window, but its intervals are wide relative to the grammaticality effect itself, so the ERP data do not exclude a language difference as large as the effect they measure. The grammaticality effect was credible in 3 of 6 cells, so the absence of a credible language difference is not an absence of any response to measure, but it is not evidence that the groups responded alike.

Property-by-property accounts allow exactly this outcome. On those accounts, Norwegian gender is available to every participant, and a learner of either miniature language can recruit it once the input shows the property to be shared (Westergaard et al., 2017; Westergaard, 2021). Their allowance for early lexical influence, and the structural resemblance of the Mini-Norwegian determiner to its Norwegian counterpart, meant that they could also have accommodated a Mini-Norwegian advantage. They did not require one, and none appeared. The data therefore fit property-level availability better than lexically cued selection of a whole grammar, which is the reading of the design's central contrast that the accounts' own predictions support.

Three limits apply to that reading, and they are properties of the design, not of the analysis. The first is that no measurement precedes training. Both groups were already highly sensitive to gender agreement at Session 2, after pretraining, training and a test, so a difference in the initial grammar that a single training session erased would leave no trace here. The Typological Primacy Model expects non-facilitative transfer to be overcome with input, so these data speak against an advantage that persists into learning more than against a difference at the very first encounter (González Alonso et al., 2025; González Alonso & Rothman, 2017). The second is that no group lacks gender in its first language. High sensitivity in both groups is what facilitation from Norwegian would produce, but it is also what learning a salient agreement pattern within a session would produce without any transfer, so the design cannot show that Norwegian helped either group. For the same reason, it cannot rule out the L2 Status Factor, which also predicts no Mini-Norwegian advantage. The third is that sensitivity in a judgement and the amplitude of a violation response index processing. Neither establishes the grammar that supports it, and the learners' responses may rest on more than linguistic computation (González Alonso et al., 2020). The design was not built to settle the transfer debate, and these findings narrow it only as far as the absence of the predicted group difference allows.

A possible lexical influence on the response criterion

Where the lexicon may have left a mark was in how readily learners accepted gender-agreement sentences. The Mini-Norwegian group's criterion became credibly more liberal ($\Delta = -1.04$, 95% CrI $[-1.27, -0.82]$, $p_d > .999$) between Sessions 2 and 6, and in the reported models the difference from the Mini-English group was credible for gender agreement at Sessions 4 and 6. Once each learner's change in criterion was modelled, the difference was credible at no session (see *Strength of evidence*), so this section interprets a suggestive result. Read as accuracy, this shift looks like an advantage, because a learner who accepts more sentences is right more often on the grammatical ones. Read as signal detection, it is a change in willingness to accept, unaccompanied by better discrimination. The shift appeared in the judgement and not in the ERP response. If it is genuine, that places the group difference at the decision more clearly than in the

online processing of the violation, although the ERP estimates were not precise enough to exclude a group difference in that processing.

Several mechanisms could produce such a shift, and the design does not separate them. One is familiarity. Sentences that are processed more fluently are more readily endorsed as grammatical in artificial grammar learning, particularly where knowledge is uncertain (Scott & Dienes, 2010; Topolinski & Strack, 2009). Words built on Norwegian stems may be processed more fluently by Norwegian speakers than words built on English stems, although we know of no study that has shown lexical familiarity to shift the criterion in this way. A second is crosslinguistic influence of a non-facilitative kind. If a Norwegian-like lexicon activates Norwegian agreement patterns that do not match those of the miniature language, some violations would look acceptable by Norwegian standards, raising false alarms and lowering sensitivity together. That is the pattern the Linguistic Proximity Model calls misanalysis (Westergaard et al., 2017), and it is consistent with the direction of the sensitivity difference, which was credible only at Session 6, and not in three of the six refits (with the reaction-time screen, with the diversity covariate and with both the added slope and the by-item term). It could not arise from gender assignment, because every noun takes the gender of its Norwegian equivalent in both languages (see *Materials*), so no violation is licensed by a Norwegian assignment. For the same reason, an item-level analysis relating false alarms to the Norwegian gender of each noun is not available here: congruency is constant by construction, and the account can be tested only in materials that cross it. A third is strategy. Mini-English learners, facing gender marking that English lacks, may have scrutinised agreement more cautiously. For differential object marking, the criterion changed between Sessions 3 and 6 in both groups (Mini-Norwegian $\Delta = 0.49$, 95% CrI $[0.26, 0.72]$, $p_d > .999$; Mini-English $\Delta = 0.60$, 95% CrI $[0.34, 0.87]$, $p_d > .999$), and the between-language difference in that change was $\Delta = -0.11$, 95% CrI $[-0.46, 0.23]$, $p_d = .74$. A familiarity account that operates on the lexicon as a whole should affect every property, so whether the other properties show the same direction of difference bears on the first mechanism.

The pattern also clarifies what the closest precedent found. González Alonso et al. (2020) observed a group difference at first exposure in the ERP response but not in judgement accuracy,

and attributed it to how attention was allocated, a precursor of selecting a source, not to grammatical processing. Their accuracy measure, like ours before we separated sensitivity from criterion, could not have shown a criterion difference as such. Here, after training, the lexical manipulation did not credibly alter the ERP grammaticality effect, and in the reported models it moved the decision criterion. In both studies, then, the lexicon left more trace in how learners engaged with the task than in their sensitivity to the grammar, although here that trace is suggestive.

Relation to the initial-exposure antecedent

Against González Alonso et al. (2020), these data replicate the general profile but not its specifics. Their learners produced no P600 to gender-agreement violations at first exposure, and neither did ours at any session. The dominant signature for gender agreement here was a violation negativity, under a comparable artificial-language paradigm. The measurement occasion was not comparable, since their ERPs come from a single first-exposure session, whereas ours begin after the first training session, whose post-training test was repeated once when a learner scored below 80%, and continue across four sessions. An early negativity to gender-agreement violations was present in both studies. Their negativity was confined to the English-lexicon group and to right-anterior electrodes, whereas ours was not credibly modulated by language in any window. Their Spanish-lexicon group showed a broad positivity to violations at 300–600 ms, whereas in our gender-agreement data the ungrammatical trials were the more negative-going.

The divergences admit two kinds of explanation, and the literature on replication cautions against reaching for the substantive kind first. Substantively, the designs differ in the first language (Spanish against Norwegian), in the realisation of gender (concord across the noun phrase against a suffixed or free determiner with predicate-adjective agreement), and in the measurement occasion. Statistically, their groups comprised 26 and 24 learners. Effects that reach significance in samples of that size systematically overestimate their true magnitude and can even carry the wrong sign (Gelman & Carlin, 2014), the significance filter makes published effects look more replicable than they are (Vasishth et al., 2018), and trial-level measurement error can inflate as well as attenuate an observed effect in small samples (Loken & Gelman, 2017). ERP effects compound the problem,

since even systematic resampling finds the magnitude, onset and scalp distribution of the P600 to vary widely across samples (Kim et al., 2024). The level at which replication should be assessed is therefore the qualitative one, where the two studies agree that early-stage learners of a miniature language respond to morphosyntactic violations without a canonical P600, and that the lexical manipulation shows itself in how learners engage with the task more than in their grammatical sensitivity. The same arguments cut against over-reading any group-specific effect in our own data, as the *Strength of evidence* section below does.

Development, retention and the control properties

Behavioural sensitivity declined across sessions (Figure 11), whereas the gender-agreement ERP grammaticality effect grew in the late window. The two measures are recorded on the same trials, so the dissociation is informative. Each session adds a property, and a judgement made against a larger grammar is harder even when the response to a given violation has strengthened. Session 6 follows about four months without exposure, which would lower sensitivity in the judgements more than a well-consolidated violation response. The design cannot separate these explanations from a drift in how learners approached the task over the sessions.

The control properties showed no credible ERP modulation by language during training. For differential object marking, the grammaticality × session × language term was credible in 0 of 6 cells, and for verb–object number agreement in 0 of 6. Neither property differed between the groups in behavioural sensitivity at any session, although the estimated decline in sensitivity to differential object marking was the smaller in the Mini-Norwegian group, a difference whose interval included zero in the reported models (see Results). The one language-conditioned ERP result for a control property concerns retention. In the refits that free the Session 6 step, the early differential-object-marking effect of the Mini-Norwegian group fell below its training trend at Session 6, and the difference between the groups in that deviation was credible in two of the six cells, both on the midline. No account predicted it, it rests on a trend estimated from two sessions, and it falls in the window where the stimulus difference before the critical word could contribute, so we treat it as a candidate for replication and not as a finding about transfer. Differences in how

easily the two lexicons were learned, and chance variation across the grid, remain possible sources.

Over the retention interval, the gender-agreement effect strengthened in the late window, where it had grown during training, and the groups did not differ credibly in that change. Because the linear specification carries the training trend across the gap, that strengthening is the trend evaluated over the interval. The refits that free the Session 6 step show that the effect at Session 6 did not depart credibly from that trend in any gender-agreement cell (midline 400–900 ms, $\Delta = -0.019$, 95% CrI $[-0.081, 0.042]$, $p_d = .73$), in either group or in the difference between them (Mini-Norwegian minus Mini-English $\Delta = 0.07$, 95% CrI $[-0.05, 0.20]$, $p_d = .88$), although they estimate the departure too imprecisely to tell continued growth from a plateau. Behavioural sensitivity, by contrast, fell over the same interval, and the groups did not differ in how far ($\Delta = -0.6065$, 95% CrI $[-1.2361, -0.0000]$, $p_d = .97$). Training studies of this kind find violation responses that hold or grow over months without exposure (Morgan-Short, Finger, et al., 2012), and the offline interval here could allow the consolidation such studies describe (M. H. Davis & Gaskell, 2009; Dudai et al., 2015), but the present data do not isolate it.

What the multivariate analysis adds

In the decoding reported here, the univariate and multivariate analyses order the three properties in broadly the same way. Gender agreement carries an amplitude effect credible in 3 of 6 cells and pattern separability that passed the maximum-cluster test but not the second correction layer, which we grade as suggestive. Verb–object number agreement carries neither. Differential object marking carries separability that its pre-stimulus baseline forbids us to read as a violation response. That agreement was not guaranteed, because the two analyses ask different questions. Window-averaged amplitude asks whether a condition shifts the mean at a set of electrodes, whereas decoding asks whether the joint sensor pattern separates single trials (Hebart & Baker, 2018; Wang & Kuperberg, 2024). A divergence in either direction would not have told us which property is better represented, and classification accuracy is in any case not a standardised effect size and cannot be compared across properties as though it were (Hebart & Baker, 2018). One limit is worth setting out before the mechanisms. Rejecting a pooled cluster null licenses only that a

grammaticality pattern is decodable in some part of the sample, leaving its prevalence across participants unestimated (Allefeld et al., 2016).

Both profiles that can set the two analyses apart have documented mechanisms. A contrast can be decodable with little univariate effect because decoding tolerates non-uniform response signs, so neighbouring sensors of opposite polarity both carry information where averaging cancels them (Hebart & Baker, 2018). Forward modelling shows that such cancellation is substantial at the scalp for distributed sources with partly opposing orientations (Ahlfors et al., 2010). The transient mean amplitude and comparatively high classification of differential object marking are therefore compatible with a single generator whose spatial configuration is stable even where its net amplitude is small, so far as the baseline difference allows that reading. Multivariate analyses of sentence processing have repeatedly recovered effects that the corresponding ERPs did not (Cho & Brennan, 2025; He et al., 2024). The converse profile is equally well characterised. Multivariate tests draw their sensitivity from electrode-level variability in the effect within participants and are insensitive to the between-participant variance in mean amplitude that governs univariate group tests (T. Davis et al., 2014, on fMRI voxels), and increased trial-to-trial variability leaves expected mean amplitude unchanged while progressively shrinking expected decoding accuracy (Carrasco et al., 2024). An effect that is consistent in sign but spatially homogeneous would therefore yield a credible amplitude effect with comparatively little for a classifier to exploit. That profile may describe gender agreement in part, since its credible amplitude effect came with separability that passed the maximum-cluster test but not the second correction layer. Decoding is usually the more sensitive test of the two, but it is not uniformly so, and comparable orderings are documented for the mismatch negativity and the N2pc (Carrasco et al., 2024).

These considerations constrain the claims available to us. A failure to decode is a failure to find linearly separable information under one analysis, and it leaves open whether the brain distinguishes the conditions (Peelen & Downing, 2023). Classification performance, high or low, is a poor guide to representational content (Ritchie et al., 2019). Nor would a divergence between the two analyses support a conclusion about the dimensionality of the underlying code (T. Davis et al.,

2014).

The weak cross-property generalisation bears on a live theoretical question. If the grammaticality response were largely a domain-general salience or integration signal (Aurnhammer et al., 2023; Sassenhagen & Fiebach, 2019), a decoder trained on one property should transfer to the others, but no direction exceeded a peak AUC of 0.54, and with no null distribution that value cannot be told apart from chance. That pattern is consistent with evidence that the late positivity fractionates topographically and functionally (Kaan & Swaab, 2003), that agreement features follow separable developmental courses (Alemán Bañón et al., 2018), that different violation types within the same learners are predicted by different individual-difference variables (E. S. Nichols & Joanisse, 2019), and that decoders of grammatical features generalise selectively (Cho & Brennan, 2025). Reliable cross-context transfer of linear EEG decoders is achievable (Zhao et al., 2025), which weighs against reading the present result as simple insensitivity. Even so, failed cross-classification is confounded with feature selection, since the sensors and latencies that best separate grammatical from ungrammatical trials within a property need not be those carrying a property-general signal (Kaplan et al., 2015). We therefore read the result as consistent with construction-specific coding, without taking it to refute any shared component. A Bayes-factor treatment of the decoding time-course would allow such nulls to be quantified as well as reported (Keysers et al., 2020; Teichmann, 2022).

Strength of evidence

The conclusions above should be read on the graded scale set out under *Inference* (Kruschke & Liddell, 2018; Makowski et al., 2019). The central finding is an absence, and absences need their precision stated. In the ERP models, the interval for the grammaticality × language term for gender agreement is wide relative to the grammaticality effect itself (no bound beyond 0.039 within-participant standard deviations; compare the first and second rows of Figure 7), so the ERP data cannot exclude a language difference of the size of the effect they measure, and they add little to the judgements on this point. In the judgements, the sensitivity difference is also imprecisely estimated (averaged over sessions, $\Delta = -0.97$, 95% CrI $[-2.52, 0.59]$, $p_d = .89$), so the

behavioural evidence against a Mini-Norwegian advantage rests more on its direction than on its precision. We did not adopt an equivalence criterion, so neither null is a demonstration that the groups are equivalent (Amrhein et al., 2019; Kruschke & Liddell, 2018).

The criterion difference for gender agreement is the result that most needs qualifying. In the reported models it was credible at Sessions 4 and 6 and grew between Sessions 2 and 6. The refit that adds a by-participant slope for the change in bias moved its estimates by no more than 0.06 on the logit scale but widened their intervals, and in that refit the difference was credible at no session, nor did it grow credibly (averaged over sessions, $\Delta = -0.55$, 95% CrI $[-1.13, 0.03]$, $p_d = .97$; growth between Sessions 2 and 6, $\Delta = -0.87$, 95% CrI $[-2.04, 0.33]$, $p_d = .93$). That slope lets each learner's criterion change at its own rate, and leaving it out can make the interval of the corresponding fixed effect too narrow (Barr et al., 2013), which is why the ERP models carry its counterpart. The direction of the difference is the same in every fit, with a posterior probability of at least .96 in each that the Mini-Norwegian criterion was the more liberal at Sessions 4 and 6, but its credibility depends on the random-effect structure, and we grade it as suggestive.

The refit with the reaction-time screen qualifies the gender-agreement criterion difference in another way. With the screen, the groups' criteria differed credibly for gender agreement at Sessions 3, 4 and 6. The difference was larger at every session, by 0.18 to 0.23 on the logit scale, but as without the screen it was not credible at Session 2 and grew credibly between Sessions 2 and 6 ($\Delta = -0.93$, 95% CrI $[-1.32, -0.53]$, $p_d > .999$). The screen therefore bears on the size of the difference more than on its course. We report the fits without it, for the reasons given under *Grammaticality-judgement task*, and grade as suggestive the six results that only one of the two fits supports: the criterion difference at Session 3, credible only with the screen ($\Delta = -0.67$, 95% CrI $[-1.27, -0.09]$, $p_d = .99$); the Mini-Norwegian group's lower sensitivity at Session 6, credible only without it ($\Delta = -2.01$, 95% CrI $[-3.93, -0.11]$, $p_d = .98$); the difference between the groups' changes in sensitivity between Sessions 2 and 6 for gender agreement, credible only without it ($\Delta = -1.8196$, 95% CrI $[-3.7083, -0.0001]$, $p_d = .97$); the difference between the groups' changes in sensitivity between Sessions 4 and 6 for gender agreement, credible only without it

($\Delta = -0.6065$, 95% CrI $[-1.2361, -0.0000]$, $p_d = .97$); the difference between the groups' changes in sensitivity between Sessions 3 and 6 for differential object marking, credible only with it ($\Delta = 3.53$, 95% CrI $[0.01, 7.10]$, $p_d = .98$); and the difference between the groups' changes in sensitivity between Sessions 4 and 6 for differential object marking, credible only with it ($\Delta = 1.766$, 95% CrI $[0.003, 3.548]$, $p_d = .98$). Each interval excluded zero, in the fit that supports it, by no more than 0.11 on the logit scale, and four of them by less than two Monte Carlo standard errors of that bound, so these runs do not settle which side of zero they fall on. The gender-agreement sensitivity differences run against a Mini-Norwegian advantage, not towards one. Those for differential object marking concern a property absent from both prior languages, for which no account predicts a group difference.

The gender-agreement ERP learning effect is credible in 2 of 6 cells under the maximal random-effect structure and in 2 of 6 without the by-item slope, and the same cells stay credible under the weak priors and without the reprocessed and recovered records. It is confined to the late window, however, and the interval of its midline estimate lies close to zero, so we grade it as clear in direction but modest in size. The one language-conditioned ERP result, the deviation at retention in two early differential-object-marking cells, is unpredicted and confined to two of the twelve cells in which it was estimated, and we grade it as suggestive. The retention estimates rest on the participants who returned for Session 6 and are graded no higher.

The decoding evidence grades separately. Gender agreement has no cluster surviving the confirmatory correction, although four clusters, and two in the robustness analysis, passed the maximum-cluster test, which by itself controls the family-wise error rate across time. Its pre-stimulus baseline sat at chance, and we grade its separability as suggestive. Differential object marking's two clusters begin before the critical word, and the word before it differs between the conditions, so no part of its separability can be attributed to a violation response. The session-resolved breakdowns enter no correction family and are descriptive. Several features of the analysed sample were fixed after the models had first been fitted, and all are set out under *Transparency, preregistration and analysis status*: the reprocessing of nine Session 3 recordings

filtered at 1 Hz, the recovery of two of three mislabelled exports and the exclusion of the third, and the exclusion of a participant trained in the wrong miniature language at Session 2. The robustness analysis that leaves out the reprocessed and recovered records changes no ERP conclusion (Table A1).

Limitations and future directions

The strongest constraint on what this design can conclude is that the miniature languages were built to probe the lexicon's role at first exposure, and the transfer accounts do not make mutually exclusive predictions for most of what the design measures once learning is under way. Only gender agreement has a first-language analogue, and for it the accounts separate only on the absence of a Mini-Norwegian advantage. Three additions would sharpen the test. A measurement before any training would capture the initial state that the Typological Primacy Model constrains. A group whose first language lacks gender, learning the same miniature languages, would show whether Norwegian facilitated gender agreement at all, and so whether the L2 Status Factor can be separated from property-by-property accounts. Crossing the lexicon with the position of the gender-marking determiner would separate the lexical cue from the structural one, which the present languages confound (Castle et al., 2026; Jensen & Westergaard, 2023).

Several further features of the design constrain the conclusions. The properties enter at different sessions, so a property's identity is confounded with its amount of exposure, and cross-property comparisons are not exposure-matched. Assignment to language followed a fixed rule on the participant identifier, so the groups were not formed by random allocation. The post-training test was repeated once for a learner who scored below 80% at the first attempt, after which every learner continued, so the first ERP session follows a variable amount of training. Additional gendered or case-marking languages reported under *Participants* were not screened out. Such experience could support the gender or marking structures, and it is only partly absorbed by the L1-dominance covariate, which enters the ERP models only. Of the 65 learners enrolled, 36 attended the retention test, so the Session 6 estimates draw on the returners, who may differ from those who left. Effects are quantified as mean amplitudes in fixed latency windows, and without a

native-speaker baseline the native-likeness of the developing response is untested. The trigger rule that segmented the exports discarded 36% of the critical words, and the check of whether the discarded words differ from the kept ones passed in three of its twelve cells (Table A3), so elsewhere it cannot exclude an omission that shifts a grammaticality contrast by more than 0.33 μV. In the one cell where the two sets differed (gender agreement, 200–500 ms, lateral), the discarded words carried the more negative difference, so the analysed contrast there understates any violation negativity. Each sentence frame belongs to one miniature language, so a difference between the groups' judgements is also a difference between two sets of frames, which a by-item term can absorb but no design that keeps the lexicons apart can remove. Cognitive individual-difference predictors are analysed in a companion paper (Bernabeu et al., 2026).

Conclusion

Norwegian–English bilinguals learning a Norwegian-like miniature language were no more sensitive to its gender agreement than learners of an English-like one, although gender is a Norwegian property that English lacks. Their judgements showed no sensitivity advantage at any session, and their ERP response showed no group difference either, though it could not exclude one. That absence is what property-by-property accounts of transfer allow and what the Typological Primacy Model, which predicts that the lexicon selects the grammar, does not. The lexicon may have left its mark on the decision instead. In the later sessions, Mini-Norwegian learners accepted gender-agreement sentences more readily, ungrammatical ones included, but that difference was not credible once each learner's change in criterion was modelled. The design cannot show that Norwegian helped either group or recover the initial state, so the transfer debate is narrowed and not settled. What the results establish is that a Norwegian-like lexicon brought no advantage in sensitivity to a Norwegian property. Whether it shaped how learners engaged with the task is a question for a design built to test it.

Data and code availability

Data extraction, model fitting and reporting are fully scripted as a reproducible research compendium (Marwick et al., 2018). Models were fitted on High-Performance Computing (HPC)

clusters at Sigma2 (Norway) and the University of Oxford's Advanced Research Computing facility, on Linux (x86_64-pc-linux-gnu) under R 4.5.1 (R Core Team, 2025), with `brms` 2.23.0 (Bürkner, 2017) driving CmdStan 2.39.0 through the `cmdstanr` 0.9.0 backend. The clusters supply that stack through environment modules, with no container. The compiler settings used on those clusters are recorded in the repository and do not change the models. Every version given here comes from the provenance record the fitting run wrote. Model fitting used the fixed seed 20260607, set in the shared Bayesian settings and recorded in that provenance record. Each chain evaluated the likelihood in parallel (`reduce_sum`), which groups its partial sums at run time, so a rerun with the same seed reproduces the posterior summaries to within Monte Carlo error, not draw for draw. For the signal-detection contrasts, the Monte Carlo standard error of an interval bound had a median of 0.012 and a maximum of 0.075 on the logit scale. The decoding permutations used the seed 20260702, exported by the job script that runs them and recorded by that script beside the decoding artefacts. This manuscript reads only the script-generated summary tables, so every statistic in the Results is populated from them and none is transcribed by hand. The pooled posterior summaries, retention contrasts, convergence diagnostics, sample-flow counts and decoding time-courses that populate the figures and tables are all published with the repository. The decoding artefacts include the participant count behind every per-cell score. Figures are drawn with `ggplot2` (Wickham, 2016) using the theme and colour system of `depictr` (Bernabeu, 2026a), whose default qualitative palette is the colourblind-safe Okabe–Ito set. The companion paper (Bernabeu et al., 2026) uses the same system, so the two share one visual identity. The literature counts in Figure 1 were retrieved from the Scopus Search API with `scopusflow` (Bernabeu, 2026b), which records the exact query beside each count, so the search is reproducible from the repository. Its coverage limits are stated in the figure note. All extraction, modelling and reporting scripts, the script-generated summary tables and the environment specification are available at https://github.com/pablobernabeu/LESS-morphosyntactic-transfer. The environment specification comprises an `renv` lockfile pinning the R packages of the fitting library and a provenance record written at fit time. That record holds the software versions in force when the ERP models were

fitted, which is what the versions above are read from. The accuracy models and the decoding analyses run in the same cluster environment under separate scripts. The accuracy script records the R, brms, cmdstanr and CmdStan versions of each fit in the pooled fit metadata that accompanies the results, and the decoding job records its seed and permutation settings beside the decoding artefacts. The raw electroencephalographic recordings, too large to distribute through an ordinary version-control repository, and the raw behavioural logs are openly available in the project's Open Science Framework repository at https://osf.io/tq7vy.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude Code (version 2.1, Anthropic), running primarily the Claude Opus and Claude Fable models, to support the research, the analyses and the writing: it drafted and revised text and code, checked the reporting code that populates this manuscript from the analysis artefacts, and checked the manuscript against those artefacts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Pablo Bernabeu: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Christina Athanasiadi**: Data curation, Project administration, Writing – review & editing. **Stella L. Pischinger**: Data curation, Writing – review & editing. **Gabriella Silva**: Conceptualization, Methodology, Investigation, Writing – review & editing. **My Ngoc Giang Hoang**: Data curation, Writing – review & editing. **Vincent DeLuca**: Conceptualization, Methodology, Supervision, Writing – review & editing. **Jason Rothman**: Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Iva Ivanova**: Conceptualization, Methodology, Writing – review & editing. **Claudia Poch**: Methodology, Validation, Writing – review & editing. **Jorge González Alonso**: Conceptualization, Methodology, Funding acquisition, Project administration, Supervision, Writing – review & editing.

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Declaration of competing interest

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Appendix A

Analytic decisions in detail

This appendix sets out the reasoning behind three analytic choices that the Method states but does not argue in full: the retention of the by-item grammaticality slope in the ERP models, the treatment of the recordings filtered at 1 Hz and of the mislabelled exports, and the definition of the false-discovery-rate correction family for the decoding analyses. It also reports a check on the trials that the segmentation's trigger rule discards.

The by-item grammaticality slope

The reason for keeping the by-item grammaticality slope is that dropping it would bias the test in the direction of the hypothesis. Barr et al. (2013) show that omitting random-effect structure that is present in the population inflates the Type I error rate. They add that, for within-items manipulations specifically, an intercept-only by-item structure can inflate it severely, whatever method is used to obtain p-values or intervals. Grammaticality is a within-item manipulation here, so an item term carrying only an intercept and a session slope is exactly that case for the effect under test.

Matuschek et al. (2017) argue the other side: a maximal model can cost power, and they recommend reducing to a structure the data support. We report the maximal structure nonetheless, for two reasons. Their argument is that a maximal structure the data do not support costs power, a result they establish in simulations of models that converge, so it does not turn on convergence failure. They themselves note that maximal models usually do converge when fitted in a Bayesian framework such as Stan, while explicitly leaving open whether fitting overparameterised maximal models makes sense there. Every cell of our reported grid converged under the maximal structure, on the same criteria as under the reduced one. The retention-free refits are diagnosed separately in Appendix D, where any fit that missed a criterion is marked. We also fit the whole grid a second time without the by-item grammaticality slope, so a reader can see which conclusions depend on it. That comparison follows.

Sensitivity to the by-item grammaticality slope

To show what the maximal structure buys, we refitted the whole grid without the by-item grammaticality slope, the structure a design in which grammaticality varied only within participants would have justified. All eighteen cells met the same convergence criteria under both.

In these data, the choice matters little. Posterior medians barely move between the two structures, and at the median the maximal structure's intervals are 1.01 times the simpler structure's width for the grammaticality main effect and 1.00 times for the grammaticality × session × language interaction. The credible cells are the same under both for the grammaticality main effect (5 of 18 cells under the maximal structure against 5 of 18 without the slope), for the grammaticality × session interaction (2 of 18 against 2 of 18) and for the three-way interaction (0 of 18 against 0 of 18). The one difference is in the grammaticality × language interaction for differential object marking in the early lateral window, which is credible without the by-item slope ($b = 0.0270$, 95% CrI $[0.0001, 0.0542]$, $p_d = .98$) and falls just short of credibility with it ($b = 0.0269$, 95% CrI $[-0.0002, 0.0543]$, $p_d = .97$), so that the count is 0 of 18 cells against 1 of 18. No conclusion rests on that cell. By-item variability in the grammaticality response is therefore small in these data, and the case for the maximal structure rests on the argument of the previous section, not on what it changes here.

The reprocessed recordings and the mislabelled exports

The Session 3 recordings of nine participants were high-pass filtered at 1 Hz where every other recording was filtered at 0.1 Hz. Four were identified when the analysis was first assembled and five more when the archived processing histories of every recording were audited. One further history, of a Session 2 recording, also reads 1 Hz, but its settings were edited after the export was written: rebuilt from the raw data, that export is reproduced only at 0.1 Hz, so it is used as it stands. We did not assess whether these participants' data look unusual, and the treatment of the 1 Hz data does not rest on that. It rests on the setting itself, which is a documented generator of artefacts in this kind of paradigm. Filtering data from a syntactic- and semantic-violation experiment at cutoffs of 0.3 Hz and above introduces spurious deflections of opposite polarity immediately before the true

N400 and P600 effects (Tanner et al., 2015), and 1 Hz lies well inside that range. The general principle, that a high-pass cutoff must be matched to the analysis window because an inappropriate one generates artefactual slow deflections, is set out by Widmann et al. (2015). Because the decision follows from the filter's known behaviour, and nothing observed in these participants bore on it, it is not a results-contingent choice.

Pooling the 1 Hz data with the rest would be the wrong remedy, because a distortion that precedes and reshapes the component of interest does not cancel in the difference between conditions (Tanner et al., 2015). Excluding them would remove about a sixth of the Session 3 ERP trials, and four or five participants from each group at that session. The raw continuous recordings of all nine sessions are archived, so they were instead reprocessed at 0.1 Hz, under a protocol fixed before any reprocessed output was inspected. The archived processing was first reconstructed by script and shown to reproduce every existing 1 Hz export to within the rounding of the export, and only then was the high-pass cutoff changed, with every other step and every retained trial kept as archived. The reprocessed data enter the primary analysis, and their exclusion is the single sensitivity analysis. That analysis can qualify a primary result but never create one, so running it adds no opportunity for a false positive. A primary ERP result counts as sample-sensitive if, without the reprocessed and recovered records, its median changes sign or its interval no longer excludes zero in the primary direction. Table A1 sets the two analyses side by side for every grammaticality term, and Table A2 does the same for the Session 4 to Session 6 contrasts. No primary ERP result is sample-sensitive by that rule. No grammaticality term becomes credible without the reprocessed and recovered records that was not credible with them, and one retention contrast does, as reported in the Results; under the protocol it cannot become a claim.

The same audit traced every exported trial to its source recording and checked every export for a stimulus-locked response, and three exports failed. The Session 3 export of one Mini-Norwegian participant was produced from another Mini-Norwegian participant's Session 3 recording, so it duplicates that participant's data: the two exports share every trial time stamp, and trials paired by time stamp correlate closely at the posterior midline, whereas trials of different

participants do not. The Session 6 export of one Mini-English participant was cut at the event times of that participant's Session 4 recording, and the Session 2 export of another Mini-English participant at the event times of a recording other than the one archived. The epochs of both are not time-locked to any stimulus: the average over their trials shows no reliable evoked response, and does not resemble the other participants' averages, whereas the same participants' other sessions show clear evoked responses. Unlike the filter setting, which distorts a real response, these exports contain no response of the participant-session they are named for, so no analysis can use them as they stand. A misaligned record can be recovered if its archived inspection and ocular correction were made on its own EEG, with only its event times wrong. The protocol tests that by first reproducing the legacy export from the archived recording, and only then re-segments the record at its own events. Both misaligned records passed. In each, the processing history had been built on the participant's own recording, within a stretch of it that the analyst had marked out, and only the event markers had come from another recording. The re-segmentation keeps that stretch, since outside it no epoch received any inspection, so 31 of the 144 condition events of the Session 2 record, which precede it, are not used. The recovered records enter the primary analysis and are left out of the robustness analysis with the reprocessed ones. The duplicated Session 3 record cannot be recovered, because every processing decision behind it was made on the other participant's recording, and it stays excluded.

Table A1

Grammaticality Terms of the ERP Models With and Without the Reprocessed and Recovered Records

	Primary			Without reprocessed		
Term	*Mdn*	*LL*	*UL*	*Mdn*	*LL*	*UL*
Gender agreement						
200–500 ms, lateral						
Grammaticality (G)	0.0197	−0.0003	0.0399	0.0184	−0.0015	0.0379
G × Language	−0.0043	−0.0245	0.0162	−0.0049	−0.0241	0.0149
G × Session	0.0055	−0.0103	0.0212	0.0061	−0.0100	0.0222
G × Session × Language	0.0019	−0.0140	0.0174	0.0011	−0.0149	0.0171
200–500 ms, midline						
Grammaticality (G)	0.0314	0.0108	0.0522	0.0293	0.0085	0.0500
G × Language	−0.0053	−0.0260	0.0156	−0.0049	−0.0254	0.0160
G × Session	−0.0005	−0.0182	0.0168	0.0003	−0.0175	0.0177
G × Session × Language	0.0053	−0.0124	0.0230	0.0047	−0.0130	0.0223
300–600 ms, lateral						
Grammaticality (G)	0.0252	0.0038	0.0472	0.0234	0.0013	0.0458
G × Language	−0.0069	−0.0285	0.0149	−0.0073	−0.0293	0.0152
G × Session	0.0114	−0.0051	0.0277	0.0127	−0.0044	0.0296
G × Session × Language	−0.0027	−0.0192	0.0137	−0.0036	−0.0207	0.0132
300–600 ms, midline						
Grammaticality (G)	0.0355	0.0129	0.0581	0.0331	0.0097	0.0565
G × Language	−0.0105	−0.0336	0.0125	−0.0096	−0.0328	0.0138
G × Session	0.0059	−0.0120	0.0237	0.0073	−0.0111	0.0257
G × Session × Language	0.0012	−0.0170	0.0189	0.0000	−0.0184	0.0184
400–900 ms, lateral						
Grammaticality (G)	0.0167	−0.0058	0.0392	0.0162	−0.0066	0.0390
G × Language	−0.0084	−0.0308	0.0140	−0.0077	−0.0303	0.0150
G × Session	0.0238	0.0059	0.0417	0.0255	0.0073	0.0439
G × Session × Language	−0.0034	−0.0216	0.0143	−0.0053	−0.0236	0.0127
400–900 ms, midline						
Grammaticality (G)	0.0198	−0.0036	0.0431	0.0187	−0.0053	0.0426
G × Language	−0.0154	−0.0389	0.0077	−0.0139	−0.0380	0.0104
G × Session	0.0215	0.0020	0.0410	0.0232	0.0035	0.0430
G × Session × Language	−0.0051	−0.0252	0.0140	−0.0071	−0.0271	0.0127
Differential object marking						
200–500 ms, lateral						
Grammaticality (G)	0.0422	0.0148	0.0696	0.0427	0.0136	0.0718
G × Language	0.0269	−0.0002	0.0543	0.0288	−0.0000	0.0582
G × Session	0.0042	−0.0159	0.0246	0.0052	−0.0165	0.0274
G × Session × Language	−0.0008	−0.0207	0.0195	−0.0044	−0.0262	0.0173

Table A1

Grammaticality Terms of the ERP Models With and Without the Reprocessed and Recovered Records

	Primary			Without reprocessed		
Term	*Mdn*	*LL*	*UL*	*Mdn*	*LL*	*UL*
200–500 ms, midline						
Grammaticality (G)	0.0433	0.0161	0.0708	0.0456	0.0161	0.0753
G × Language	0.0179	−0.0096	0.0457	0.0198	−0.0098	0.0493
G × Session	0.0011	−0.0224	0.0251	−0.0005	−0.0254	0.0252
G × Session × Language	0.0089	−0.0148	0.0325	0.0065	−0.0186	0.0318
300–600 ms, lateral						
Grammaticality (G)	0.0221	−0.0051	0.0503	0.0234	−0.0064	0.0535
G × Language	0.0206	−0.0073	0.0483	0.0222	−0.0077	0.0524
G × Session	0.0091	−0.0110	0.0289	0.0091	−0.0121	0.0305
G × Session × Language	−0.0010	−0.0206	0.0191	−0.0042	−0.0260	0.0174
300–600 ms, midline						
Grammaticality (G)	0.0224	−0.0045	0.0497	0.0261	−0.0034	0.0555
G × Language	0.0206	−0.0067	0.0478	0.0231	−0.0068	0.0526
G × Session	0.0065	−0.0160	0.0293	0.0039	−0.0201	0.0277
G × Session × Language	0.0059	−0.0168	0.0284	0.0039	−0.0197	0.0276
400–900 ms, lateral						
Grammaticality (G)	−0.0188	−0.0488	0.0103	−0.0206	−0.0527	0.0113
G × Language	0.0162	−0.0130	0.0451	0.0170	−0.0153	0.0493
G × Session	0.0044	−0.0175	0.0258	0.0081	−0.0153	0.0304
G × Session × Language	−0.0035	−0.0250	0.0183	−0.0061	−0.0288	0.0177
400–900 ms, midline						
Grammaticality (G)	−0.0217	−0.0501	0.0070	−0.0228	−0.0542	0.0088
G × Language	0.0263	−0.0027	0.0549	0.0289	−0.0024	0.0600
G × Session	0.0010	−0.0229	0.0243	0.0020	−0.0238	0.0268
G × Session × Language	0.0017	−0.0211	0.0256	0.0006	−0.0242	0.0258
Verb–object number agreement						
200–500 ms, lateral						
Grammaticality (G)	0.0115	−0.0125	0.0355	0.0134	−0.0108	0.0377
G × Language	0.0094	−0.0142	0.0334	0.0078	−0.0163	0.0320
G × Session	−0.0027	−0.0266	0.0211	−0.0007	−0.0248	0.0229
G × Session × Language	−0.0000	−0.0238	0.0233	−0.0019	−0.0259	0.0219
200–500 ms, midline						
Grammaticality (G)	0.0040	−0.0214	0.0291	0.0059	−0.0192	0.0314
G × Language	0.0074	−0.0177	0.0329	0.0056	−0.0195	0.0313
G × Session	−0.0074	−0.0325	0.0172	−0.0055	−0.0304	0.0195
G × Session × Language	0.0012	−0.0235	0.0257	−0.0009	−0.0258	0.0238

Table A1

Grammaticality Terms of the ERP Models With and Without the Reprocessed and Recovered Records

	Primary			Without reprocessed		
Term	*Mdn*	*LL*	*UL*	*Mdn*	*LL*	*UL*
300–600 ms, lateral						
Grammaticality (G)	0.0073	−0.0183	0.0329	0.0085	−0.0164	0.0349
G × Language	0.0199	−0.0059	0.0456	0.0184	−0.0071	0.0446
G × Session	0.0029	−0.0227	0.0278	0.0045	−0.0213	0.0293
G × Session × Language	−0.0016	−0.0268	0.0235	−0.0031	−0.0287	0.0221
300–600 ms, midline						
Grammaticality (G)	−0.0028	−0.0294	0.0235	−0.0014	−0.0279	0.0251
G × Language	0.0160	−0.0104	0.0428	0.0145	−0.0116	0.0411
G × Session	0.0011	−0.0251	0.0267	0.0027	−0.0236	0.0284
G × Session × Language	0.0024	−0.0238	0.0284	0.0007	−0.0256	0.0264
400–900 ms, lateral						
Grammaticality (G)	−0.0200	−0.0503	0.0114	−0.0182	−0.0487	0.0131
G × Language	0.0184	−0.0118	0.0490	0.0167	−0.0143	0.0475
G × Session	0.0174	−0.0118	0.0452	0.0195	−0.0096	0.0476
G × Session × Language	0.0142	−0.0146	0.0427	0.0122	−0.0164	0.0405
400–900 ms, midline						
Grammaticality (G)	−0.0257	−0.0590	0.0083	−0.0245	−0.0576	0.0093
G × Language	0.0164	−0.0180	0.0499	0.0147	−0.0182	0.0488
G × Session	0.0187	−0.0084	0.0465	0.0208	−0.0074	0.0483
G × Session × Language	0.0160	−0.0121	0.0438	0.0142	−0.0144	0.0426

Note. Values are posterior summaries of the four grammaticality terms of every ERP cell under the reported specification (maximal random-effect structure, informative priors), fitted to the primary sample (Primary) and refitted without the nine reprocessed Session 3 recordings and the two recovered records (Without reprocessed), in within-participant standard-deviation units of single-trial amplitude. G = grammaticality, coded grammatical = +0.5 before standardisation, so a positive coefficient is a relative negativity for violations; language is coded Mini-Norwegian = +0.5. *Mdn* = posterior median; *LL* and *UL* = lower and upper limits of the 95% equal-tailed credible interval. An em dash marks an estimate not available.

Table A2

Change in the ERP Grammaticality Effect From Session 4 to Session 6 With and Without the Reprocessed and Recovered Records

	Primary			Without reprocessed		
Contrast	*Mdn*	*LL*	*UL*	*Mdn*	*LL*	*UL*
Gender agreement						
200–500 ms, lateral						
Overall	0.0052	−0.0096	0.0199	0.0056	−0.0092	0.0204
Mini-Norwegian	0.0070	−0.0143	0.0286	0.0067	−0.0147	0.0280
Mini-English	0.0035	−0.0166	0.0237	0.0045	−0.0158	0.0248
Group difference	0.0036	−0.0262	0.0327	0.0021	−0.0274	0.0315
200–500 ms, midline						
Overall	−0.0005	−0.0170	0.0158	0.0003	−0.0162	0.0163
Mini-Norwegian	0.0046	−0.0194	0.0286	0.0047	−0.0188	0.0275
Mini-English	−0.0053	−0.0277	0.0171	−0.0039	−0.0266	0.0185
Group difference	0.0099	−0.0233	0.0430	0.0086	−0.0239	0.0411
300–600 ms, lateral						
Overall	0.0107	−0.0048	0.0260	0.0117	−0.0041	0.0273
Mini-Norwegian	0.0081	−0.0146	0.0304	0.0082	−0.0146	0.0307
Mini-English	0.0131	−0.0083	0.0343	0.0150	−0.0067	0.0368
Group difference	−0.0051	−0.0361	0.0257	−0.0067	−0.0382	0.0244
300–600 ms, midline						
Overall	0.0055	−0.0112	0.0222	0.0068	−0.0102	0.0237
Mini-Norwegian	0.0066	−0.0177	0.0311	0.0068	−0.0177	0.0312
Mini-English	0.0043	−0.0184	0.0278	0.0067	−0.0169	0.0305
Group difference	0.0023	−0.0318	0.0354	0.0000	−0.0340	0.0339
400–900 ms, lateral						
Overall	0.0222	0.0056	0.0390	0.0235	0.0067	0.0404
Mini-Norwegian	0.0189	−0.0060	0.0436	0.0186	−0.0057	0.0424
Mini-English	0.0254	0.0024	0.0485	0.0283	0.0052	0.0520
Group difference	−0.0065	−0.0406	0.0269	−0.0098	−0.0435	0.0233
400–900 ms, midline						
Overall	0.0201	0.0019	0.0384	0.0214	0.0033	0.0397
Mini-Norwegian	0.0151	−0.0118	0.0410	0.0148	−0.0116	0.0409
Mini-English	0.0248	−0.0001	0.0507	0.0277	0.0030	0.0532
Group difference	−0.0097	−0.0473	0.0262	−0.0131	−0.0500	0.0234
Differential object marking						
200–500 ms, lateral						
Overall	0.0054	−0.0205	0.0318	0.0069	−0.0216	0.0358
Mini-Norwegian	0.0044	−0.0329	0.0423	0.0012	−0.0391	0.0422
Mini-English	0.0063	−0.0298	0.0427	0.0125	−0.0269	0.0532
Group difference	−0.0020	−0.0533	0.0504	−0.0116	−0.0687	0.0452

Table A2

Change in the ERP Grammaticality Effect From Session 4 to Session 6 With and Without the Reprocessed and Recovered Records

	Primary			Without reprocessed		
Contrast	*Mdn*	*LL*	*UL*	*Mdn*	*LL*	*UL*
200–500 ms, midline						
Overall	0.0015	−0.0289	0.0324	−0.0007	−0.0333	0.0330
Mini-Norwegian	0.0129	−0.0301	0.0577	0.0078	−0.0390	0.0559
Mini-English	−0.0100	−0.0519	0.0330	−0.0091	−0.0550	0.0371
Group difference	0.0229	−0.0381	0.0839	0.0169	−0.0487	0.0833
300–600 ms, lateral						
Overall	0.0118	−0.0141	0.0373	0.0119	−0.0159	0.0399
Mini-Norwegian	0.0105	−0.0263	0.0472	0.0065	−0.0346	0.0471
Mini-English	0.0128	−0.0229	0.0484	0.0177	−0.0218	0.0566
Group difference	−0.0025	−0.0532	0.0494	−0.0111	−0.0680	0.0455
300–600 ms, midline						
Overall	0.0084	−0.0206	0.0378	0.0051	−0.0263	0.0362
Mini-Norwegian	0.0159	−0.0260	0.0585	0.0102	−0.0341	0.0556
Mini-English	0.0009	−0.0393	0.0414	0.0000	−0.0435	0.0439
Group difference	0.0153	−0.0434	0.0733	0.0103	−0.0515	0.0723
400–900 ms, lateral						
Overall	0.0057	−0.0226	0.0333	0.0106	−0.0200	0.0398
Mini-Norwegian	0.0011	−0.0391	0.0413	0.0027	−0.0405	0.0453
Mini-English	0.0102	−0.0290	0.0484	0.0185	−0.0242	0.0597
Group difference	−0.0090	−0.0645	0.0473	−0.0159	−0.0754	0.0463
400–900 ms, midline						
Overall	0.0013	−0.0296	0.0314	0.0026	−0.0311	0.0350
Mini-Norwegian	0.0036	−0.0407	0.0473	0.0035	−0.0440	0.0494
Mini-English	−0.0009	−0.0446	0.0400	0.0021	−0.0455	0.0470
Group difference	0.0045	−0.0544	0.0660	0.0014	−0.0635	0.0677
Verb–object number agreement						
200–500 ms, lateral						
Overall	−0.0055	−0.0545	0.0432	−0.0014	−0.0510	0.0472
Mini-Norwegian	−0.0055	−0.0766	0.0627	−0.0052	−0.0752	0.0634
Mini-English	−0.0053	−0.0737	0.0631	0.0026	−0.0673	0.0718
Group difference	−0.0002	−0.0975	0.0957	−0.0077	−0.1066	0.0901
200–500 ms, midline						
Overall	−0.0153	−0.0667	0.0353	−0.0112	−0.0626	0.0402
Mini-Norwegian	−0.0125	−0.0852	0.0582	−0.0131	−0.0849	0.0586
Mini-English	−0.0180	−0.0897	0.0541	−0.0092	−0.0818	0.0628
Group difference	0.0051	−0.0963	0.1055	−0.0035	−0.1060	0.0980

Table A2

Change in the ERP Grammaticality Effect From Session 4 to Session 6 With and Without the Reprocessed and Recovered Records

	Primary			Without reprocessed		
Contrast	*Mdn*	*LL*	*UL*	*Mdn*	*LL*	*UL*
300–600 ms, lateral						
Overall	0.0060	−0.0465	0.0569	0.0092	−0.0438	0.0603
Mini-Norwegian	0.0026	−0.0717	0.0760	0.0027	−0.0721	0.0757
Mini-English	0.0094	−0.0642	0.0814	0.0156	−0.0583	0.0904
Group difference	−0.0064	−0.1098	0.0963	−0.0128	−0.1180	0.0908
300–600 ms, midline						
Overall	0.0022	−0.0514	0.0548	0.0055	−0.0486	0.0585
Mini-Norwegian	0.0072	−0.0688	0.0820	0.0068	−0.0695	0.0815
Mini-English	−0.0027	−0.0781	0.0719	0.0038	−0.0727	0.0807
Group difference	0.0099	−0.0975	0.1164	0.0029	−0.1054	0.1085
400–900 ms, lateral						
Overall	0.0358	−0.0243	0.0927	0.0400	−0.0197	0.0979
Mini-Norwegian	0.0644	−0.0207	0.1470	0.0645	−0.0193	0.1457
Mini-English	0.0062	−0.0773	0.0875	0.0140	−0.0688	0.0973
Group difference	0.0582	−0.0599	0.1752	0.0502	−0.0675	0.1666
400–900 ms, midline						
Overall	0.0383	−0.0173	0.0954	0.0428	−0.0153	0.0993
Mini-Norwegian	0.0709	−0.0094	0.1513	0.0707	−0.0103	0.1522
Mini-English	0.0050	−0.0752	0.0853	0.0125	−0.0703	0.0968
Group difference	0.0656	−0.0497	0.1796	0.0585	−0.0593	0.1753

Note. Values are posterior summaries of the change in the ERP grammaticality effect from Session 4 to Session 6, as in the table of Session 4 to Session 6 changes, from the reported models fitted to the primary sample (Primary) and refitted without the nine reprocessed Session 3 recordings and the two recovered records (Without reprocessed), in within-participant standard-deviation units of amplitude. Positive values indicate an effect that strengthened over the interval. Overall = averaged over the two groups; Group difference = Mini-Norwegian minus Mini-English. *Mdn* = posterior median; *LL* and *UL* = lower and upper limits of the 95% equal-tailed credible interval. An em dash marks a contrast not available.

The trials the trigger rule discards

The two-step segmentation described in the Method keeps a critical word only when its condition trigger followed its property trigger by exactly five samples, and about a third of the words failed that rule. Fewer trials lower precision, but they bias the grammaticality contrast only if

the discarded trials differ from the kept ones within a condition. The reprocessing protocol fixed a check of that assumption before any reprocessed output was inspected, on the seventeen recordings whose discarded epochs could be rebuilt from the raw data: the nine reprocessed Session 3 records and eight 0.1 Hz recordings rebuilt for the filter validation, four from Session 2 and four from Session 3. Each recording was first rebuilt and checked against its archived exports. Every grammatical and ungrammatical epoch was then locked to the word-onset trigger (−96 to 1,096 ms), and each window mean was adjusted for the pre-stimulus baseline within each recording, as the ERP models adjust it. In each property, window and macroregion, the check estimates D, the difference between discarded and kept trials in the ungrammatical-minus-grammatical contrast, weighting the recordings equally. Omitting the discarded trials shifts the analysed contrast by the discarded share times D, so the check passes where the 90% interval of D lies within ±0.91 μV (two one-sided tests at $\alpha = .05$), which keeps that shift within about 0.33 μV. Three of the twelve cells passed. Across the 17 recordings, 36% of the eligible grammatical and ungrammatical words had been discarded. The interval excluded zero in gender agreement, 200–500 ms, lateral ($D = -0.55$ μV, 90% CI $[-1.07, -0.03]$), where the discarded trials carried the more negative ungrammatical-minus-grammatical difference. Across twelve cells, 1.2 such intervals would be expected on average at the 90% level if the two sets of trials did not differ. The interval reaching furthest from zero was that of gender agreement, 200–500 ms, midline ($D = -0.86$ μV, 90% CI $[-1.88, 0.16]$), which would shift the analysed contrast by −0.31 μV at its median. Every cell is given in Table A3. The check has about 80% power for a single cell, and much less for all of them jointly, so a pass bounds gross differences without establishing that the two sets of trials are equivalent. It admits no trial to any analysis.

The decoding correction family

The Method states how the second layer of multiplicity control on the decoding analyses is defined. Correcting instead across the pooled, per-session and per-session-by-language cells of one property would span nested, overlapping re-analyses of the same trials. Across those, the number of hypotheses is not well defined, and the positive-dependence condition that licenses

Table A3

Discarded Against Kept Trials in the Ungrammatical-Minus-Grammatical Contrast

Window and macroregion	*k*	Kept	Discarded	*D*	90% interval *LL*	*UL*	Within bound
Gender agreement							
200–500 ms, lateral	17	1,040	580	−0.55	−1.07	−0.03	No
200–500 ms, midline	17	1,040	580	−0.86	−1.88	0.16	No
300–600 ms, lateral	17	1,040	580	−0.33	−0.89	0.23	Yes
300–600 ms, midline	17	1,040	580	−0.59	−1.53	0.34	No
400–900 ms, lateral	17	1,040	580	−0.35	−0.97	0.28	No
400–900 ms, midline	17	1,040	580	−0.57	−1.47	0.33	No
Differential object marking							
200–500 ms, lateral	13	805	440	0.03	−0.75	0.81	Yes
200–500 ms, midline	13	805	440	−0.32	−1.51	0.87	No
300–600 ms, lateral	13	805	440	0.02	−0.70	0.74	Yes
300–600 ms, midline	13	805	440	−0.44	−1.69	0.82	No
400–900 ms, lateral	13	805	440	0.16	−0.64	0.95	No
400–900 ms, midline	13	805	440	−0.09	−1.59	1.40	No

Note. *D* = difference between the trials the five-sample trigger rule discards and the trials it keeps in the ungrammatical-minus-grammatical contrast of the window mean (μV), locked to the word-onset trigger and adjusted for the pre-stimulus baseline within each recording, averaged over recordings with equal weight. *k* = recordings; Kept and Discarded = grammatical and ungrammatical trials in each set. *LL* and *UL* = limits of the 90% interval of the two one-sided tests; a cell passes where both lie within ±0.91 μV. Gender agreement is recorded at Sessions 2 and 3 and differential object marking at Session 3, so the two properties draw on different recordings.

Benjamini–Hochberg under dependence cannot be established. A false-discovery rate computed over such a family is not interpretable. The three properties' pooled time-courses rest on disjoint sets of trials, and correcting each property's family separately keeps every family self-contained.

Within a property, the second layer adds no error control to the maximum-cluster test. That test controls the family-wise error rate across the time-course, in the sense the Results describe, and the false-discovery rate cannot exceed it. A Benjamini–Hochberg adjustment never lowers a *p* value, so applied to these values it can only remove clusters. The layer was specified, and stated in the Method, before the decoding of the corrected sample was run. That it adds no error control was noted in an audit of the analysis code before that run, and the decision to keep it was taken once the

corrected results were available. We keep it so that no decision rests on a correction chosen after those results were seen. Its cost is visible wherever it matters, because the clusters that pass the maximum-cluster test are reported beside those that survive both layers. For gender agreement, it is what separates four clusters that pass the first test from none that survive both. With its 17 confirmatory clusters, the smallest cluster-level $p = .003$ becomes $q = .051$. The family itself was narrowed to the pooled time-course in July 2026, after a run under the wider family had been inspected, for the reasons given above.

Appendix B

Retention contrasts in full

The Results report the contrasts that carry the argument. The three tables below give the full record: every Session 4 to Session 6 contrast of the reported ERP models (Table B1), every retention-free contrast from the refits that carry an explicit retention indicator (Table B2), and the sensitivity and criterion contrasts derived from the judgement models (Table B3). All three are read from the pooled summary artefacts and reproduce with them.

Table B1

Change in the ERP Grammaticality Effect From Session 4 to Session 6

	Lateral				Midline			
Contrast	*Mdn*	*LL*	*UL*	p_d	*Mdn*	*LL*	*UL*	p_d
Gender agreement								
200–500 ms								
Overall	0.0052	−0.0096	0.0199	.76	−0.0005	−0.0170	0.0158	.52
Mini-Norwegian	0.0070	−0.0143	0.0286	.74	0.0046	−0.0194	0.0286	.65
Mini-English	0.0035	−0.0166	0.0237	.63	−0.0053	−0.0277	0.0171	.68
Group difference	0.0036	−0.0262	0.0327	.59	0.0099	−0.0233	0.0430	.72
300–600 ms								
Overall	0.0107	−0.0048	0.0260	.91	0.0055	−0.0112	0.0222	.74
Mini-Norwegian	0.0081	−0.0146	0.0304	.76	0.0066	−0.0177	0.0311	.70
Mini-English	0.0131	−0.0083	0.0343	.89	0.0043	−0.0184	0.0278	.65
Group difference	−0.0051	−0.0361	0.0257	.62	0.0023	−0.0318	0.0354	.55
400–900 ms								
Overall	0.0222	0.0056	0.0390	.995	0.0201	0.0019	0.0384	.98
Mini-Norwegian	0.0189	−0.0060	0.0436	.93	0.0151	−0.0118	0.0410	.87
Mini-English	0.0254	0.0024	0.0485	.99	0.0248	−0.0001	0.0507	.97
Group difference	−0.0065	−0.0406	0.0269	.64	−0.0097	−0.0473	0.0262	.70
Differential object marking								
200–500 ms								
Overall	0.0054	−0.0205	0.0318	.66	0.0015	−0.0289	0.0324	.54
Mini-Norwegian	0.0044	−0.0329	0.0423	.59	0.0129	−0.0301	0.0577	.72
Mini-English	0.0063	−0.0298	0.0427	.63	−0.0100	−0.0519	0.0330	.68
Group difference	−0.0020	−0.0533	0.0504	.53	0.0229	−0.0381	0.0839	.77
300–600 ms								
Overall	0.0118	−0.0141	0.0373	.82	0.0084	−0.0206	0.0378	.72
Mini-Norwegian	0.0105	−0.0263	0.0472	.71	0.0159	−0.0260	0.0585	.78
Mini-English	0.0128	−0.0229	0.0484	.76	0.0009	−0.0393	0.0414	.52
Group difference	−0.0025	−0.0532	0.0494	.54	0.0153	−0.0434	0.0733	.70

Table B1

Change in the ERP Grammaticality Effect From Session 4 to Session 6

	Lateral				Midline			
Contrast	*Mdn*	*LL*	*UL*	p_d	*Mdn*	*LL*	*UL*	p_d
400–900 ms								
Overall	0.0057	−0.0226	0.0333	.66	0.0013	−0.0296	0.0314	.54
Mini-Norwegian	0.0011	−0.0391	0.0413	.52	0.0036	−0.0407	0.0473	.57
Mini-English	0.0102	−0.0290	0.0484	.70	−0.0009	−0.0446	0.0400	.52
Group difference	−0.0090	−0.0645	0.0473	.63	0.0045	−0.0544	0.0660	.56
Verb–object number agreement								
200–500 ms								
Overall	−0.0055	−0.0545	0.0432	.59	−0.0153	−0.0667	0.0353	.73
Mini-Norwegian	−0.0055	−0.0766	0.0627	.56	−0.0125	−0.0852	0.0582	.64
Mini-English	−0.0053	−0.0737	0.0631	.56	−0.0180	−0.0897	0.0541	.69
Group difference	−0.0002	−0.0975	0.0957	.50	0.0051	−0.0963	0.1055	.54
300–600 ms								
Overall	0.0060	−0.0465	0.0569	.59	0.0022	−0.0514	0.0548	.53
Mini-Norwegian	0.0026	−0.0717	0.0760	.53	0.0072	−0.0688	0.0820	.57
Mini-English	0.0094	−0.0642	0.0814	.60	−0.0027	−0.0781	0.0719	.53
Group difference	−0.0064	−0.1098	0.0963	.55	0.0099	−0.0975	0.1164	.57
400–900 ms								
Overall	0.0358	−0.0243	0.0927	.89	0.0383	−0.0173	0.0954	.91
Mini-Norwegian	0.0644	−0.0207	0.1470	.93	0.0709	−0.0094	0.1513	.96
Mini-English	0.0062	−0.0773	0.0875	.56	0.0050	−0.0752	0.0853	.55
Group difference	0.0582	−0.0599	0.1752	.84	0.0656	−0.0497	0.1796	.87

Note. Values are posterior summaries of the change in the ERP grammaticality effect from Session 4 to Session 6 in the reported models, in within-participant standard-deviation units of amplitude. Positive values indicate an effect that strengthened over the interval. Overall = averaged over the two groups; Group difference = Mini-Norwegian minus Mini-English. *Mdn* = posterior median; *LL* and *UL* = lower and upper limits of the 95% equal-tailed credible interval; p_d = probability of direction. An em dash marks a contrast not yet available. The judgement models' retention quantities are reported in signal-detection terms with the sensitivity and criterion contrasts in this appendix.

Table B2

Deviation of the ERP Grammaticality Effect at Session 6 From the Training Trend (Retention-Free Refits)

	Lateral				Midline			
Contrast	*Mdn*	*LL*	*UL*	p_d	*Mdn*	*LL*	*UL*	p_d
Gender agreement								
200–500 ms								
Overall	0.014	−0.040	0.069	.69	0.016	−0.042	0.074	.71
Mini-Norwegian	0.039	−0.040	0.117	.83	0.046	−0.038	0.128	.86
Mini-English	−0.009	−0.087	0.069	.59	−0.011	−0.094	0.070	.61
Group difference	0.047	−0.062	0.159	.80	0.057	−0.062	0.174	.83
300–600 ms								
Overall	0.005	−0.052	0.062	.57	0.007	−0.053	0.066	.59
Mini-Norwegian	0.042	−0.039	0.125	.85	0.052	−0.034	0.138	.88
Mini-English	−0.029	−0.111	0.051	.76	−0.035	−0.119	0.049	.79
Group difference	0.071	−0.044	0.189	.89	0.087	−0.035	0.208	.92
400–900 ms								
Overall	−0.016	−0.077	0.044	.70	−0.019	−0.081	0.042	.73
Mini-Norwegian	0.015	−0.070	0.102	.64	0.019	−0.072	0.107	.66
Mini-English	−0.045	−0.130	0.040	.85	−0.055	−0.143	0.033	.89
Group difference	0.061	−0.061	0.181	.84	0.074	−0.052	0.199	.88
Differential object marking								
200–500 ms								
Overall	−0.017	−0.090	0.056	.68	−0.033	−0.110	0.043	.80
Mini-Norwegian	−0.073	−0.174	0.030	.92	−0.127	−0.235	−0.019	.99
Mini-English	0.037	−0.068	0.142	.76	0.059	−0.049	0.168	.86
Group difference	−0.110	−0.256	0.037	.93	−0.187	−0.336	−0.034	.99
300–600 ms								
Overall	−0.026	−0.101	0.048	.76	−0.047	−0.125	0.033	.87
Mini-Norwegian	−0.072	−0.176	0.032	.91	−0.139	−0.250	−0.031	.99
Mini-English	0.018	−0.091	0.125	.63	0.045	−0.068	0.158	.78
Group difference	−0.091	−0.239	0.061	.88	−0.185	−0.341	−0.027	.99

Table B2

Deviation of the ERP Grammaticality Effect at Session 6 From the Training Trend (Retention-Free Refits)

	Lateral				Midline			
Contrast	*Mdn*	*LL*	*UL*	p_d	*Mdn*	*LL*	*UL*	p_d
400–900 ms								
Overall	0.004	−0.076	0.084	.54	−0.016	−0.097	0.065	.65
Mini-Norwegian	−0.027	−0.138	0.084	.68	−0.095	−0.206	0.016	.95
Mini-English	0.034	−0.079	0.148	.72	0.061	−0.055	0.175	.85
Group difference	−0.062	−0.218	0.098	.78	−0.156	−0.316	0.003	.97

Note. Values are posterior summaries of the deviation of the grammaticality effect at Session 6 from the trend established across the training sessions, from refits that carry an explicit retention indicator (coded +0.5 at Session 6), in within-participant standard-deviation units of amplitude. Positive values mean that the grammaticality coefficient at Session 6 exceeds the value the training trend predicts; because a positive grammaticality coefficient is a relative negativity for violations, a positive deviation shifts the effect towards violation negativity. Verb–object number agreement is omitted because its indicator is not identifiable. Overall = averaged over the two groups; Group difference = Mini-Norwegian minus Mini-English. *Mdn*, *LL*, *UL* and p_d are as in the table of Session 4 to Session 6 changes.

Table B3

Group Differences and Changes in Sensitivity and Criterion From the Judgement Models

	Sensitivity (d_L)				Criterion (c_L)			
Contrast	*Mdn*	*LL*	*UL*	p_d	*Mdn*	*LL*	*UL*	p_d
Gender agreement								
Group difference (Mini-Norwegian minus Mini-English)								
Session 2	−0.1860	−1.9260	1.5396	.58	−0.1294	−0.6665	0.4036	.68
Session 3	−0.7967	−2.3483	0.7646	.84	−0.4571	−0.9747	0.0524	.96
Session 4	−1.4052	−3.0447	0.2135	.96	−0.7833	−1.3061	−0.2706	.999
Session 6	−2.0106	−3.9262	−0.1055	.98	−1.1089	−1.6611	−0.5691	> .999
Averaged over sessions	−0.9691	−2.5172	0.5863	.89	−0.5488	−1.0640	−0.0414	.98
Change from Session 2 to Session 6								
Mini-Norwegian	−3.4798	−4.8128	−2.1514	> .999	−1.0421	−1.2675	−0.8181	> .999
Mini-English	−1.6652	−2.9456	−0.3076	.99	−0.0635	−0.3303	0.1996	.68
Group difference	−1.8196	−3.7083	−0.0001	.97	−0.9785	−1.3237	−0.6381	> .999

Table B3

Group Differences and Changes in Sensitivity and Criterion From the Judgement Models

	Sensitivity (d_L)				Criterion (c_L)			
Contrast	*Mdn*	*LL*	*UL*	p_d	*Mdn*	*LL*	*UL*	p_d
Differential object marking								
Group difference (Mini-Norwegian minus Mini-English)								
Session 3	−1.7677	−4.9228	1.4172	.86	−0.2588	−1.1064	0.5722	.73
Session 4	−0.4503	−2.3619	1.5505	.68	−0.3171	−1.1371	0.4908	.78
Session 6	0.8595	−1.0592	2.9003	.81	−0.3762	−1.2089	0.4417	.82
Averaged over sessions	−0.6612	−2.7045	1.4855	.73	−0.3076	−1.1305	0.5017	.78
Change from Session 3 to Session 6								
Mini-Norwegian	−4.3521	−6.8253	−1.8189	> .999	0.4886	0.2644	0.7158	> .999
Mini-English	−6.9881	−9.3845	−4.4559	> .999	0.6029	0.3434	0.8688	> .999
Group difference	2.6369	−0.9124	6.1872	.93	−0.1127	−0.4596	0.2306	.74
Verb–object number agreement								
Group difference (Mini-Norwegian minus Mini-English)								
Session 4	0.9378	−1.4338	3.3221	.78	−0.1140	−0.7343	0.5115	.64
Session 6	−0.5665	−2.0112	0.8474	.79	−0.2170	−0.8468	0.4198	.75
Averaged over sessions	0.3399	−1.2727	1.9299	.66	−0.1550	−0.7665	0.4554	.69
Change from Session 4 to Session 6								
Mini-Norwegian	−2.9194	−4.7792	−1.0899	.999	−0.0874	−0.2982	0.1221	.80
Mini-English	−1.4154	−3.2011	0.3879	.94	0.0136	−0.1903	0.2207	.55
Group difference	−1.4899	−4.1514	1.0771	.88	−0.1023	−0.3960	0.1874	.75

Note. Values are posterior summaries derived from the draws of the reported accuracy models (see *Signal-detection reading of the judgement models*): sensitivity d_L on the logit scale, and criterion c_L, negative for a readiness to answer grammatical. Group differences are Mini-Norwegian minus Mini-English, at each session and averaged over sessions. Changes run from the property's first session to Session 6, for each group and as the group difference in that change. *Mdn* = posterior median; *LL* and *UL* = lower and upper limits of the 95% equal-tailed credible interval; p_d = probability of direction.

Appendix C

Prior specifications

The reasoning behind each prior is given in the Method, under *Priors*. What follows is the specification itself. Every mean and standard deviation is on the standardised scale, so a coefficient is in standard-deviation units, within participant for the ERP models. Prior means encode a direction and a regularised magnitude only, never a literature effect size, because the sources do not supply a transferable one. A directional prior is placed only where the literature supports a sign. Everything else, including the grammaticality slope in the 200–500 ms window, differential object marking at 300–600 ms and the language-conditioned interaction under test, takes the model's zero-centred default.

Table C1
Prior Distributions of the ERP and Judgement-Accuracy Models

	Prior	
Parameter	Informative	Weak
ERP models (Gaussian likelihood)		
Intercept	$N(0, 0.5)$	$N(0, 1)$
Grammaticality, agreement properties, 300–600 ms	$N(-0.1, 0.3)$	$N(0, 1)$
Grammaticality, agreement properties, 400–900 ms	$N(-0.2, 0.3)$	$N(0, 1)$
Grammaticality, differential object marking, 400–900 ms	$N(-0.2, 0.4)$	$N(0, 1)$
All other coefficients	$N(0, 0.5)$	$N(0, 1)$
Residual standard deviation	half-$t(3, 0, 2.5)$	half-$t(3, 0, 2.5)$
Random-effect standard deviations	half-$t(3, 0, 2.5)$	half-$t(3, 0, 2.5)$
Random-effect correlations	LKJ(2)	LKJ(2)
Judgement-accuracy models (Bernoulli likelihood, logit link)		
Intercept	$N(1.5, 1)$	$N(0, 1.5)$
Session	$N(0.3, 0.5)$	$N(0, 1.5)$
All other coefficients	$N(0, 1)$	$N(0, 1.5)$
Random-effect standard deviations	half-$t(3, 0, 2.5)$	half-$t(3, 0, 2.5)$
Random-effect correlations	LKJ(2)	LKJ(2)

Note. Informative = the literature-derived priors of the reported models; Weak = the weakly informative baseline of the prior-sensitivity refits. Zero-centred entries carry no directional claim, and the standard deviation is what bounds a prior's influence. $N(\mu, \sigma)$ = normal distribution with mean μ and standard deviation σ; half-$t(\nu, 0, \sigma)$ = Student-t distribution with ν degrees of freedom and scale σ, truncated at zero; LKJ(η) = Lewandowski–Kurowicka–Joe distribution over correlation matrices with shape η. All coefficients are on the standardised scale.

Appendix D

Convergence diagnostics

This appendix reports the realised sampling diagnostics for every fitted model. Each of the 18 ERP cells was fitted under two random-effect structures (base and maximal) and two prior sets, the literature-derived informative priors reported throughout and the weakly informative baseline used for the sensitivity analysis, giving 72 grid fits. Each of the 3 accuracy models was fitted under the two prior sets with its reported by-participant structure, giving 6 fits. The retention-free refits add 24 fits, the 12 gender-agreement and differential-object-marking cells under both structures with the informative priors. A further 15 accuracy refits were run as sensitivity analyses, each property refitted with a by-participant grammaticality × session slope, with the crossed by-item term, with both, with the questionnaire's multilingual language diversity score as a between-participant covariate and with the reaction-time screen of the earlier preprocessing. Estimates from all but the diversity refit appear alongside the signal-detection results. The diversity score is not the covariate the ERP models carry, which is the L1 dominance score of the same questionnaire, so this refit does not remedy the asymmetry noted in the Method. The robustness analysis of the reprocessing protocol adds 18 ERP fits, the 18 cells refitted without the reprocessed and recovered records under the maximal structure and the informative priors. The data-side sensitivity variants are identified in the tables' Fit column. In all, 135 models were fitted, and the five groups above account for every one of them. Table D1 (ERP models) and Table D2 (judgement-accuracy models) give the realised diagnostics for every one, so a model that converged comfortably can be told from one that converged marginally.

Table D1

Convergence Diagnostics for the ERP Models

			Maximal structure			Base structure		
Window (ms)	Region	Fit	$\widehat{R}$	Bulk	Tail	$\widehat{R}$	Bulk	Tail
Gender agreement								
200–500	Lateral	Informative prior	1.006	1,051	2,308	1.002	1,007	2,814
		Informative prior, reprocessed excluded	1.002	1,360	3,292	—	—	—
		Weak prior	1.006	953	2,253	1.001	2,145	5,050
		Retention-free	1.005	1,139	3,364	1.002	1,035	2,157
	Midline	Informative prior	1.003	1,276	2,890	1.006	1,171	2,464
		Informative prior, reprocessed excluded	1.002	1,521	3,162	—	—	—
		Weak prior	1.006	1,086	2,856	1.002	1,389	3,792
		Retention-free	1.004	1,322	3,489	1.003	1,125	3,188
300–600	Lateral	Informative prior	1.009	832	1,900	1.004	1,147	2,198
		Informative prior, reprocessed excluded	1.004	847	2,212	—	—	—
		Weak prior	1.003	872	1,918	1.002	1,146	2,527
		Retention-free	1.002	883	2,714	1.003	785	2,598
	Midline	Informative prior	1.004	1,239	2,650	1.006	1,023	2,138
		Informative prior, reprocessed excluded	1.001	1,273	2,617	—	—	—
		Weak prior	1.001	1,321	2,410	1.003	1,416	2,936
		Retention-free	1.004	1,186	2,733	1.003	1,115	2,662
400–900	Lateral	Informative prior	1.002	1,213	2,487	1.003	868	2,381
		Informative prior, reprocessed excluded	1.004	1,166	3,013	—	—	—
		Weak prior	1.002	1,149	2,812	1.005	1,130	2,663
		Retention-free	1.005	1,194	3,342	1.002	1,174	2,948
	Midline	Informative prior	1.004	1,165	2,204	1.005	1,370	3,484
		Informative prior, reprocessed excluded	1.004	1,368	3,558	—	—	—
		Weak prior	1.005	1,427	3,324	1.005	1,067	2,945
		Retention-free	1.005	1,265	3,484	1.005	1,188	2,909

Table D1

Convergence Diagnostics for the ERP Models

			Maximal structure			Base structure		
Window (ms)	Region	Fit	$\widehat{R}$	Bulk	Tail	$\widehat{R}$	Bulk	Tail
Differential object marking								
200–500	Lateral	Informative prior	1.002	2,248	6,221	1.002	1,519	3,265
		Informative prior, reprocessed excluded	1.002	2,368	3,742	—	—	—
		Weak prior	1.001	2,356	6,394	1.003	1,229	2,765
		Retention-free	1.001	2,218	5,724	1.001	2,220	6,669
	Midline	Informative prior	1.003	2,233	5,674	1.003	1,486	4,311
		Informative prior, reprocessed excluded	1.003	2,056	4,873	—	—	—
		Weak prior	1.001	2,225	4,979	1.001	2,206	5,806
		Retention-free	1.002	1,956	5,895	1.003	1,597	3,851
300–600	Lateral	Informative prior	1.002	2,611	5,110	1.003	2,507	6,209
		Informative prior, reprocessed excluded	1.001	1,968	3,015	—	—	—
		Weak prior	1.001	2,535	4,680	1.001	2,377	4,799
		Retention-free	1.001	3,422	8,269	1.002	1,965	4,995
	Midline	Informative prior	1.001	2,774	7,022	1.003	2,039	4,036
		Informative prior, reprocessed excluded	1.001	2,834	5,407	—	—	—
		Weak prior	1.003	2,880	7,234	1.001	2,439	6,097
		Retention-free	1.001	3,710	8,753	1.001	2,797	6,038
400–900	Lateral	Informative prior	1.003	2,338	4,643	1.001	2,743	3,998
		Informative prior, reprocessed excluded	1.001	3,107	7,229	—	—	—
		Weak prior	1.001	3,236	5,170	1.001	2,543	4,312
		Retention-free	1.001	2,562	5,069	1.002	2,101	2,580
	Midline	Informative prior	1.001	3,635	8,332	1.002	2,345	5,239
		Informative prior, reprocessed excluded	1.001	2,743	5,873	—	—	—
		Weak prior	1.001	2,556	5,191	1.002	2,731	6,049
		Retention-free	1.001	3,694	9,400	1.001	3,172	7,862

Table D1

Convergence Diagnostics for the ERP Models

			Maximal structure			Base structure		
Window (ms)	Region	Fit	$\widehat{R}$	Bulk	Tail	$\widehat{R}$	Bulk	Tail
Verb–object number agreement								
200–500	Lateral	Informative prior	1.001	3,957	7,261	1.001	3,133	6,795
		Informative prior, reprocessed excluded	1.002	3,145	6,516	—	—	—
		Weak prior	1.001	3,409	6,169	1.001	3,890	5,966
	Midline	Informative prior	1.001	3,762	7,636	1.001	3,157	7,538
		Informative prior, reprocessed excluded	1.001	3,788	8,151	—	—	—
		Weak prior	1.003	3,740	7,410	1.002	2,546	6,834
300–600	Lateral	Informative prior	1.002	4,004	7,754	1.002	3,846	5,466
		Informative prior, reprocessed excluded	1.001	4,657	8,103	—	—	—
		Weak prior	1.001	4,263	7,242	1.002	3,307	4,329
	Midline	Informative prior	1.001	3,491	6,895	1.001	3,561	7,869
		Informative prior, reprocessed excluded	1.001	4,950	10,506	—	—	—
		Weak prior	1.001	3,576	7,750	1.002	2,796	5,904
400–900	Lateral	Informative prior	1.004	2,391	4,020	1.001	2,390	3,990
		Informative prior, reprocessed excluded	1.001	3,183	7,474	—	—	—
		Weak prior	1.002	2,136	3,960	1.001	3,150	6,315
	Midline	Informative prior	1.001	2,503	6,448	1.002	1,949	4,476
		Informative prior, reprocessed excluded	1.002	2,201	5,005	—	—	—
		Weak prior	1.002	2,552	6,572	1.002	2,541	5,651

Note. Maximal structure = the reported by-participant and by-item random-effect structure; Base structure = the same without the by-item grammaticality slope. Informative prior = the reported priors; Weak prior = the weakly informative baseline; Retention-free = the refit that frees the Session-6 step from the training trend (informative priors). Reprocessed excluded = the robustness refit that leaves out the nine reprocessed Session 3 recordings and the two recovered records. $\widehat{R}$ = largest rank-normalised $\widehat{R}$; Bulk and Tail = smallest bulk and tail effective sample size (ESS), each over all parameters of the model. $\widehat{R}$ is rounded to three decimals; the criteria ($\widehat{R} < 1.01$, both ESS > 400, no divergent transitions) were applied before rounding. An em dash marks a fit whose diagnostics are not yet available.

Table D2

Convergence Diagnostics for the Judgement-Accuracy Models

Fit	$\widehat{R}$	ESS	
		Bulk	Tail
Gender agreement			
Informative prior	1.001	3,538	7,350
Weak prior	1.002	3,621	6,624
Grammaticality × session slope added	1.003	2,689	6,001
Language-diversity covariate added	1.001	4,002	7,151
By-item term added	1.001	2,721	5,249
Reaction-time screen applied	1.002	3,239	6,727
Grammaticality × session slope and by-item term added	1.006	2,516	4,960
Differential object marking			
Informative prior	1.001	5,270	8,365
Weak prior	1.001	5,419	8,477
Grammaticality × session slope added	1.001	6,636	11,292
Language-diversity covariate added	1.001	6,190	11,800
By-item term added	1.003	1,134	2,399
Reaction-time screen applied	1.001	4,467	7,943
Grammaticality × session slope and by-item term added	1.004	1,024	2,156
Verb–object number agreement			
Informative prior	1.000	6,679	11,649
Weak prior	1.001	6,496	11,858
Grammaticality × session slope added	1.001	6,817	11,813
Language-diversity covariate added	1.001	6,472	11,751
By-item term added	1.002	3,534	7,225
Reaction-time screen applied	1.001	5,884	10,443
Grammaticality × session slope and by-item term added	1.002	3,432	6,983

Note. Informative prior = the reported model; Weak prior = the weakly informative baseline. The other refits, all under the informative priors, add a by-participant grammaticality × session slope, the LHQ3 Multilingual Language Diversity Score as a covariate, the crossed by-item term of the ERP models or both the slope and that term, or apply the reaction-time screen of the earlier preprocessing. $\widehat{R}$ = largest rank-normalised $\widehat{R}$; Bulk and Tail = smallest bulk and tail effective sample size (ESS), each over all parameters of the model. $\widehat{R}$ is rounded to three decimals; the criteria ($\widehat{R} < 1.01$, both ESS > 400, no divergent transitions) were applied before rounding.

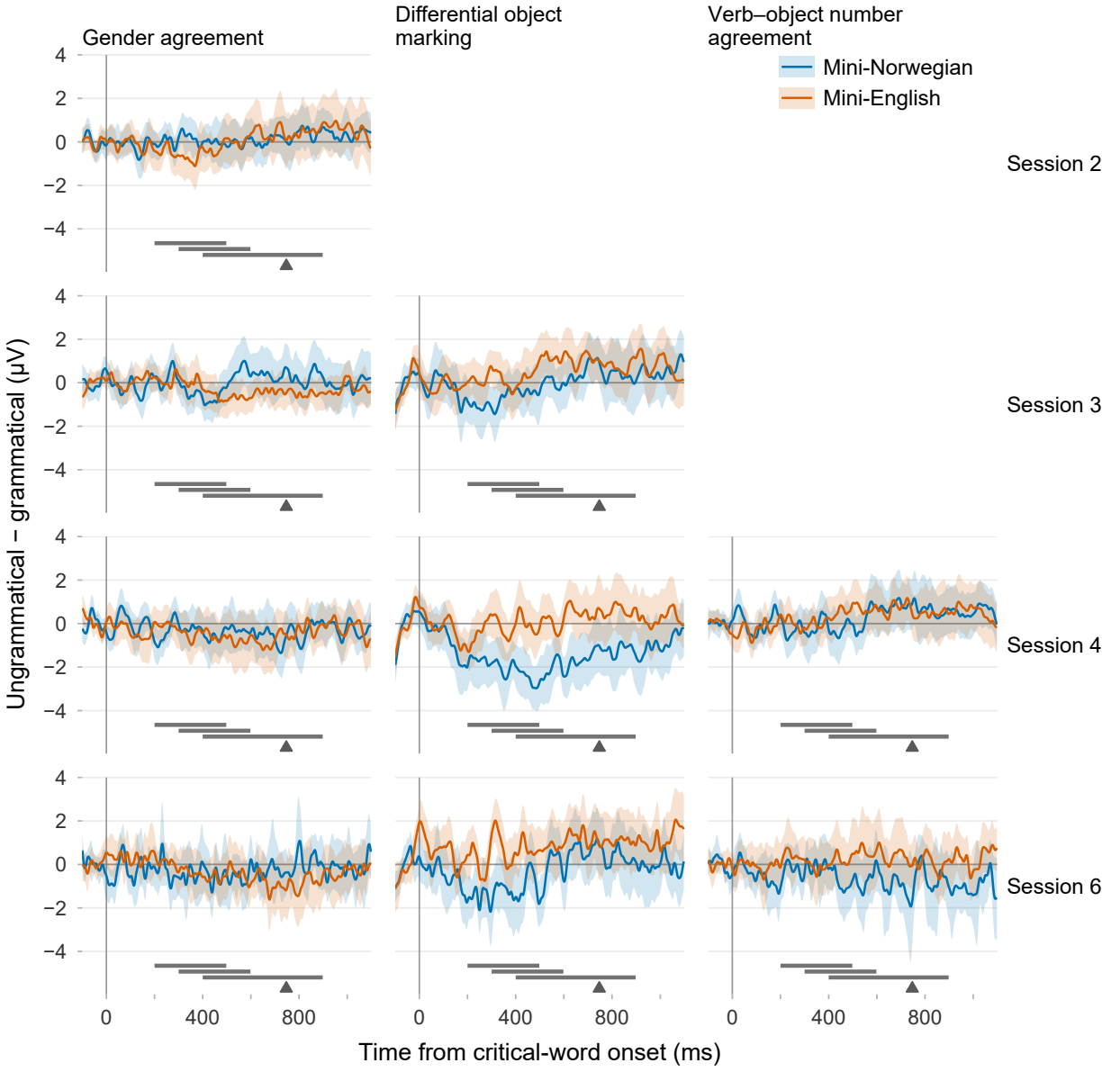
Appendix E

Group displays by session

The main-text group displays pool each property's sessions. This appendix resolves them by session, so that the development the models estimate can be inspected in each group. Like the pooled displays, they are descriptive and were not used to select any analysis. Figure E1 gives the difference waves at the midline posterior cluster and Figure E2 the late-window maps. The observed judgement accuracies behind Figure 11 are given in Figure E3.

Figure E1

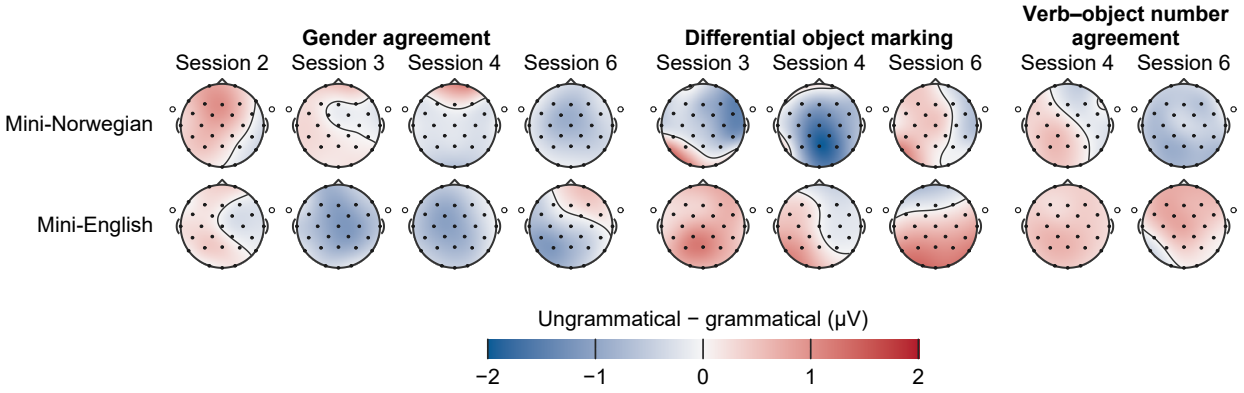
Difference Waves (Ungrammatical Minus Grammatical) at the Midline Posterior Cluster by Session, Miniature Language and Property



Note. Each line is the participant-paired difference between ungrammatical and grammatical baseline-corrected waveforms at the midline posterior cluster at one session, averaged over participants, for the Mini-Norwegian (blue) and Mini-English (vermilion) groups. Ribbons are pointwise 95% normal intervals from the participant-level standard error. Cells are blank at sessions before a property was introduced, since differential object marking entered the design at Session 3 and verb–object number agreement at Session 4. Sessions 2 to 4 are a week apart. Session 6 is the retention test, about four months after Session 4 without exposure, and draws on fewer participants (see the participant flow figure). Polarity, window bars and the next-word triangle are as in the pooled difference-wave figure in the Results.

Figure E2

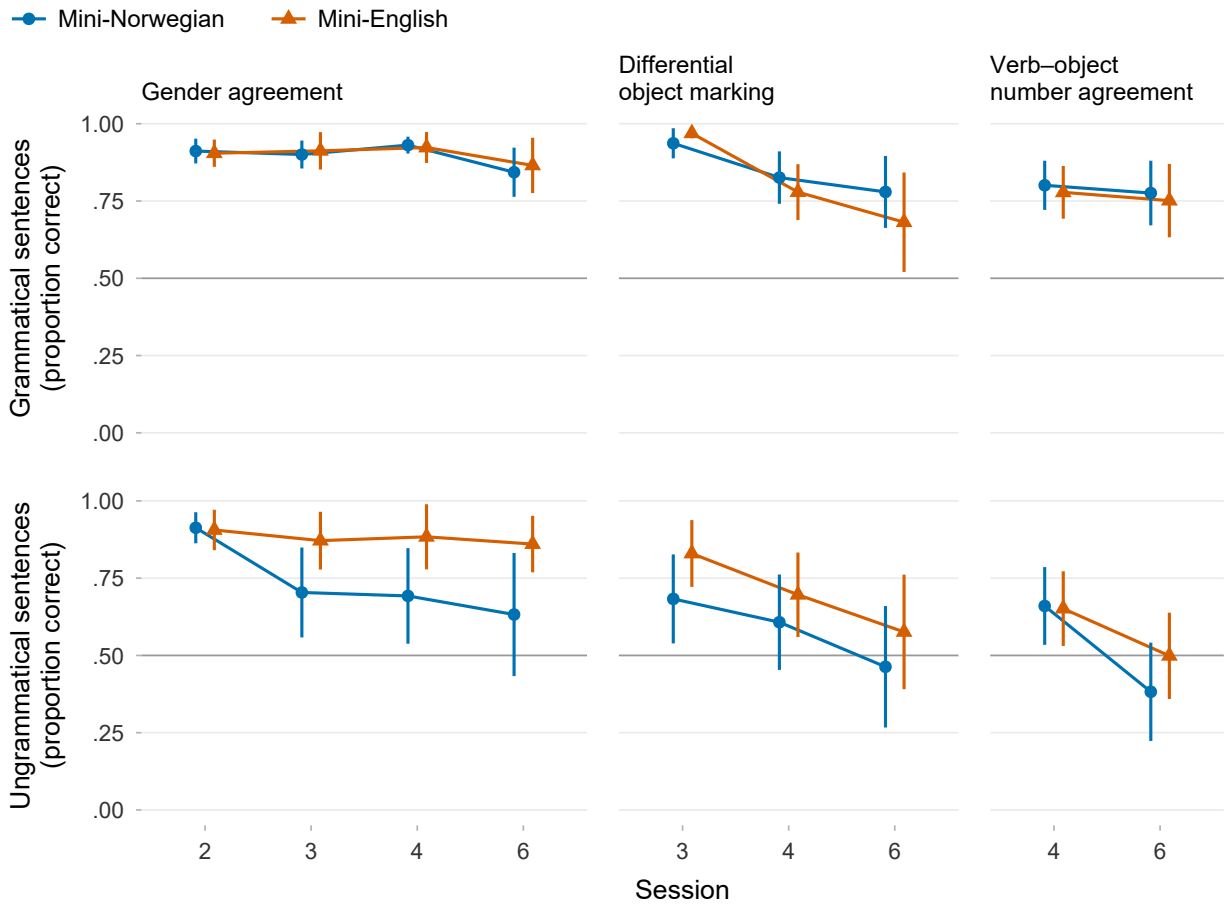
Scalp Topography of the Grammaticality Effect by Property, Session and Miniature Language, 400–900 ms



Note. Maps show the participant-paired ungrammatical-minus-grammatical mean amplitude in the 400–900 ms window at each session, baseline-corrected within participant and averaged over participants. Each block is one property, with sessions in columns and the Mini-Norwegian and Mini-English groups in rows. Only the sessions at which a property was tested are shown: gender agreement from Session 2, differential object marking from Session 3, and verb–object number agreement at Sessions 4 and 6. Sessions 2 to 4 are a week apart. Session 6 is the retention test, about four months after Session 4 without exposure, and draws on fewer participants (see the participant flow figure). Surface estimation, montage, symbols, polarity and the colour scale are those of the scalp-topography figure in the Results, so colours can be compared directly between the two figures.

Figure E3

Observed Judgement Accuracy by Sentence Type, Miniature Language, Property and Session



Note. Points are group means of participant-level proportion correct on grammatical sentences (upper row) and on ungrammatical sentences (lower row), and every participant receives equal weight. Vertical lines are 95% *t* intervals over participants, and the horizontal line marks chance (.50). Each property is plotted from the session at which it entered the design. Accuracy on the two sentence types moves in opposite directions when a group's criterion shifts, which is why the Results analyse sensitivity and criterion and not accuracy on either sentence type. Sessions are equally spaced, as in the models, although Sessions 2 to 4 are a week apart and Session 6 follows about four months without exposure.